# Reinforcement Learning for Lean Proofs with a Small Language Model

Midterm research report

Manas Ravulapalli

Revised 25 October 2026

**Abstract**

We test whether reinforcement learning (RL) improves Lean 4 proof generation by Qwen2.5-0.5B-Instruct after supervised fine-tuning (SFT). From the same SFT checkpoint, prerequisite-ordered RL accepts 255 of 4,048 training attempts, compared with 164 for shuffled RL, with one training run per arm. The unchanged, SFT and both RL models each prove 0 of 8 DEV theorems. On a separate selection set, SFT proves 1 of 163 assessed theorems, shuffled RL 1 of 164, and RL on 62 partially solvable training theorems 2 of 164. No arm meets the registered promotion rule. As in earlier experiments, training gains do not establish better performance on held-out proofs. Exploratory analyses find sparse rewards and narrower tactic choices, but no controlled run tests whether these explain the missing transfer.

**Reading guide.** Section 4 reports the main study; Appendix A gives the detailed exploratory analyses. Earlier experiments and protocols are in Appendices B, E and F; unrun follow-ups are in Appendix G. The larger evaluation review pack in Appendix H is separate from the main DEV and TEST sets. Experiments ended on 20 October 2026. Later analyses use saved records. The README gives current status; `docs/EXPERIMENTS.md` indexes the protocols and evidence in `docs/EXPERIMENT_HISTORY.md`.

**Models and measures.** B is the unchanged model; S is B after SFT. R is the shuffled-RL model; C is the prerequisite-ordered-RL model. Both start from S. These names refer to the 18 and 19 October study. Earlier experiments define their own model letters. TRAIN supplies learning examples, DEV supports choices, and TEST is reserved for final evaluation. The separate EXP-028 selection set supports exploratory comparisons. Filtered RL (`RL_filtered`, E6) trains on the 62 TRAIN theorems that S proves in 1 to 3 of 4 reference attempts. An *action* is one model response containing Lean tactics; an *attempt* is a sequence of actions on one theorem. An accepted attempt contains a complete proof that Lean accepts. A proved theorem has at least one accepted attempt.

# Contents


1 Research question and scope 3

2 Formulation as an MDP 4

3 Earlier datasets and training stages 7

4 Main study: proof completion after SFT and RL 8

5 Limitations 20

A Post-hoc analyses of 20 and 21 October 2026 23

B Earlier experiments 39

C Reward-verifier exploits and data exposure 43

D Training and verification implementation defects 43

E Plan recorded on 13 October 44

F Historical protocols: generalization and curriculum 49

G Follow-up designs that did not run 55

H Evaluation review pack: statements and expected answers 57

# 1 Research question and scope

We test whether RL helps a small language model write Lean 4 proofs in school and early-university number theory: parity, divisibility, gcd and lcm, primes, and congruences. The main comparison asks what RL adds beyond SFT and whether theorem order changes that outcome. Training acceptance and held-out proof completion are separate endpoints. Earlier pilots examine initialization, rewards and task feasibility; they are not one matched algorithm comparison. `docs/EXPERIMENTS.md` lists their titles and motivations.

The midterm goal was to produce some complete proofs that Lean accepts. The longer-term goal is proof strategy on competition-style theorems. We seek more proofs within a fixed compute budget by studying which theorems the model practices, their order, and how rewards are assigned across proof actions.

**Rationale.** Lean checks proofs during learning and supplies the proof reward without a human grader. This setting is called RL with verifiable rewards (RLVR); the name describes the reward, not the optimizer. A model trained without a checker can learn false statements. Here, the terminal proof reward is sparse: it arrives only after a complete accepted proof. The model writes informal reasoning and a Lean tactic in one response, avoiding the errors and compute of a separate translator.

**Lean proof verification.** A Lean theorem consists of a statement and a proof. Tactics transform the remaining *goals* until none remain. Table 1 gives an illustrative proof; Figure 1 shows the acceptance checks.

Table 1: A tactic proof that $a \mid b$ and $b \mid c$ imply $a \mid c$ (here $a \mid b$ means $b = ak$). This example explains the idea; it is not a measured proof.

| Action | Goal before the action | Tactic (the action) | Effect |
|---|---|---|---|
| 1 | $a \mid c$, given $a \mid b$, $b \mid c$ | `obtain ⟨k, hk⟩ := h1` | Adds $b = ak$ |
| 2 | $a \mid c$, given $b = ak$, $b \mid c$ | `obtain ⟨l, hl⟩ := h2` | Adds $c = bl$ |
| 3 | $a \mid c$, given $b = ak$, $c = bl$ | `exact ⟨k * l, by rw [hl, hk, Nat.mul_assoc]⟩` | $c = (ak)l = a(kl)$. No goals remain |

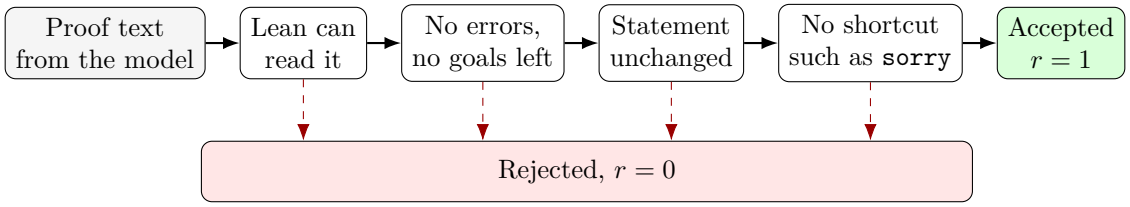



Figure 1: When Lean accepts a proof. Every check must pass. One wrong character or one invented lemma name causes a rejection.

**Experimental setup.** We use `Qwen/Qwen2.5-0.5B-Instruct` [1] (494,032,768 parameters), Lean `v4.10.0-rc1` and a fixed mathlib revision. Runs use one rented NVIDIA L40S GPU and a shared CPU server.

## 2 Formulation as an MDP

We model proof generation as a Markov decision process (MDP) [3]. At step $t$, the model (policy $\pi_\theta$) observes state $s_t$, chooses action $a_t$, receives reward $r_t$, and reaches state $s_{t+1}$. Lean is the environment. The two formulations differ in action size: a complete proof or an interactive tactic block (Figure 2).

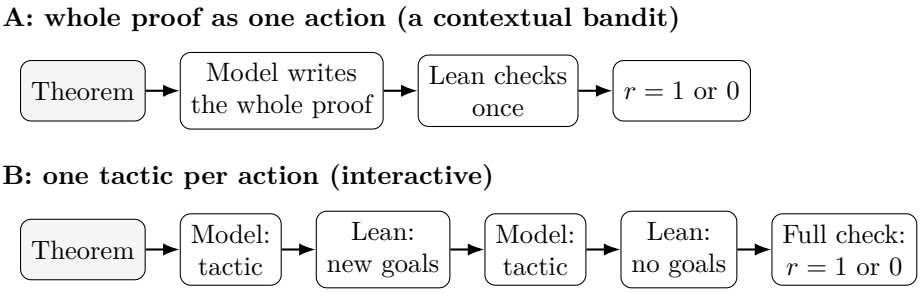



Figure 2: The two formulations. In A, the model receives no feedback while it writes (Experiments 4 to 6, 20 to 22). In B, Lean runs each tactic first (Experiments 19 and 23).

Table 2 defines Formulation B, and Figure 3 shows it as a graph.

Table 2: The interactive MDP (environment version 1).

| Part | Definition |
|---|---|
| State $s_t$ | The current Lean goals. At $t = 0$, the theorem. In an *assisted* attempt, part of a known proof is applied first |
| Action $a_t$ | A short tactic block (at most 64 tokens in the main configuration of this version) |
| Transition | Lean runs the block. If it succeeds, the new goals are $s_{t+1}$ |
| End of episode | No goals remain (then Lean checks the full proof), Lean reports an error, or the action budget is used (4 actions) |
| Reward | 1 at the end if Lean accepts the full proof, otherwise 0 |
| Discount | $\gamma = 1$ |

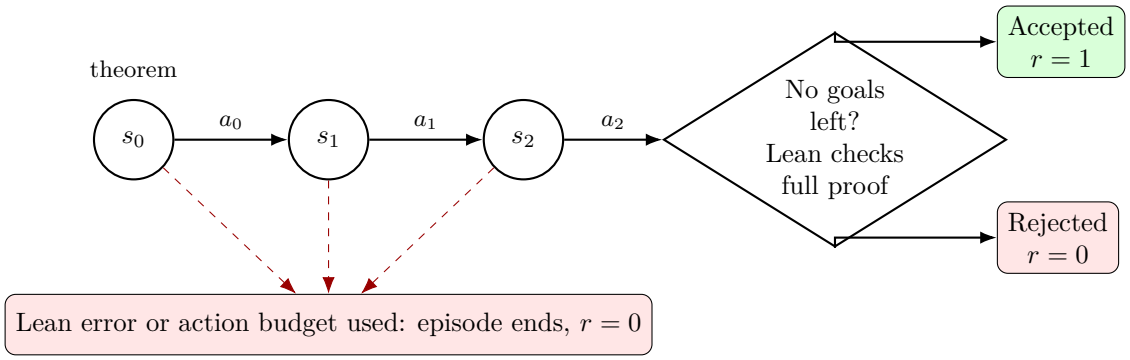



Figure 3: Formulation B for the proof of Table 1. Solid arrows: the tactic succeeds. Dashed arrows: an error ends the episode with reward 0.

Version 1 hides the theorem, proof so far and remaining action budget; the unobserved budget makes the process not exactly Markov. It also ends the episode after one error. Version 2 adds a fuller state and replaces error termination with a self-loop; version 2.1 adds tactic chunks and backtracking (Tables 12 and 13). Version 2.2 fixes the deployed interface (Table 6); Section E explains these changes.

### 2.1 REINFORCE and GRPO estimators

We maximize the expected return $J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\big[\sum_t \gamma^t r_t\big]$. The only reward is at the end, so the return of an episode $\tau$ is $R(\tau) \in \{0, 1\}$. For one theorem, we sample $G$ episodes (one episode is one attempt) and use the REINFORCE estimator [4]

$$\nabla_\theta J(\theta) \approx \frac{1}{G}\sum_{i=1}^{G} A_i \, \nabla_\theta \log \pi_\theta(\tau_i). \tag{1}$$

The *advantage* $A_i$ compares attempt $i$ with the other attempts. The interactive runs use REINFORCE with a leave-one-out baseline (RLOO) [5]. The whole-proof runs use group relative policy optimization (GRPO) [6, 7]:

$$A_i^{\text{RLOO}} = R_i - \frac{1}{G-1}\sum_{j\neq i} R_j, \qquad A_i^{\text{GRPO}} = \frac{R_i - \bar{R}}{s_R + \varepsilon}, \tag{2}$$

where $\bar{R}$ and $s_R$ are the mean and standard deviation of the $G$ rewards.

### 2.2 Reward variation within sampled groups

If all $G$ attempts fail, every $R_i = 0$, so every $A_i = 0$, and the update in Equation (1) is exactly zero. For this undiscounted, terminal-only reward, the same is true when Lean accepts all attempts. The later discounted runs can update when all proofs succeed at different lengths (Section 4.1). The model learns only from *mixed groups*, which contain at least one accepted and one rejected attempt (Figure 4).

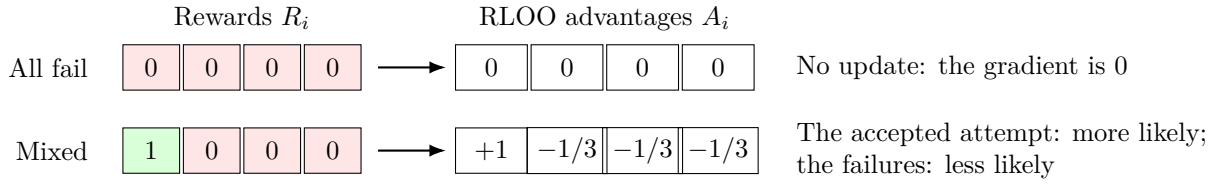



Figure 4: A group needs both an accepted and a rejected attempt ($G = 4$).

If Lean accepts each attempt with probability $p$, a group of $G$ independent attempts is mixed with probability

$$P_{\text{mix}}(p, G) = 1 - p^G - (1-p)^G. \tag{3}$$

Figure 5 shows that this probability is small when $p$ is small. This equation explains most of our failures.

### 2.3 Target-likelihood reward (RLP)

Some experiments used the RLP reward [8] instead of Lean. For a question $x$, reasoning $c$, and known answer $y$, the reward is $\log p_\theta(y \mid x, c) - \log p_{\bar{\theta}}(y \mid x)$, where $p_{\bar{\theta}}$ is a slowly updated copy of the model. The reward is high when the reasoning makes the correct answer more probable. It does not check whether the reasoning is correct (Figure 6). We call it a *heuristic* reward.

**Measures.** *Accepted attempts* counts sampled proofs that Lean accepts, for example 26/48. *Theorem coverage* counts theorems with at least one accepted attempt. One standard tactic proves 28 of the 420 validation statements without a model, so the main validation set is the other 392 (*non-trivial*) statements.

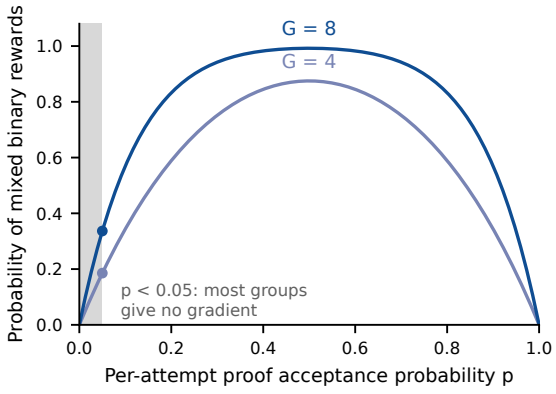



Figure 5: Probability that a group is mixed (Equation 3). At $p = 0.05$ it is 0.185 for $G = 4$ and 0.337 for $G = 8$.

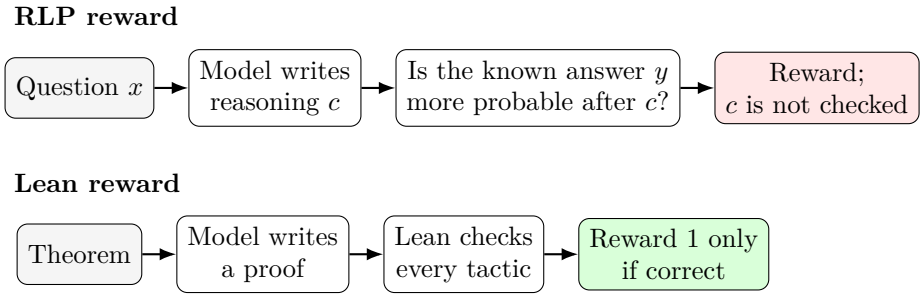



Figure 6: The two rewards. RLP rewards reasoning that makes a known answer more probable, correct or not. Lean rewards only a correct proof.

# 3 Earlier datasets and training stages

Table 3: Data sources and their use.

| Source | Content | Use |
|---|---|---|
| LeanDojo Benchmark 4 [2] | Theorems and tactic proofs from mathlib; number-theory subset of 706 training, 17 validation, 15 test theorems | Supervised training; short proofs for interactive RL |
| Lean Workbook [9] | Competition-style theorems as Lean statements | Pool of 4,306 number-theory statements that Lean checks |
| Number-theory SFT set | 5,812 examples: Lean-accepted proofs, mathlib syntax examples, proof variants | Second supervised stage |
| Hand-written Lean theorems | Logic, arithmetic, divisibility, induction | Experiments 5, 6; DEV of Experiment 19 |
| Hand-written arithmetic questions | 448 questions on gcd, inverses, primes, congruences | RLP (Experiment 9) |
| Open textbook (Judson) | Chapters of *Abstract Algebra: Theory and Applications* | RLP (Experiment 7) |
| Curriculum templates | 192 training theorems in 5 stages and 152 held-out theorems, all checked by Lean | Final run (Section E) |

Table 3 and Figure 7 describe the earlier datasets and splits. Renamed copies of a theorem stay together, and DEV is excluded from training. The main study uses a later release (Section 4.1; Appendix F.5).

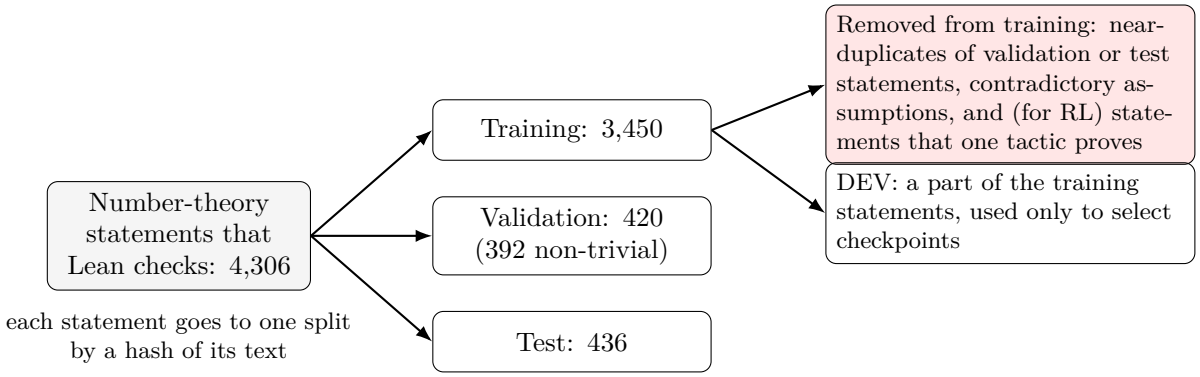



Figure 7: Data splits. Two statements are near-duplicates when their token sets overlap by at least 85%.

**Training stages.** The earlier checkpoint chain (Figure 8) starts with SFT v2: one epoch on LeanDojo proofs, with 4,526 optimizer updates of 64 examples. Files containing validation or test theorems are excluded. Common SFT adds supervised training on the number-theory set. SFT learns from reference proofs; RL learns from the model's own attempts and therefore needs some accepted proofs.

**Curricula.** We use curricula to give the model tasks it can sometimes solve: supply knowledge, supply a proof prefix (a *reverse curriculum* [10]; Figure 9), or ask for one action rather than a whole

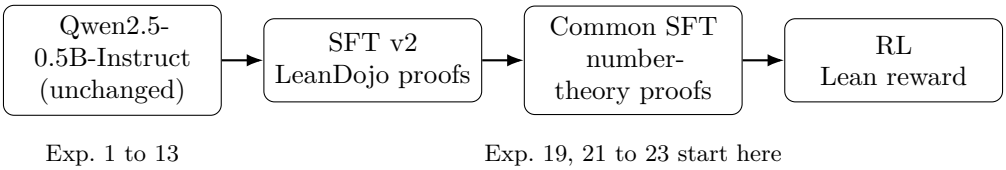



Figure 8: The training chain.

proof. Reverse curricula offer few levels here: 85% of the LeanDojo pool's linear tactic proofs have one or two tactics, and only 24 have five or more. To isolate ordering, ordered and shuffled arms receive each theorem equally often. Table 4 lists the comparisons.

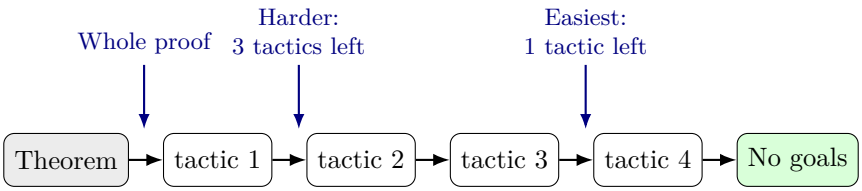



Figure 9: A reverse curriculum on one known proof. Each blue arrow is a start point; the earlier tactics are given. Training moves the start point toward the theorem.

Table 4: Curricula by experiment.

| Exp. | Training data | Order and support | Comparison |
|---|---|---|---|
| 4 | LeanDojo number-theory theorems | Grouped by topic | Random order |
| 5 | Hand-written Lean theorems | Logic, arithmetic, divisibility, induction; support decreases | Random order |
| 6 | Theorems with part of the proof given | Missing expression, then tactic, then body | Random order |
| 9 | 448 arithmetic questions | Prerequisites first, with replay | None |
| 13 | 72 questions in 5 versions | Support decreases from full information to none | Shuffled order |
| 20 | Number-theory proofs | Short proof endings, longer endings, whole proofs | SFT parent |
| 19 | 3 practice proof endings | Fixed mixture | SFT parent |
| 23 | 22 Lean Workbook theorems | Structure, lemma use, combined steps, whole proof | Shuffled order |
| Final | S0 to S4 templates | Identity, one lemma, two steps, chain of subgoals, 3 to 6 steps | Shuffled; adaptive |

## 4 Main study: proof completion after SFT and RL

The 18–19 October study evaluates training stages (EXP-026), prerequisite-ordered versus shuffled RL (EXP-027), and exploratory interventions (EXP-028). EXP-028 compares S, its derived arms and two SFT runs from B on a separate selection set, using S (E0) as baseline (Table 5). B, S, R and C refer to the main study (Section F.5), not the EXP-023 arms in Table 15.

Counts are recounted from saved records, without new Lean or model runs. Costs and selection-set construction counts come from the dated records in `docs/EXPERIMENT_HISTORY.md`, EXP-028. The sign-flip $p$-values are exact, as recorded in the 19 October recount; earlier entries used Monte Carlo estimates.

**Selection set and decision rule.** The EXP-028 selection set comes from the existing corpus. Its construction excludes the TRAIN theorems and their aliases, every SFT source, and the DEV and TEST sets. Of 216 candidates, admission removes 14 that a single fixed tactic proves and 1 with known exposure of the parent. The selection set holds 164 of the 201 admitted theorems. Of the 164 theorems, 79 fall in the four TRAIN topics and 85 in other topics. The median reference proof has 29 lines. Each arm makes 2 attempts per theorem, and each comparison pairs the theorems that both arms assess. Promotion requires a paired gain of at least 5 points over E0 and a bootstrap 95% interval excluding 0. E3 instead uses its SFT seed control, E5. The bootstrap resamples theorems with 10,000 replicates; TEST remains unused.

**Unexecuted and failed study arms.** E4 was to test SFT from B on the 1,880 base SFT examples plus 64 SFT examples mined from step search on TRAIN. Its SFT stage failed after 4 s, because the mining script wrote rows without the `format` field that the trainer requires. The frozen lease program could not rerun the stage, so E4 needs a later approved lease. E1, step search without the automation fallback, was dropped before any paid run. Automation runs only after a failed policy attempt, so E1's model-only outcome equals E2's up to Lean timing.

**Compute cost.** EXP-028 used an estimated USD 52.28 of instance compute against USD 165.32 reserved. The main run, which trained S, R and C, used an estimated USD 33.77 against USD 102.31 reserved. Both figures use the quoted on-demand rate and exclude storage and transfer. They are estimates, not invoices. The EXP-028 total uses USD 6.23 for Lease 3; another dated entry gives USD 6.59 for the same lease.

Table 5: Runs reported in this section. S is the SFT checkpoint of the main run. B is its pinned upstream model. The selection set has 164 theorems. E0 and E2 assess 163, because one group is censored.

| Run | Start | What changed | Population | Attempts |
|---|---|---|---|---|
| S (E0) | S | Nothing: one candidate action per state, model-only; the baseline | Selection set (163 assessed) | 2 per theorem |
| E2, step search | S | 4 candidate actions per state, with an automation fallback | Selection set (163 assessed) | 2 per theorem |
| E3, lemma-section SFT | B | SFT on the SFT examples with a lemma section; evaluated with retrieval | Selection set | 2 per theorem |
| E5, SFT seed 2 | B | SFT on the SFT examples without the lemma section, with a second seed | Selection set | 2 per theorem |
| R (E7) | S | REINFORCE on all 506 TRAIN theorems in shuffled order | Selection set | 2 per theorem |
| `RL_filtered` (E6) | S | REINFORCE on 62 TRAIN theorems with S acceptance in $(0, 0.75]$; 1,012 groups of 4 | Selection set | 2 per theorem |
| Kimina-Prover-Distill-0.6B, two draws | Released model | A different model; whole proofs with long reasoning; 8,096-token limit per attempt | Selection set | 8 per theorem per draw |
| `pass_rates` | S | Nothing; measures S on TRAIN | 506 TRAIN theorems | 4 per theorem |
| R and C training | S | Same theorem multiset; R in shuffled order, C in prerequisite order run twice | 506 TRAIN theorems | 1,012 groups of 4 |
| `dev_R`, `dev_C` | R, C | Nothing; final weights | 8 DEV theorems | 32 per model |

### 4.1 Interactive proof environment and policy-gradient estimator

Every run in this section uses environment version 2.2 (Section E). Table 6 defines it with the settings of these runs. Figure 10 shows how the environment processes one action. The settings come from the recorded run configurations. The EXP-030 record in `docs/EXPERIMENT_HISTORY.md` names the keys in which its configuration differs from R's. The status names in Figure 10 are the action categories of Figure 14 and Section 4.6.

The learner uses the REINFORCE estimator of Equation (1) with a leave-one-out baseline on discounted returns. The loss of action $t$ in attempt $i$ is $-A_{i,t} \log \pi_\theta(a_{i,t} \mid s_{i,t})/G$, and the gradient norm is clipped at 1.0. For attempt $i$ of a group of $G$ attempts, the advantage of action $t$ is

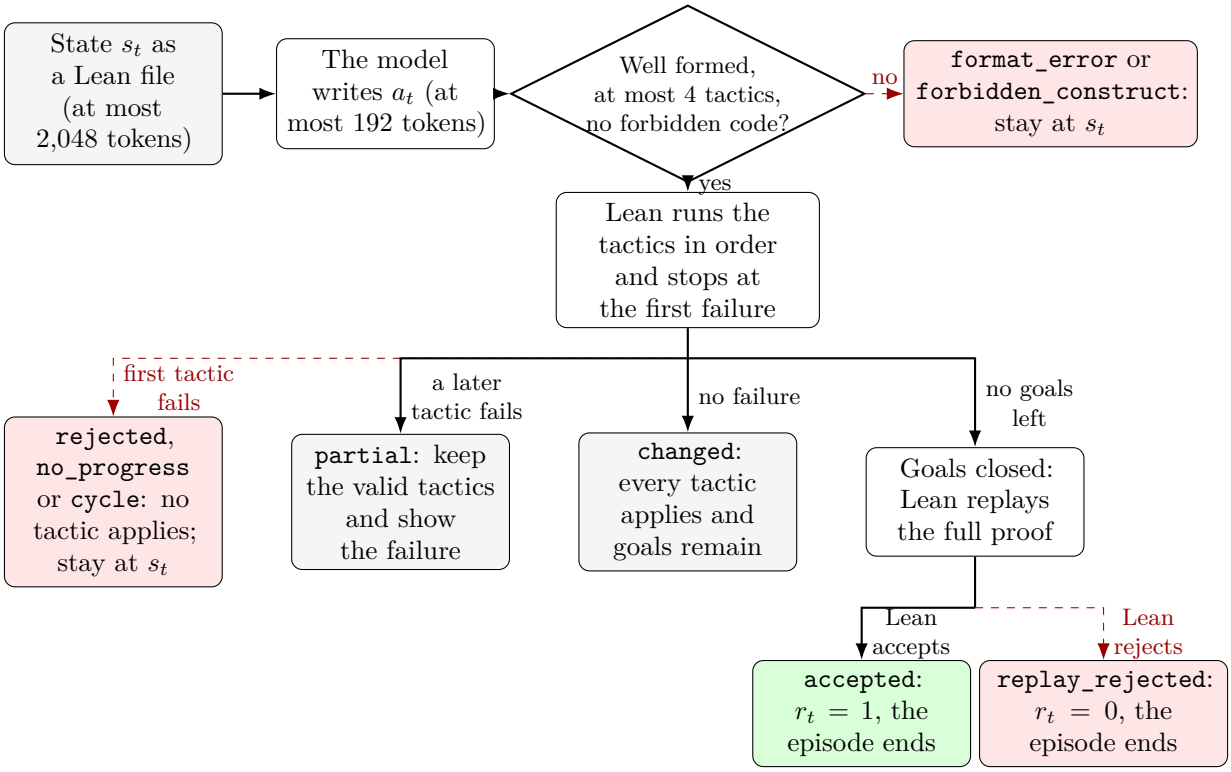
$$A_{i,t} = \sum_{k \geq t} \gamma^k r_{i,k} - \frac{1}{G-1} \sum_{j \neq i} \sum_{k \geq 0} \gamma^k r_{j,k}, \tag{4}$$

where $r_{i,k}$ is the reward after action $k$ of attempt $i$. The reward comes only at the end, so every action of an attempt gets the same advantage. If all $G$ returns of a group are equal, every advantage is 0, and the group gives no update (Section 2.2). With $\gamma < 1$, a group in which Lean accepts every attempt can still update, because proofs that close at different actions have different returns.

Table 6: The interactive MDP, environment version 2.2, with the settings of the runs in this section.

| Part | Definition |
|---|---|
| Theorem | One statement, from TRAIN or a held-out set. No proof lines are supplied |
| State $s_t$ | The theorem, the policy actions that Lean accepted, the current goals, the result of the last action, the tactics that already failed at the current goals, and the number of actions left |
| Observation | The state written as a Lean file, with the goals in a `/- tactic state -/` comment. The prompt holds at most 2,048 tokens. If the prompt is longer, the oldest policy actions are hidden. If it is still longer after an action, the episode ends |
| Action $a_t$ | Lean tactic text of at most 192 generated tokens, bare or in one code fence labelled `lean` or `lean4` in any letter case, with at most 4 top-level tactics and optional `--` comment lines. Text around a fence, or a second fence, is a format error |
| Transition | Lean runs the tactics in order and keeps the valid tactics before the first failure (Figure 10). A failure is a Lean error, unchanged goals, or a repeat of an earlier state on the current proof path |
| Harness revert | After 2 distinct tactics fail at one state, the harness removes the last accepted action and records it as failed at the earlier state. It reverts at most once per action |
| End of episode | Lean replays a closed proof; 6 actions are used; a Lean call exceeds the 20 s timeout; or the next prompt exceeds 2,048 tokens with every policy action hidden. A REPL crash, or a first prompt over the limit, censors the whole group |
| Reward | $r_t = 1$ if Lean accepts the replayed full proof after action $a_t$, otherwise 0. The reward is Lean-verified and has no heuristic term |
| Discount | $\gamma = 0.95$. A proof closed by $a_t$, with $t = 0, \ldots, 5$, has return $\gamma^t$: 1 for the first action and 0.774 for the sixth |
| Learner | Equation (4) with $G = 4$ attempts per group. AdamW with learning rate $10^{-6}$ and no weight decay; gradient norm clipped at 1.0. No KL, entropy or reference-model term |
| Evaluation | The same environment with no update, 2 attempts per selection theorem. A theorem counts as proved if Lean accepts any attempt |

Table 7 isolates the change. Version 2.2 is not a new reward and not a new state idea: the reward remains terminal and Lean-verified, and the state keeps the same fields. The changes are the deployed interface: the action cap, the prompt cap, the failure set, the harness revert, the end conditions, and the discount. Section 4.6 reports what the interface still rejects, and Section A.3 reports which reference proofs it cannot hold.




Each action uses one of the 6 actions of an episode. Outcomes other than a closed proof give the next state $s_{t+1}$; after a self-loop, $s_{t+1}$ has the same goals and a new failure message. After 2 distinct tactics fail at one state, the harness removes the last accepted action, at most once per action. After 6 actions without a proof, a Lean timeout of 20 s, or a next prompt over 2,048 tokens with every policy action hidden, the episode ends with reward 0. A REPL crash censors the group.

Figure 10: How environment version 2.2 processes one action $a_t$. Solid arrows lead to progress or a proof; dashed arrows lead to a self-loop or a failed replay. The status names are the action categories of Figures 14 and 16.

Table 7: What version 2.2 changed relative to version 2.1. State fields, the terminal Lean-verified reward, and the discount value are unchanged.

| Part | Version 2.1 | Version 2.2 |
|---|---|---|
| Action | Up to 4 top-level tactics, each optionally preceded by `--` comment lines | At most 192 generated tokens, bare or in one fence labelled `lean` or `lean4` in any letter case; at most 4 top-level tactics and optional comments. Text around a fence, or a second fence, is a format error |
| Observation | A Lean file with the accepted steps and the tactic state | Prompt at most 2,048 tokens. The oldest policy actions are hidden first; if the prompt is still too long after an action, the episode ends |
| Failure | A Lean error, unchanged goals, or a cycle gives a self-loop | A Lean error, unchanged goals, or a repeat of an earlier state on the current proof path gives a self-loop |
| Harness revert | After 2 distinct tactics fail at one state, revert one step; at most once per action | Same, with the accepted action removed from the state and recorded as failed at the earlier state |
| End of episode | Closed proof replayed; action or time budget; timeout | Closed proof replayed; 6 actions; 20 s timeout; prompt cap. A REPL crash or a first prompt over the limit censors the group |
| Reward | 1 for a verified complete proof, otherwise 0 | Same; Lean-verified, with no heuristic term |
| Discount | $\gamma = 0.95$ at the configured value | $\gamma = 0.95$; a proof closed by action $t$ has return $\gamma^t$ |

### 4.2 Selection-set proof completion

S (E0) proves 1 of 163 assessed selection theorems in 1 of 326 attempts (Figure 11a). Group 143 was censored after a root automation probe hit the 20 s Lean timeout; E2 loses the same group. Comparisons with E0 therefore pair 163 of the 164 theorems. Later runs record this as a failed automation outcome and continue.

Figure 11a gives all arm counts. None meets the promotion rule, and no paired interval excludes 0. The largest gain is E6 against E0: +0.92 points, with a bootstrap 95% interval of $[-0.61, +3.07]$. E3 against its control E5 gains +0.30 points, with an interval of $[0.00, +0.91]$. E2 and E3 prove the same theorem; E0 and E7 each prove a different one. This pattern fits sampling noise at the floor.

**External prover evaluation.** To assess whether the selection set is beyond a small prover's capability, we test Kimina-Prover-Distill-0.6B, described as a distillation of Kimina-Prover-72B [17]. We use its model-card sampling settings and S's verifier path. In draw 1, Kimina proves 47 of 164 theorems within 8 attempts. In draw 2, which changes only the batching, it proves 40 of 164. An exact McNemar test on the 25 theorems proved in only one draw gives $p = 0.23$.

*Limits*: the probe is not an arm matched to S. Kimina generates whole proofs with long reasoning, not S's action format. In draw 1, Kimina generates 6,444 tokens per attempt on average, with a limit of 8,096. In E0, S generates about 574 tokens per attempt. Kimina's 8 attempts therefore use far more generated tokens than S's 2 attempts. In draw 1, 853 of 1,312 attempts reach the token limit without a closed proof and count as failures. The pinned Lean and mathlib are probably older than Kimina's training environment, so some lemma names may not exist. The selection theorems have no informal problem text, so the problem field of Kimina's prompt is empty. These three effects bias Kimina's score down. A public prover may have trained on these statements, and this unknown exposure could bias the score up.

Figure 11a shows Kimina's unbiased pass@2 estimates from 8 attempts per theorem. The estimate

is 11.35% in draw 1 and 11.67% in draw 2. S proves 0.61% of its 163 theorems within 2 attempts. The panel compares models at an equal number of attempts, not at equal generated tokens. A 0.6B prover with whole-proof generation and a larger generation budget therefore proves many of these theorems. The probe does not separate S's model and training data from its action format and generation budget.

Kimina proves a smaller fraction of theorems with longer reference proofs in both draws (Figure 11b). Reference length describes one recorded proof, not its minimum difficulty.

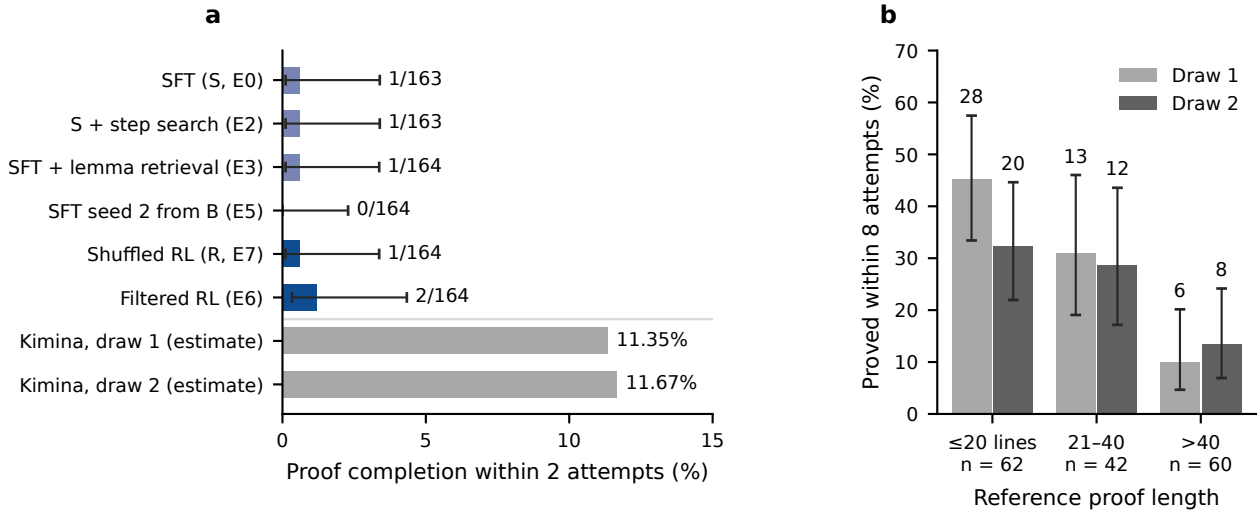



Figure 11: Held-out results on the EXP-028 selection set of 164 theorems. (a) Theorems proved within 2 attempts per theorem: E0 1/163, E2 1/163, E3 1/164, E5 0/164, E7 1/164 and E6 2/164. Kimina draws 1 and 2 appear as unbiased pass@2 estimates from 8 attempts per theorem, 11.35% and 11.67%, with no interval. Blue-grey bars are SFT models, dark blue bars are RL from S, and grey bars are Kimina. The panel holds attempts equal, not generated tokens. (b) Kimina theorems proved within 8 attempts, by reference-proof length, for draws 1 and 2: at most 20 lines, 28/62 and 20/62; 21 to 40 lines, 13/42 and 12/42; over 40 lines, 6/60 and 8/60. Error bars in both panels: 95% Wilson score intervals over theorems, which treat theorems as independent.

### 4.3 Training acceptance under pass-rate-filtered RL

`pass_rates` measures S with 4 attempts on each of the 506 TRAIN theorems. S accepts 84 of 2,024 attempts and proves 63 of 506 theorems. Of the 506 theorems, 443 have no accepted attempt. The registered filter keeps the 62 theorems with an S acceptance rate in $(0, 0.75]$. `RL_filtered` runs REINFORCE from S on these 62 theorems, with the resource caps of R. It completes all 1,012 groups of 4 attempts, with 337 optimizer updates and no censored group.

Acceptance rises across training quarters (Figure 12a). Each theorem is visited 16 or 17 times. The mean first-visit acceptance per theorem is 0.306. The mean last-visit acceptance is 0.597. Figure 12b pairs the first and last visit for every theorem. An exact paired sign-flip test gives $p < 0.001$ ($p = 2.4 \times 10^{-6}$). On the same 62 theorems, S's `pass_rates` acceptance is 0.323. The first visit is close to this value, so regression to the mean after selection does not explain the rise. R, trained on all 506 theorems, shows no detected rise. Its first visits accept 81 of 2,024 attempts and its second visits 83 of 2,024 ($p = 0.93$).

E6 evaluates the final `RL_filtered` weights on the selection set. It proves 2 of 164 theorems, each in both attempts; E0 proves 1 of 163. One of its theorems is the one that E2 and E3 prove, and the

other is the one that E7 proves. Against R (E7), E6 gains +0.91 points, with an interval of [0.00, +2.44]. The training gain on 62 theorems therefore does not transfer measurably to the selection set at this set size.

*Limits*: better fit to these 62 training theorems may reflect memorization. SFT uses TRAIN reference proofs, with the same theorem exclusions as the RL bank. The 19 October audit finds an SFT example from the theorem's own reference proof for 59 of the 62 theorems. Recall of these proofs is a possible explanation; no run tests it. R and `RL_filtered` also differ in theorem set (506 against 62 theorems), visits per theorem (2 against 16 or 17) and order. No run separates these differences.

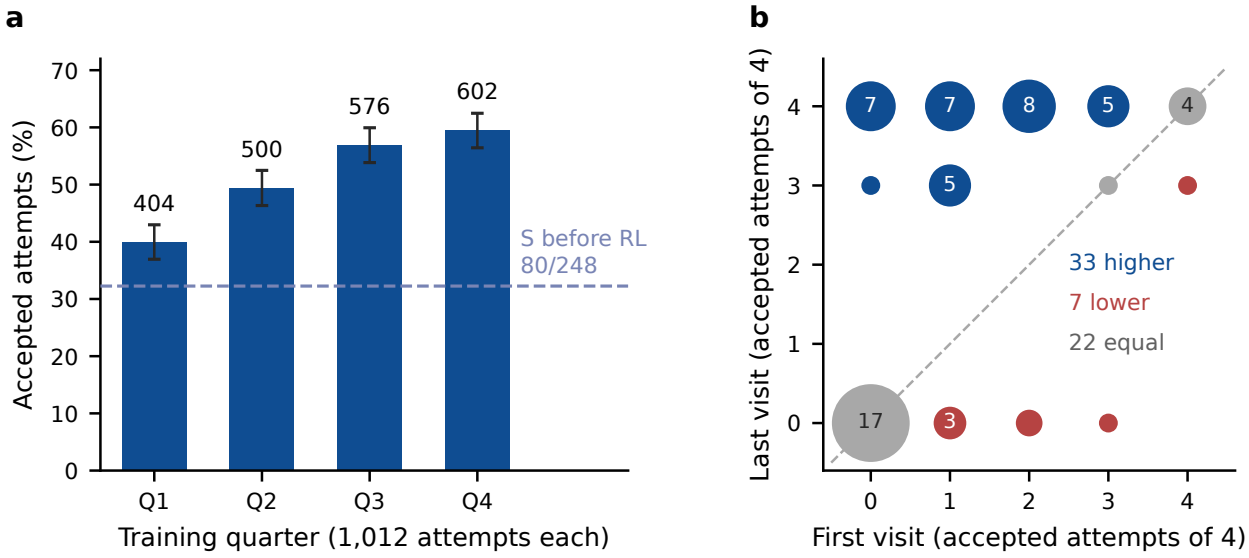



Figure 12: `RL_filtered` on its 62 TRAIN theorems. (a) Accepted attempts by training quarter: 404, 500, 576 and 602 of 1,012 attempts. The dashed line marks S before RL on the same 62 theorems: 80/248 attempts in `pass_rates`, with 4 attempts per theorem. Error bars: 95% Wilson score intervals, which treat correlated attempts as independent and so understate the uncertainty. (b) Accepted attempts of 4 per theorem at the first visit against the last visit, for 62 theorems. Bubble area shows the number of theorems, and the dashed line is $y = x$. Of the 62 theorems, 33 are higher at the last visit, 7 lower and 22 equal. Panel (b) has no error bars.

### 4.4 Training acceptance under prerequisite-ordered RL

R and C start from S and complete 1,012 groups of 4 attempts on the same 506 TRAIN theorems, with no censored groups. Their configurations differ only in identifiers and their commands only in task order: shuffled for R, twice through the prerequisite order for C. C accepts 255 of 4,048 attempts, and R accepts 164 of 4,048. C proves 93 of the 506 theorems, and R proves 63. Per theorem, C minus R is positive for 73 theorems and negative for 32 (paired sign-flip $p = 0.0002$). Each arm has one training run. The sign-flip test assumes independent theorems, so it omits run-to-run variation and dependence within theorem families.

**Training acceptance by epoch.** S's `pass_rates` run gives a measurement before RL on the same theorems. C's first visits accept 130 of 2,024 attempts, against 84 for S ($p = 0.0019$). R's first visits accept 81 of 2,024 ($p = 0.87$ against S). C's second epoch accepts 125 of 2,024 attempts ($p = 0.79$

against its first epoch). C's gain therefore appears within its first epoch and does not grow in the second. Figure 13a breaks down the first-epoch comparison by quarter.

**Exploratory automation check.** This unregistered check measures how much of the late-quarter gain comes from automation-only proofs. Under the strict rule, an accepted attempt is *automation-only* if every action is exactly one automation tactic name with no arguments. Under the argument rule, a tactic with arguments, for example `simp [h]`, also counts. In quarter 3, 3 of C's 33 accepted first-visit attempts are automation-only under the strict rule. Under the argument rule, 15 of the 33 are automation-only. S has 0 of 12 under both rules. In quarter 4, C has 0 of 6 under the strict rule and 4 of 6 under the argument rule. S has 0 of 3 in quarter 4. The dated count did not record its tactic set; the recount uses a set that reproduces the recorded counts.

**Action length and tactic concentration.** Here a quarter is one quarter of a run's training actions, in training order. C's actions narrow over training (Figure 13b and c). C's format-error rate falls from 4.7% of actions in the first quarter to 0.3% in the last. R's format-error rate is 18.1%, 6.7%, 18.6% and 47.9% by quarter. It therefore rises to 47.9% in the last quarter, after a dip in the second. C's mean action length falls from 56.3 tokens in the first quarter to 10.4 in the last. In C's last quarter, actions that are exactly `omega`, `simp` or `simp_all` make 4,597 of 5,845 actions. Under the strict rule, 50 of C's 255 accepted attempts are automation-only, against 1 of R's 164. Under the argument rule, the counts are 105 for C and 2 for R. Outside automation-only attempts, C accepts 205 against R's 163 under the strict rule. Under the argument rule, the remainders are 150 for C and 162 for R. So under the argument rule, C's TRAIN advantage over R does not remain outside automation-only attempts. These action analyses are exploratory, not registered metrics.

**Development results and selection-set evaluation coverage.** B, S, R and C each prove 0 of 8 DEV theorems. `dev_R` and `dev_C` each accept 0 of 32 attempts. C has no selection-set evaluation in these records. Both arms are at 0 on DEV, so the R/C contrast has no held-out difference to measure, and the TRAIN gains do not show transfer. Table 18 needs a familiar-panel result and a generalization-panel result. TRAIN acceptance is training accuracy, and the familiar-family panel uses held-out theorems instead. No row of that table applies yet.

### 4.5 Selection-set action validity and format errors

Figure 14 classifies every model action on the selection set for S (E0), R (E7) and `RL_filtered` (E6). Format errors are 373 of 1,953 actions for S. They are 1,288 of 1,963 for R and 18 of 1,948 for `RL_filtered`. Mean generated tokens per action are 95.8 for S, 34.6 for R and 30.8 for `RL_filtered`. In 1,636 of `RL_filtered`'s actions, Lean rejects the first tactic, against 1,411 for S and 468 for R. R mostly writes actions that fail the format check. `RL_filtered` mostly writes short tactics that Lean rejects. On DEV, inspected R format errors repeat one tactic beyond the 4-tactic limit, for example seven consecutive `simp` lines. `dev_R` makes 113 format errors in 192 actions, against 37 for S. `dev_C` makes 1 format error in 192 actions. It generates about 12 tokens per action, against 93.7 for S. This action analysis is exploratory, not a registered metric. The dated record names REINFORCE without a KL or entropy term as a likely contributor; no run tests it.

### 4.6 Possible explanations for the missing held-out gain

RL raises acceptance on the theorems that it trains on (Sections 4.3 and 4.4). No arm shows a detected held-out gain (Section 4.2). We examine five possible explanations using the saved records: the update, the reward, the starting policy S, the held-out measurement, and changes in tactic choice. These

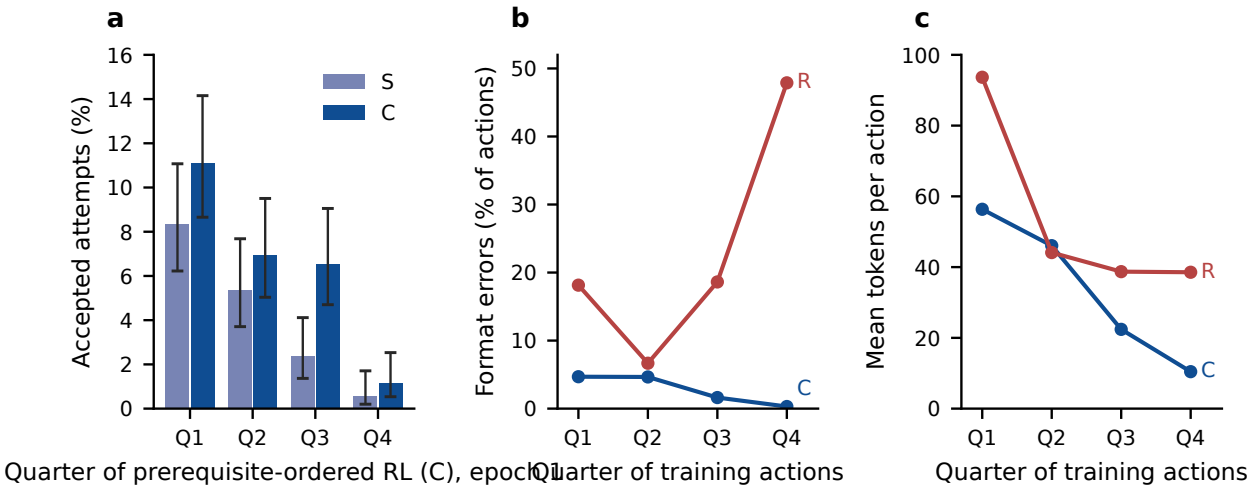



Figure 13: Curriculum order on the 506 TRAIN theorems, with R and C from the same S. (a) First-visit accepted attempts of 4 per theorem, by quarter of C's first-epoch order. S (`pass_rates`): 42/504, 27/504, 12/504 and 3/512. C: 56/504, 35/504, 33/504 and 6/512. Error bars: 95% Wilson score intervals, which treat correlated attempts as independent and so understate the uncertainty. (b) Format errors as a percentage of actions, by quarter of training actions, for C and R. (c) Mean generated tokens per action, by the same quarters. Each quarter holds 5,845 C actions or 5,935 R actions; the last of C's 23,381 actions falls outside the four quarters. Panels (b) and (c) have no error bars.

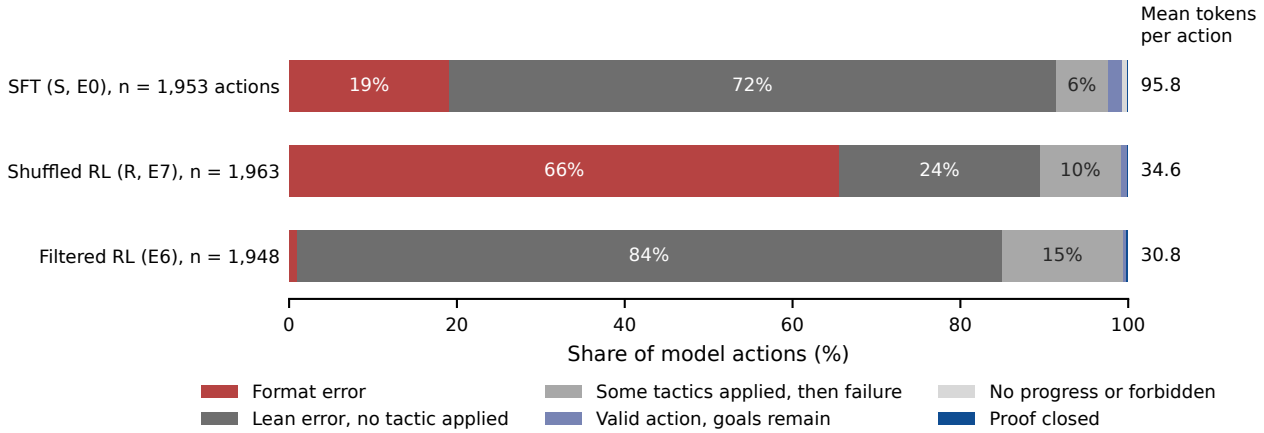



Figure 14: Status of every model action on the 164-theorem selection set, with 2 attempts per theorem: S (E0, 1,953 actions, 163 theorems assessed), R (E7, 1,963) and `RL_filtered` (E6, 1,948). Categories: format error; Lean error with no tactic applied (status `rejected`); some tactics applied, then a Lean error, unchanged goals or a repeated state (`partial`); a valid action with goals remaining (`changed`); no progress or a forbidden construct; proof closed. The right column gives mean generated tokens per action. Bars are 100% stacked shares, with no error bars. This analysis is exploratory, not a registered metric.

analyses are exploratory and were chosen after observing the outcomes. They use no new Lean or model runs. The EXP-028 entry of 20 October in `docs/EXPERIMENT_HISTORY.md` lists the scripts and their outputs.

**Analysis design.** A *visit* is one group of 4 attempts on one TRAIN theorem. R and C visit each of the 506 TRAIN theorems twice, and `RL_filtered` visits each of its 62 theorems 16 or 17 times. The *S reference* of a theorem is the number of S's 4 attempts in `pass_rates` that Lean accepts. For theorem $\tau$ and visit $v$, let $a_\tau^{(v)}$ be the arm's accepted attempts, $s_\tau$ the S reference, and $d_\tau = a_\tau^{(v)} - s_\tau$. The test is an exact two-sided paired sign-flip test on $\sum_\tau d_\tau$. We split the theorems by whether the arm's first visit has at least one accepted attempt. If the policy equalled S, the second visit and the S reference would be exchangeable given the theorem, so selecting on the first visit would not bias their difference. The split does not isolate the update from each theorem's own first visit, because the policy also changes through training on other theorems.

**Acceptance conditional on initial proof success.** Figure 15 shows the split. On the 72 theorems where C's first visit has a proof, C's second visit accepts 95 of 288 attempts, against 54 for S ($p = 0.011$; 0.087 after a Bonferroni factor of 8). On the 46 such theorems for R, R's second visit accepts 55 of 184 attempts, against 42 for S ($p = 0.199$). `RL_filtered` trains only on theorems that S proves in 1 to 3 of 4 attempts. Its mean acceptance per theorem rises from 0.306 at the first visit to 0.597 at the last (Section 4.3). At the last visit, 31 of its 62 theorems have 4 of 4 accepted attempts, and 23 still have 0. C's first visits already beat S over all 506 theorems (130 against 84 accepted attempts), so part of C's second-visit excess can come from training on earlier theorems. The rise of `RL_filtered` within its 62 theorems is the cleaner evidence.

**Frequency of nonzero proof rewards.** S has no accepted attempt on 443 of the 506 TRAIN theorems. R's first visit has a proof on 46 of the 506 theorems, and C's has one on 72. On the other 460 theorems, R's second visit accepts 28 of 1,840 attempts, against 42 for S ($p = 0.136$). On the other 434 theorems, C's second visit accepts 30 of 1,736 attempts, the same count as S ($p = 1.0$). A group with no proof gives every attempt an advantage of 0 (Section 2.2), so the first visit to these theorems gives no update.

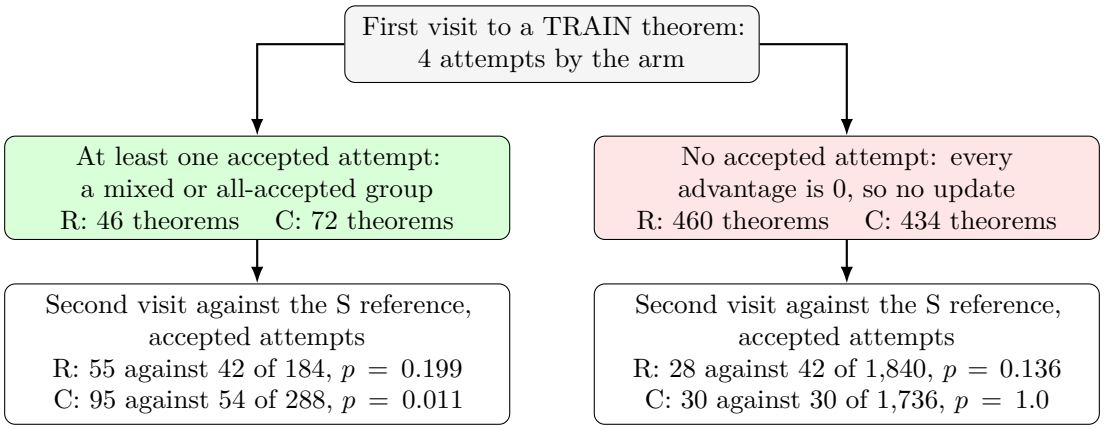



Figure 15: TRAIN theorems of R and C split by the outcome of the arm's first visit. The S reference is S's accepted attempts of 4 per theorem in `pass_rates`, a separate run on the same theorems. The $p$-values come from exact two-sided paired sign-flip tests on the per-theorem difference, with no correction for the 8 tests. The split puts all-accepted first visits with the mixed groups. This analysis is exploratory.

**Initial-policy tactic validity across data splits.** S applies at least one tactic in 105 of 326 held-out attempts (E0). On TRAIN, it applies one in 1,199 of 2,024 attempts (Figure 16a). By action, the counts are 156 of 1,953 held-out actions and 2,106 of 11,890 TRAIN actions. The TRAIN theorems also carry SFT exposure. Of the 506 TRAIN theorems, 486 have at least one SFT example from their own reference proof, and so do 59 of the 62 `RL_filtered` theorems. No SFT example comes from a selection-set statement. On the selection set, 161 of S's 326 first actions start with `have` and have more than one line. Lean rejects 147 of them at the first tactic.

**Action-template concentration.** Of `RL_filtered`'s 2,082 accepted training attempts, 1,627 close the proof with the first action (Figure 16c). S's 84 accepted TRAIN attempts close at every action from 1 to 6. Of the 1,627 closing first actions, 2 are a single fixed tactic without arguments. The 12 most frequent closing texts all start with `simp_all` and a lemma list, and they cover 1,076 of the 1,627. The most frequent text, `simp_all [Int.ModEq, Int.emod_eq_emod_iff_emod_sub_eq_zero]`, closes 451 attempts. In the last quarter of training, 40 theorems have a proof; 14 of them have one distinct proof text, and 10 have two. On the selection set, 326 of `RL_filtered`'s 328 first actions start with `simp_all` and a lemma list. Of these 326, Lean rejects 224 at the first tactic, 94 apply some tactics and then fail, 4 close the proof, and 4 are format errors. The learner has no KL, entropy or reference-model term that would keep the policy near S or keep its actions varied.

**Action-limit violations in SFT and shuffled RL.** Figure 16b splits the selection-set format errors by cause. Of R's 1,288 format errors, 1,257 have more than 4 tactics, so the environment refuses them before Lean runs them. Of S's 373 format errors, 320 reach the 192-token limit. `RL_filtered` makes 18 format errors. On TRAIN, 893 of S's 1,416 format errors reach the token limit, and 465 more have more than 4 tactics.

**Power analysis for held-out comparisons.** We compute the power of the exact two-sided sign test at $\alpha = 0.05$ on theorems that one arm proves and the other does not. The computation uses two dependence assumptions: in the nested case, the arm proves every theorem that S proves; in the independent case, the two outcomes are independent. Negative dependence, where the arms prove disjoint theorems, is not computed, and it can give a lower power. At the observed rates (S 1 of 163, `RL_filtered` 2 of 164) and 164 theorems, the power is 0.001 (nested) to 0.006 (independent). A grid search (every 10 theorems below 2,000, every 250 above) finds that the same gain needs 1,310 (nested) to 4,250 (independent) theorems for a power of 0.8. At the registered 5-point bar, 164 theorems give a power of 0.693 to 0.833. These values are for the sign test, not for the registered bootstrap rule.

**Interpretation.** The records show no sign of a fault in the update. On the 62 `RL_filtered` theorems, which all have a nonzero S acceptance rate, acceptance rises, and the rise is not regression to the mean. The reward is Lean-verified, and it is rarely nonzero: S proves 63 of the 506 TRAIN theorems in 4 attempts. On held-out theorems, S applies a tactic in fewer attempts than on TRAIN. RL reinforces `simp_all` actions with a lemma list. On the selection set, 326 of `RL_filtered`'s 328 first actions start this way, and Lean rejects 224 of them at the first tactic. The missing held-out gain is therefore consistent with a rare reward under a weak starting policy and with narrow reinforcement, more than with a fault of the update. No run varies the reward, the starting policy, a KL or entropy term, or the action limits, so this attribution is not tested.

*Alternative explanations*, not controlled:

- Theorems with a first-visit proof are easier. On R's split, S accepts 42 of 184 attempts on them and 42 of 1,840 on the others. A general drift toward short actions could help easy theorems without a theorem-specific update.

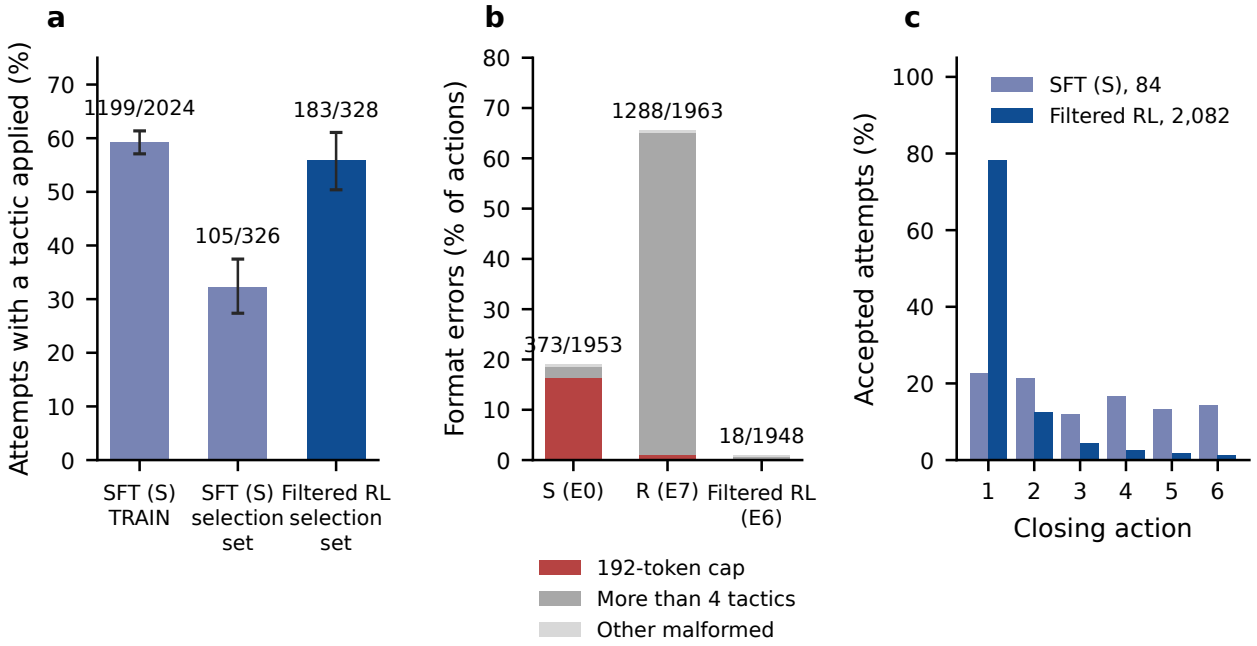



Figure 16: Three measurements behind Section 4.6. (a) Attempts in which at least one tactic applies: S on TRAIN (`pass_rates`) 1,199/2,024, S on the selection set (E0) 105/326, and `RL_filtered` on the selection set (E6) 183/328. Error bars: 95% Wilson score intervals, which treat correlated attempts as independent and so understate the uncertainty. (b) Selection-set format errors as a share of all model actions, split by cause in this order of checks: the 192-token limit, more than 4 tactics, and other malformed actions. S: 320, 44 and 9 of 1,953 actions. R: 20, 1,257 and 11 of 1,963. `RL_filtered`: 1, 10 and 7 of 1,948. (c) Accepted attempts by the action that closes the proof, as a share of each run's accepted attempts. S on TRAIN: 19, 18, 10, 14, 11 and 12 of 84. `RL_filtered` training: 1,627, 258, 90, 50, 34 and 23 of 2,082. Panels (b) and (c) have no error bars. This analysis is exploratory.

- S's TRAIN advantage can come from SFT exposure, from the exclusion of theorems that one fixed tactic proves from the selection set, or from difficulty. No count separates them.
- The `simp_all` concentration changes how `RL_filtered` fails. S does not show it and also proves 1 of 163 held-out theorems, so it is not the only cause.

*Limits*: each arm is one training run. The visit comparisons use 8 sign-flip tests, 4 per arm. Under a Bonferroni factor of 8, C's first-visit gain over S stays below 0.05 (0.015), and C's split result does not (0.087).

### 4.7 Findings and interpretation

Filtered RL improves acceptance on its 62 training theorems. C has higher training acceptance than R overall and than S in its first epoch; R shows no detected training gain. With one run per arm, these comparisons do not establish a held-out gain: the selection-set intervals include zero, and all four main models prove 0 of 8 DEV theorems.

RL also shortens actions and narrows tactic choices. R makes more format errors; C favors automation; filtered RL favors `simp_all` with lemma lists. C's second-visit gain on initially successful theorems does not survive the stated multiple-test correction. These exploratory observations do not identify a cause of the missing transfer.

Two hypotheses remain untested: RL reinforces behaviour S already has, and training gains reflect recall of SFT proofs. Filtered RL uses only theorems S proves at least once in 4 attempts; 59 of its 62 theorems have reference-proof examples in SFT. The run cannot distinguish recall from new skill. Yue et al. report higher pass@$k$ at small $k$ and lower pass@$k$ at large $k$ after RL with verifiable rewards, relative to the base model [18]. This study does not measure pass@$k$ at large $k$.

**Measurement floor.** S proves 1 of 163 selection theorems and 0 of 8 DEV theorems. The promotion rule needs a paired gain of at least 5 points, so a smaller held-out gain cannot pass it. At the observed rates and 164 theorems, an exact sign test has a power of 0.001 to 0.006 (Section 4.6). A transfer gain could remain undetected.

## 5 Limitations

- The evaluation sets are small (3 DEV theorems in Experiment 19; 4 to 8 items in Experiment 11), and every run used one seed.
- Several sets were reused across experiments, so they now act as development sets.
- We do not know how much Lean or mathlib the model saw in pretraining.
- AI reviewers graded the natural-language answers in Experiments 9, 10, 11, and 13.
- The sample-size values come from simulations with assumed parameters.
- The numbers come from recorded run outputs; we did not repeat experiments for this report.
- Low baseline acceptance and the promotion threshold limit sensitivity to smaller held-out gains (Section 4.7).
- The explanations in Section 4.6 come from one training run per arm. No controlled run tests them.

**Reproduction.** Full reproduction requires archived data, later-study weights and the pinned Linux Lean runtime. Several dataset builders and the original run configurations remain private. The public helper checks supplied inputs and prepares commands from reconstructed settings; it has not been validated by a rerun. Appendix A.9 describes the public tools and their limits; `scripts/README.md` gives the commands.

**Sources.** Each experiment has a record in `docs/EXPERIMENT_HISTORY.md`. The earlier detailed report (Git commit `e3b7747`) contains every count of Experiments 1 to 13 and the full checker audit. The reward exploits are in `docs/REWARD_HACKING.md`. The script `scripts/report/make_report_figures.py` rebuilds 12 plots from counts recorded in its source, with pointers to their records. This includes the dense-reward plot, reconstructed from the counts in its previously published PDF. The other `fig_oct21_*` PDFs retain archived plotted data, with editorial title and display-name corrections. Their original builders and recount inputs remain in the private directory named in Appendix A.

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# A Post-hoc analyses of 20 and 21 October 2026

These post-hoc analyses were conducted on 20 and 21 October 2026 using saved records from the 18–20 October runs (Section 4). They measure proof acceptance, tactic choice and responses to Lean errors, without new Lean, model or GPU runs. Section 4.6 discusses possible explanations. The scripts and their outputs are in the private recount directory `.private/ops/report_recount_20261019_v1/`, in `scripts/` and `out/`. The analysis counts come from JSON files in `out/`. Software notes appear at the end of the appendix.

## A.1 Analysis scope and definitions

- Every result is an exploratory analysis of existing records. None is a registered metric.
- Each arm is one training run with one seed. Section A.5 shows that seed noise on the selection set is not small.
- A *valid step* is an action in which Lean applies at least one tactic. A *valid attempt* is an attempt with at least one valid step. A valid step is partial progress. It is not a proof.
- A *visit* is one group of 4 attempts on one TRAIN theorem (Section 4.6). A *decile* or *quarter* is one tenth or one quarter of a run's 1,012 groups, in training order.
- On the selection set, E0 is S, E5 is S with SFT seed 2, E6 is `RL_filtered` and E7 is R (Table 5).
- An *unknown-name error* is a Lean error that reports an unknown identifier or constant. Section A.3 counts how many of these names are invented lemma names.
- The *template* is a first action that starts with `simp_all` and a lemma list (Section 4.6).

Section A.5 states which tests survive a correction for multiple tests. It uses two families from `out/p8_findings.json`. The *headline family* holds 11 tests, one primary test per finding. The *278-test family* holds every test in that file. Results from the later scripts `p9_*`, `p10_*`, `p11_regressions` and `p12_positives` are outside both families.

## A.2 Changes in proof acceptance and tactic choice

RL raised acceptance on the TRAIN theorems it trained on (Sections 4.3 and 4.4). The rise came with one short action template. No arm shows a detected gain in proved held-out theorems (Section 4.2). Table 8 lists the measured changes that this section adds.

## A.3 Failure patterns in the saved runs

**Frequency of nonzero sampled advantages.** A group gives a learning signal only if its 4 attempts get different returns. Of 1,012 groups, R has 81 mixed groups and 920 groups in which all 4 attempts fail. C has 103 mixed groups and 886 all-fail groups. `RL_filtered` has 207 mixed groups and 398 all-fail groups. Groups with a nonzero advantage are 91 for R, 122 for C and 319 for `RL_filtered` (`p6_signal`, `A.*.total`). In all three runs, the last decile has fewer such groups than the first: R 11 against 6, C 25 against 0, and `RL_filtered` 61 against 21 (Figure 17). The fall is not monotone, and it has no test. This rare signal is consistent with the missing held-out gain. No run changes the reward or the start model, so this cause is not tested.

**Reference-proof compatibility with action limits.** Section 4.6 shows that the token and tactic limits cause most format errors. A greedy split of each proof into actions also shows how many known proofs fit the limits (Figure 18). Under the current limits, 314 of 516 TRAIN reference proofs fit.

Table 8: Summary of exploratory analyses of the saved runs. "Before" is S, or the start of training. "After" is the RL model, or the end of training. All entries are exploratory and come from one run per arm. Files are in `out/`.

| Change | Measure | Before | After | n, test and file |
|---|---|---|---|---|
| Better | Selection theorems with a valid step, S against `RL_filtered` | 84 | 116 | 163 paired; McNemar 55 gains, 23 losses, $p = 0.00038$; `p11_regressions` |
| Better | Next visit repeats the first action with positive advantage, `RL_filtered` | negative advantage 0.023 | positive advantage 0.286 | 51 theorems; sign-flip $p = 1.0 \times 10^{-10}$; `p9_update_follow` |
| Better | TRAIN theorems proved where S had no accepted attempt; R and C had 8 attempts, S had 4 | S 0 | R 29; C 54 | 443 theorems; `p12_positives` |
| Worse | `RL_filtered` TRAIN theorems proved, early half to late half | 50 | 40 | 62; McNemar 1 against 11, $p = 0.0063$; `p6_coverage` |
| Worse | R TRAIN theorems proved, S against R's first visit, 4 attempts each | 63 | 46 | 506; McNemar 20 gains, 37 losses, $p = 0.033$; `p11_regressions` |
| Worse | Unknown-name first errors per 100 selection actions | S 23.91 | `RL_filtered` 45.02 | 1,953 and 1,948 actions; `p10_remedies` |
| Worse | Identical repeat after an error, C, quarter 1 to quarter 4 | 72 of 4,557 | 1,872 of 4,190 | error actions; `p7_feedback` |
| Worse | `RL_filtered` actions with a comment, quarter 1 to quarter 4 | 208 of 4,480 | 0 of 3,149 | actions; `p7_feedback` |
| Worse | C distinct first actions, decile 1 to decile 10 | 403 of 408 | 84 of 404 | first actions; `p5_diversity` |
| Worse | Jensen-Shannon distance of tactic heads to Kimina | S 0.477 | R 0.814; `RL_filtered` 0.918 | 1,953; 1,963; 1,948 action texts; `p7_heldout` |
| Same | R TRAIN theorems proved, S with 4 attempts against R with 8 | 63 | 63 | 506; 29 gains, 29 losses, $p = 1.0$; `p11_regressions` |
| Same | Accepted TRAIN proofs equal to an SFT reference | S 1 of 84 | `RL_filtered` 0 of 2,082 | accepted attempts; `p5_recall` |

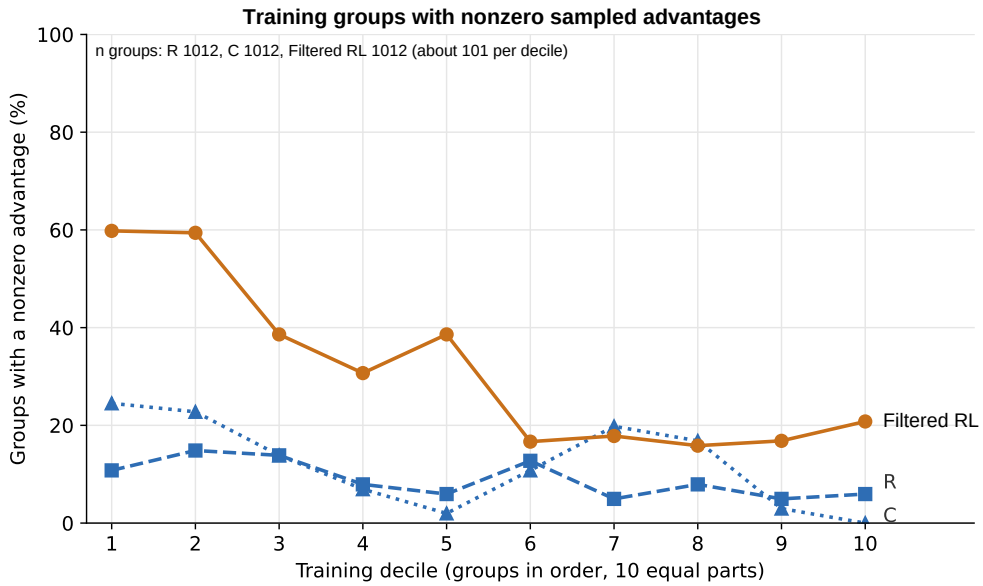



Figure 17: Groups with a nonzero advantage, as a percentage of the groups in each training decile. x-axis: training decile, groups in training order in 10 equal parts. y-axis: percentage of groups with a nonzero advantage. n: 1,012 groups per run, about 101 per decile. Totals: R 91, C 122, `RL_filtered` 319 of 1,012. No error bars. Source: `p6_signal`, `A.*.deciles`.

Under relaxed limits of 12 actions, 8 tactics and 384 tokens, 416 of 516 fit. For Kimina's accepted proofs, 118 of 180 fit the current limits and 166 of 180 fit the relaxed limits (`p6_limits`). The greedy split does not show that a model would find these proofs under relaxed limits.

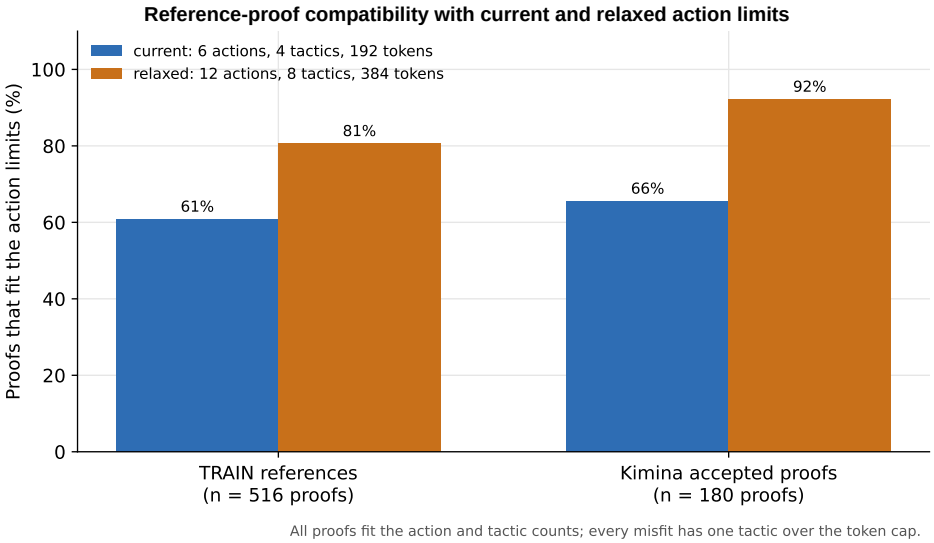



Figure 18: Proofs that fit the action limits after a greedy split into actions. x-axis: proof source, TRAIN SFT reference proofs ($n = 516$) and Kimina accepted proofs ($n = 180$). y-axis: percentage of proofs that fit. Bars: current limits (6 actions, 4 tactics, 192 tokens) and relaxed limits (12 actions, 8 tactics, 384 tokens). Counts: 314 and 416 of 516; 118 and 166 of 180. No test. Source: `p6_limits`.

**First-action repetition after unequal-return groups.** In quarter 1 of `RL_filtered`, the feature "1 action" gets an advantage mass of +46.96. The feature "6 actions" gets −100.84, and "0 tactics applied" gets −52.18 (`p7_reinforced`). The mean closing action of accepted attempts falls from 2.08 in quarter 1 to 1.14 in quarter 4, over 404 and 602 accepted attempts. After a mixed group, the next visit of `RL_filtered` repeats the first action with positive advantage in a mean share of 0.286 of its first actions, and an action with negative advantage in 0.023. Of 51 theorems, 37 are positive and 1 is negative (sign-flip $p = 1.0 \times 10^{-10}$; `p9_update_follow`). R (46 theorems, $p = 1.0$) and C (71 theorems, $p = 0.3125$) show no detected effect of this kind (Figure 19). Updates from other theorems happen between two visits, so this result is an association, not an isolated effect of one update.

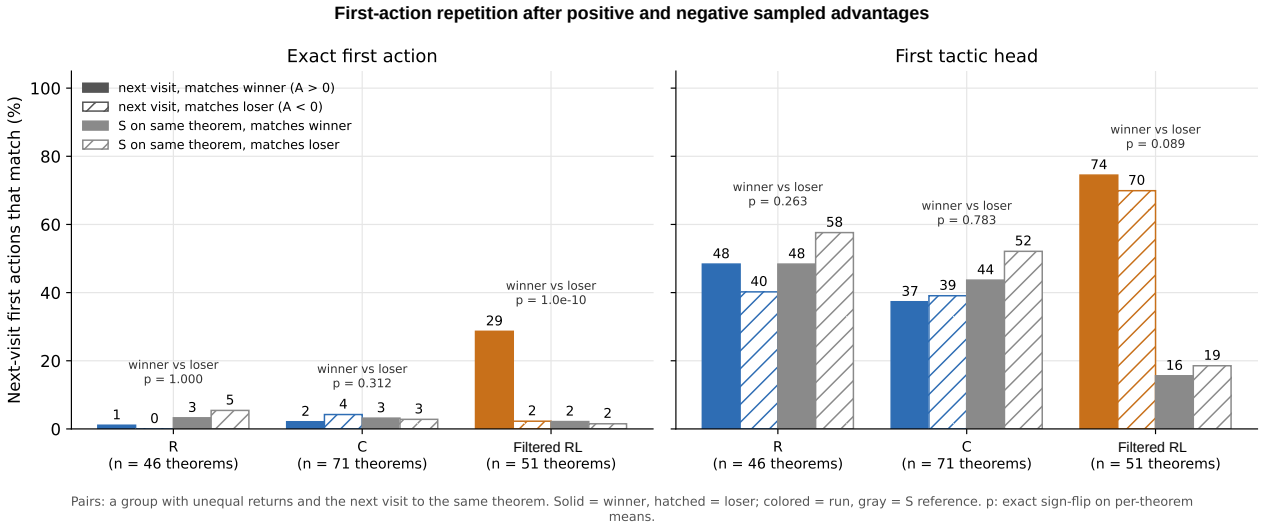



Figure 19: First-action repetition after mixed-reward groups. Each pair is a group with unequal returns and the next visit to the same theorem. x-axis: run, with R ($n = 46$ theorems), C ($n = 71$) and `RL_filtered` ($n = 51$). y-axis: percentage of next-visit first actions that match an action with positive advantage (solid) or negative advantage (hatched). Gray bars: S's samples on the same theorems. Left: exact first action. Right: first tactic head. $p$: exact sign-flip test on per-theorem means, positive- versus negative-advantage actions. No correction. Source: `p9_update_follow`.

**Automation-template concentration during training.** Section 4.6 reports the template on the selection set. During training, its share of `RL_filtered`'s first actions rises from 0.252 in decile 1 to 1.000 in decile 10 (Figure 20; `p5_diversity`). On a theorem, the template first becomes the majority of first actions after the first accepted visit on 33 theorems, and before it on 3 (binomial $p = 2.3 \times 10^{-7}$; `p7_spillover`). The test leaves out 15 theorems where both happen in the same visit and 11 theorems with no accepted attempt. Of 1,935 accepted attempts with a `simp_all` lemma list, 1,380 use a list that also occurs in an accepted attempt on another theorem, earlier or later in training. The most used list closes 814 attempts on 22 theorems (`p10_lemmas`). C changes too, on theorems that gave it no signal. On 423 such TRAIN theorems, the share of C's first actions that are one automation tactic rises from 0.031 at visit 1 to 0.517 at visit 2, against 0.0006 for S on the same theorems (sign-flip $p = 2.8 \times 10^{-91}$; Figure 21). The learner has no KL or entropy term (Table 6). No run adds one, so we cannot say that the missing term caused the collapse.

**Accepted-attempt rate and theorem coverage during filtered RL.** From the early half to the late half of training, the mean pass@1 per theorem rises from 0.446 to 0.583 (36 theorems higher, 14

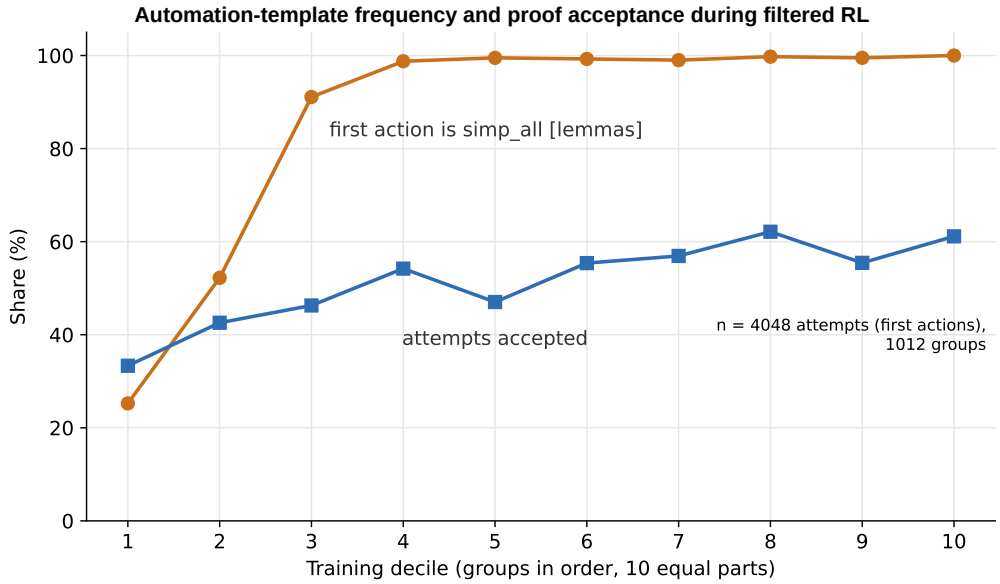



Figure 20: The template in `RL_filtered` training. x-axis: training decile, groups in order in 10 equal parts. y-axis: percentage of first actions. Orange: first actions that start with `simp_all` and a lemma list. Blue: accepted attempts. n: 4,048 attempts (one first action each) in 1,012 groups. No error bars. Source: `p5_diversity`, `runs.RL_filtered.deciles`.

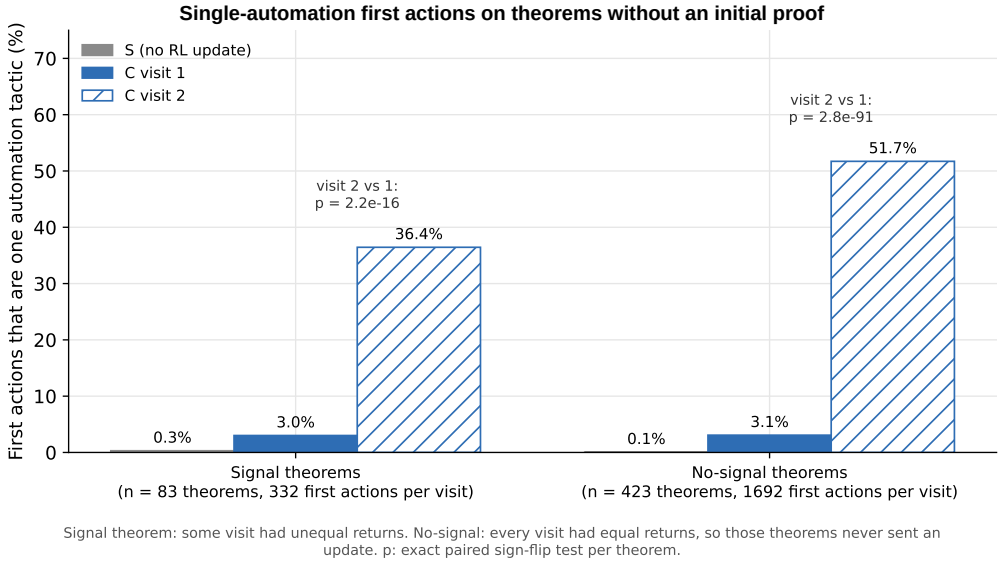



Figure 21: C moves to one automation tactic on theorems with and without a learning signal. A signal theorem had unequal returns in some visit ($n = 83$ theorems, 332 first actions per visit). A no-signal theorem never had unequal returns ($n = 423$, 1,692 first actions per visit). x-axis: theorem class. y-axis: percentage of first actions that are one automation tactic. Bars: S on the same theorems, C visit 1 and C visit 2. $p$: exact paired sign-flip test per theorem, visit 2 against visit 1. Source: `p7_spillover`, `part1.C`.

lower, sign test $p = 0.0026$). Here pass@1 is the chance that one attempt is accepted. Theorems proved fall from 50 to 40 of 62 (McNemar 1 against 11, $p = 0.0063$; Figure 22). Each half mixes policies, so pass@$k$ here is an approximation. A rule fixed before the results sorts the 62 theorems into 27 learned, 11 lost, 13 stable and 11 never proved (Figure 23; `p10_trajectories`). "Lost" means proved in visits 1 to 8 and never in visits 9 to 16. For the 11 lost theorems (`p11_regressions`, `B_rl_filtered_lost`):

- S's 14 accepted attempts on them include 0 that start with the template. 13 of the 14 use a tactic other than `simp_all` or `omega`, mostly `have` and `interval_cases`.
- The template's mean share of first actions on these theorems rises from 0.724 in visits 1 to 8 to 0.991 in visits 9 to 16.

This pattern is consistent with the template pushing out the `have` and `interval_cases` proofs, but no test isolates that.

R and C also lose TRAIN theorems that S proves (`p11_regressions`, `A_train`). With 4 attempts each, R's first visit loses 37 S-proved theorems and gains 20 ($p = 0.033$). C's first visit loses 29 and gains 38 ($p = 0.33$). 16 of the 63 S-proved theorems are proved by neither R nor C in 8 attempts. S proves each of these 16 in only 1 of its 4 attempts, so they can be sampling noise. No count separates noise from forgetting.

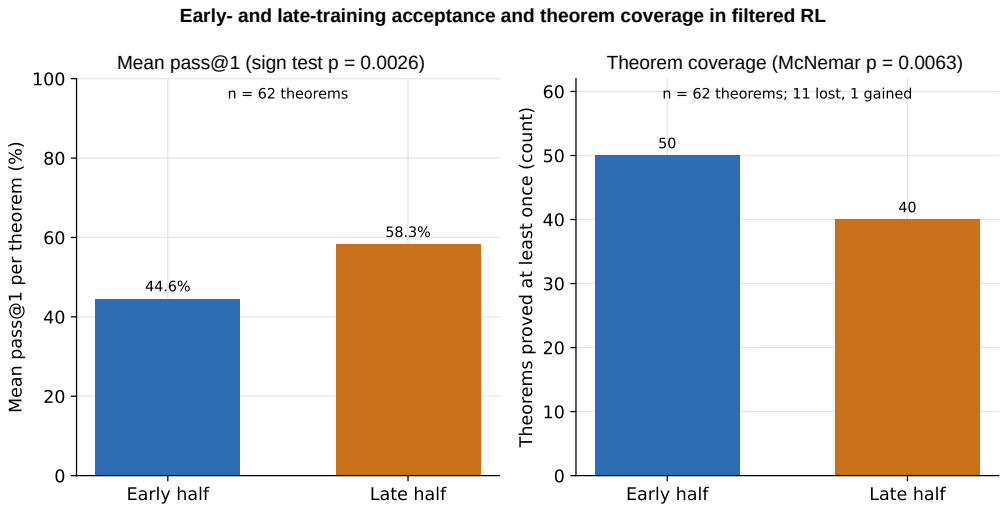



Figure 22: `RL_filtered` on its 62 TRAIN theorems, early half against late half of training. Left: y-axis is the mean pass@1 per theorem (%); 44.6% against 58.3%; sign test $p = 0.0026$. Right: y-axis is the number of theorems proved at least once; 50 against 40; McNemar $p = 0.0063$, 11 lost and 1 gained. $n = 62$ theorems. Each half mixes policies, so pass@1 is an approximation. Source: `p6_coverage`, `B`.

**Action repetition after Lean errors.** In `RL_filtered`, the next action applies a tactic after an error in 662 of 3,260 cases in quarter 1, and in 181 of 2,120 in quarter 4 (`p7_feedback`). In C, an identical repeat of the failed action follows an error in 72 of 4,557 cases in quarter 1, and in 1,872 of 4,190 in quarter 4. In `RL_filtered`, it rises from 122 of 3,260 to 420 of 2,120 (Figure 24). After a failed named tactic, `RL_filtered`'s next action starts with the same tactic in 4,663 of 5,953 pairs. After a valid action, the baseline is 213 of 885 (`p10_readerror`). After an unknown-name error, `RL_filtered` keeps the same name in 871 of 2,260 next actions, against a baseline of 103 of 808 (Figure 25). The Fisher tests in Figure 25 treat pairs inside one attempt as independent. They are not, so these $p$-values are optimistic.

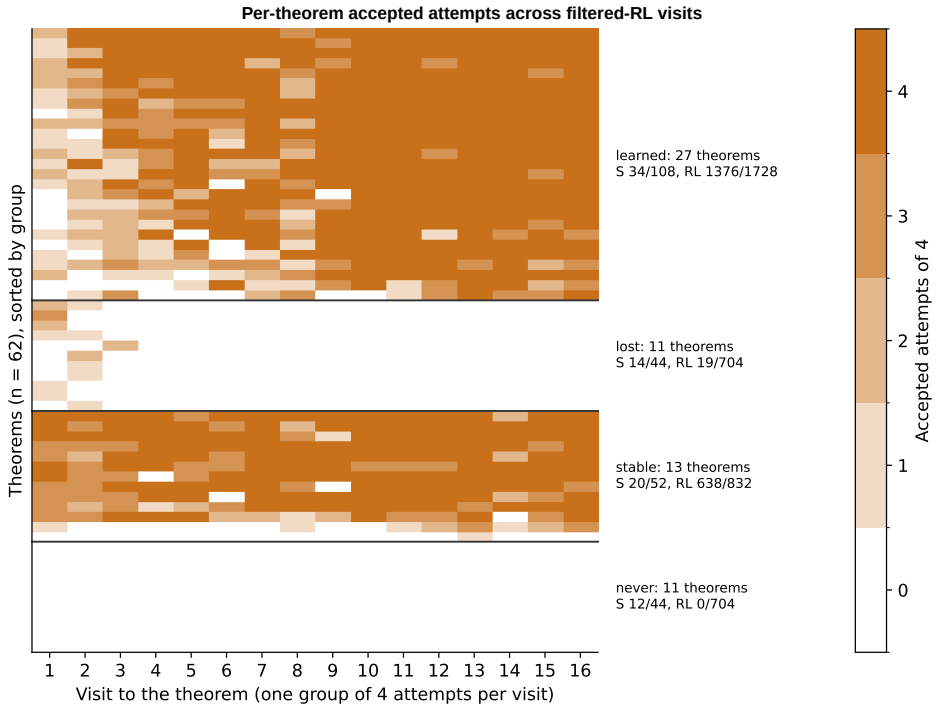



Figure 23: Accepted attempts of 4 at each visit, for each of the 62 `RL_filtered` theorems. x-axis: visit to the theorem (one group of 4 attempts), visits 1 to 16. y-axis: theorems sorted by group. Color: accepted attempts of 4. Groups follow a rule fixed before the results: learned (27), lost (11), stable (13) and never proved (11). The text beside each block gives S's and `RL_filtered`'s accepted attempts of all attempts on those theorems. Source: `p10_trajectories`, `A`.

**Comment frequency in generated actions.** Actions with a comment fall in all three runs from quarter 1 to quarter 4. R falls from 728 of 5,905 to 30 of 5,955, C from 350 of 5,765 to 22 of 5,869, and `RL_filtered` from 208 of 4,480 to 0 of 3,149 (`p7_feedback`). Comments do not change what Lean checks. We did not test whether their loss changes proof success.

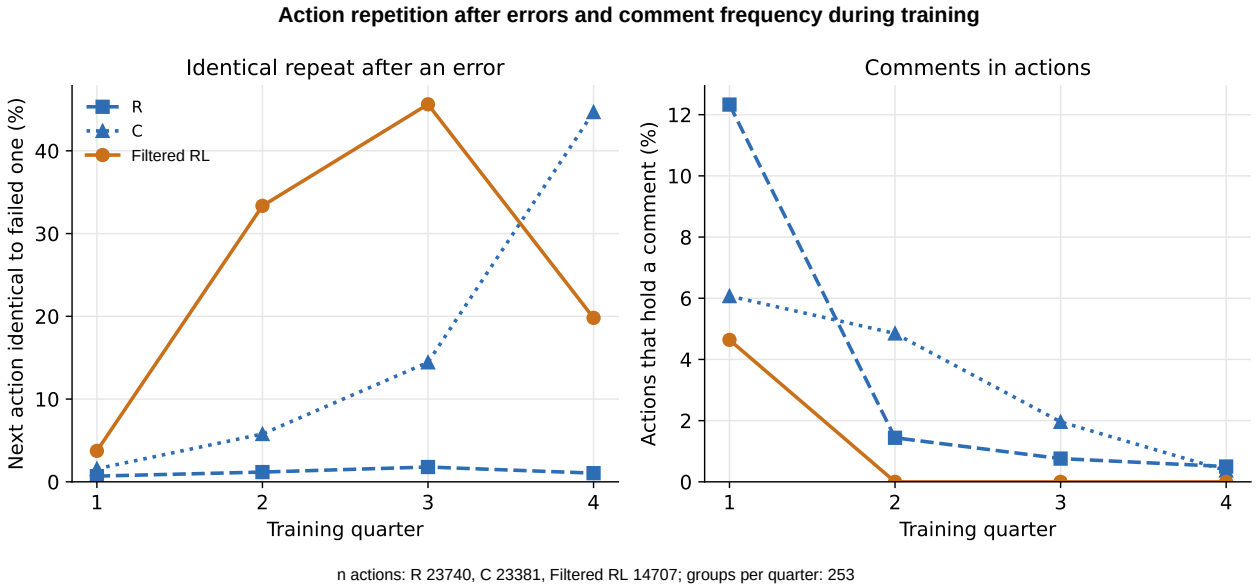



Figure 24: Feedback use and comments during training. x-axis: training quarter, 253 groups per quarter. Left: y-axis is the percentage of error actions whose next action in the same attempt is identical. Right: y-axis is the percentage of actions that hold a comment. n: R 23,740, C 23,381 and `RL_filtered` 14,707 actions. No error bars and no test. Source: `p7_feedback`, `runs.*.quarters`.

**Unknown lemma identifiers on the selection set.** On the selection set, `RL_filtered` makes an unknown-name first error in 45.02 of every 100 actions. S makes one in 23.91 of every 100 (`p10_remedies`). Of `RL_filtered`'s 1,919 failed selection actions, 877 are an unknown-name error and 948 are a failed named tactic, mostly `simp_all` (614) and `omega` (298). For S, 467 of 1,534 failed actions are an unknown-name error (Figure 26; `p5_errors`). Of `RL_filtered`'s distinct unknown names, 174 of 274 are wholly invented lemma names. For S, it is 88 of 178 (`p10_lemmas`). A name counts as wholly invented if its normalized edit distance to the nearest real name is above 0.2, for the full name and for its last component.

**Tactic-distribution distance from Kimina proofs.** Kimina proves 56 of 164 selection theorems with 16 samples each. S proves 1 of 163 (`p6_coverage`). The Jensen-Shannon distance of top-level tactic heads to Kimina's is 0.473 for the SFT examples and 0.477 for S. It is 0.814 for R and 0.918 for `RL_filtered` (Figure 27; `p7_heldout`). The distance is 0 for the same distribution and 1 for no overlap. Kimina proves theorems with shorter reference proofs: the median difference is $-16.5$ lines ($p = 7.0 \times 10^{-5}$). Kimina writes a whole proof in one output, with a larger token budget. The distance compares tactic names only.

**Late-training acceptance and theorem order.** C accepts 39 of 408 attempts in decile 1 and 4 of 404 in decile 10 (`p5_diversity`). On about the same theorems, its first pass accepts 16 of 808 attempts and its second pass 9 of 808 (`p11_regressions`). These counts are consistent with theorem

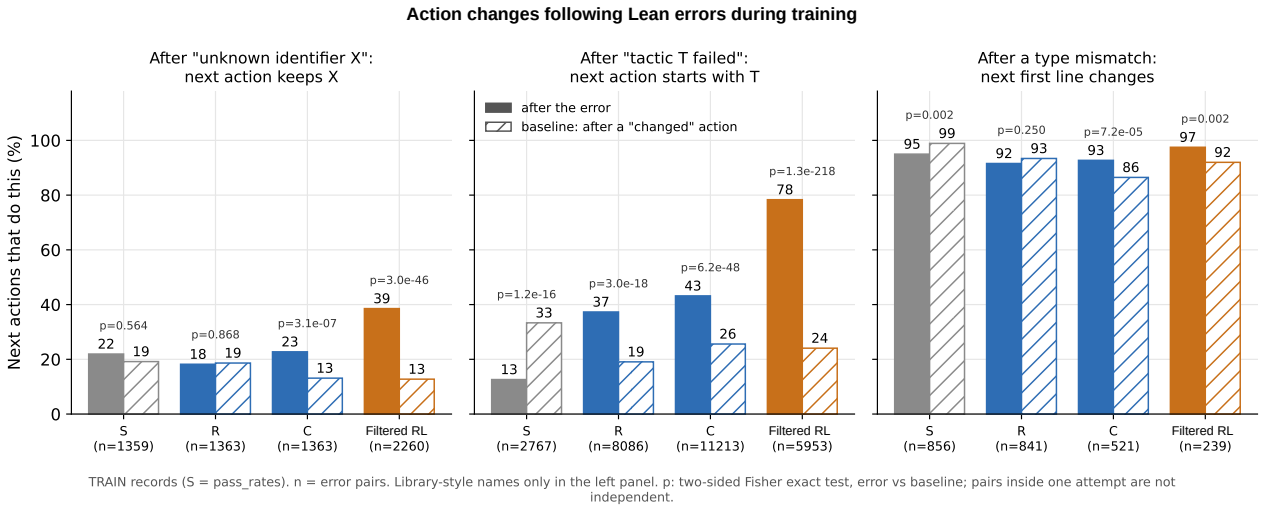



Figure 25: The next action after three Lean errors, on TRAIN records (S is `pass_rates`). y-axis: percentage of next actions that keep the unknown name (left), start with the failed tactic (middle), or change the first line after a type mismatch (right). Solid bars: after the error. Hatched bars: the baseline, after a valid (`changed`) action. n under each run: error pairs. $p$: two-sided Fisher exact test, error against baseline. Pairs inside one attempt are not independent, so the $p$-values are optimistic. Source: `p10_readerror`.

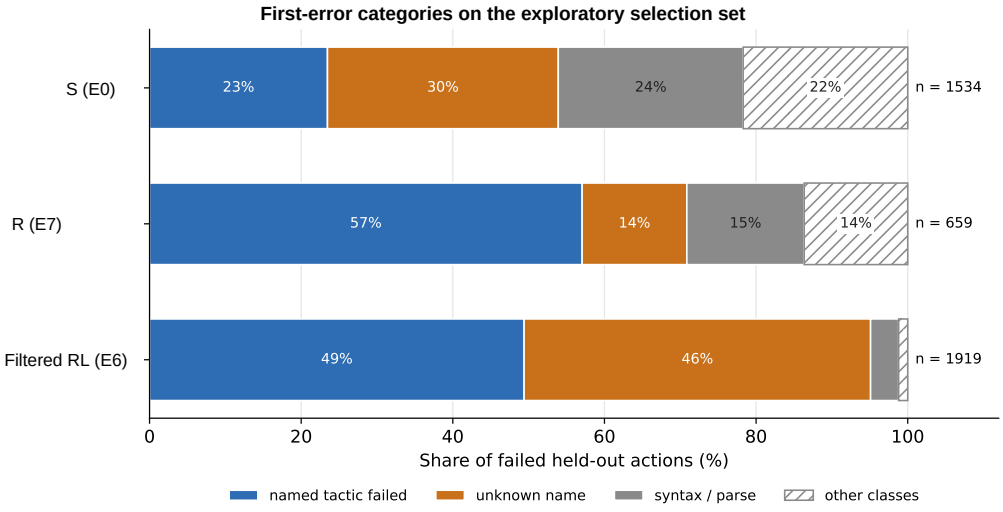



Figure 26: First-error categories of failed actions on the selection set. x-axis: share of failed held-out actions (%). y-axis: run. Classes: failed named tactic, unknown name, syntax or parse error, and other classes. n: failed actions, S (E0) 1,534, R (E7) 659 and `RL_filtered` (E6) 1,919. No test. Source: `p5_errors`, `runs.*.classes`.

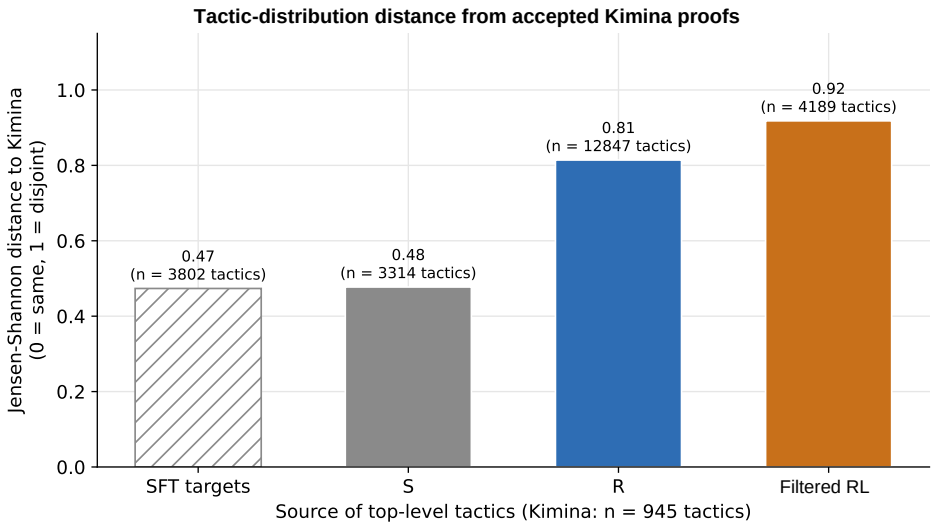



Figure 27: Jensen-Shannon distance between the top-level tactic heads of each source and those of Kimina's accepted proofs. x-axis: source. y-axis: distance, 0 for the same distribution and 1 for disjoint ones. n: SFT targets 3,802 tactics, S 3,314, R 12,847, `RL_filtered` 4,189 and Kimina 945. No interval. Source: `p7_heldout`, `vocabulary.top_level`.

order as the main source of the decile-10 drop. The same-theorem drop is smaller. Neither comparison has a test.

**Sampled advantages under counterfactual error penalties (EXP-030).** EXP-030 did not run (Section G). A re-scoring of the recorded actions under its reward (Table 20) finds positive per-action weights on many penalised actions. The baseline counts all actions of the other attempts, but the reward-to-go of a late action drops the early penalties of its own attempt. A late penalised action can therefore get a positive weight. Of R's 22,962 penalised actions, 19,750 get a positive weight. For `RL_filtered`, it is 8,306 of 11,656 (Figure 28; `p9_counterfactual`). The script reproduces the recorded advantages with a maximum absolute difference of 0.0. An advantage compares a return with a baseline; its sign alone does not establish an error in the estimator. This re-scoring does not establish estimator bias or predict the outcome of a dense-reward run. The records do not show that this finding caused the stop of EXP-030.

## A.4 Partial progress and training gains

Table 9 pairs each training or partial-progress gain with its control and limitation. Section 4 reports the complete-proof comparisons.

**Selection-set partial progress and SFT-seed variability.** Figure 29 shows valid steps on the selection set. The control is two S models: E0, and E5, a fresh SFT from B with SFT seed 2. On the 163 paired theorems, valid attempts are 105 for E0, 130 for E5, 183 for E6 and 146 for E7 (`p12_positives`). E6 beats both S models, and E5 against E0 measures the noise of S's seed and sampling. R (E7) against E5 gives 58 theorems up and 48 down ($p = 0.27$). Held-out theorems similar to the 27 learned theorems do not show more E6 progress. Only 6 of 163 are similar (token Jaccard at least 0.5). Their mean E6-minus-E0 difference is 0.0, against 0.4968 for the other 157 (exact permutation $p = 0.34$). With 6 similar theorems, this check cannot detect a small effect.

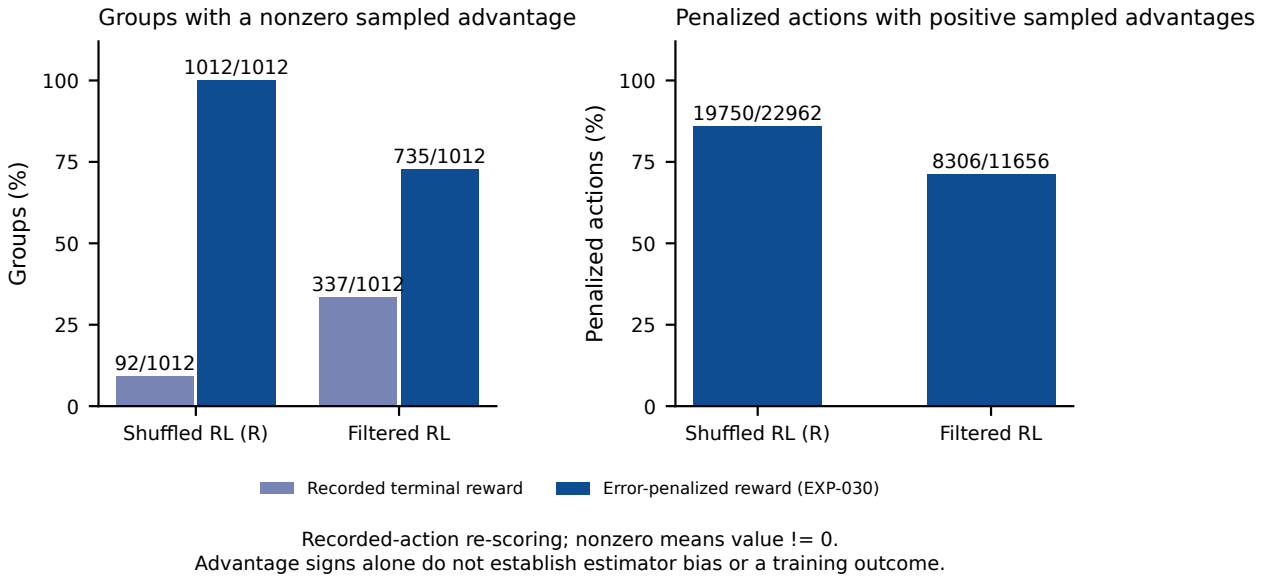



Figure 28: Re-scoring of recorded actions under the EXP-030 dense reward. Left: y-axis is the percentage of groups with a nonzero advantage, under the recorded terminal reward (solid) and the dense reward (hatched); counts of 1,012 groups. This panel uses the original figure's exact nonzero comparison. Right: y-axis is the percentage of penalised actions with a positive per-action weight; R 19,750 of 22,962 and `RL_filtered` 8,306 of 11,656. x-axis: run. These sampled weights do not establish estimator bias or predict a dense-reward training outcome. Source: `p9_counterfactual`; the public builder preserves the counts from the earlier PDF.

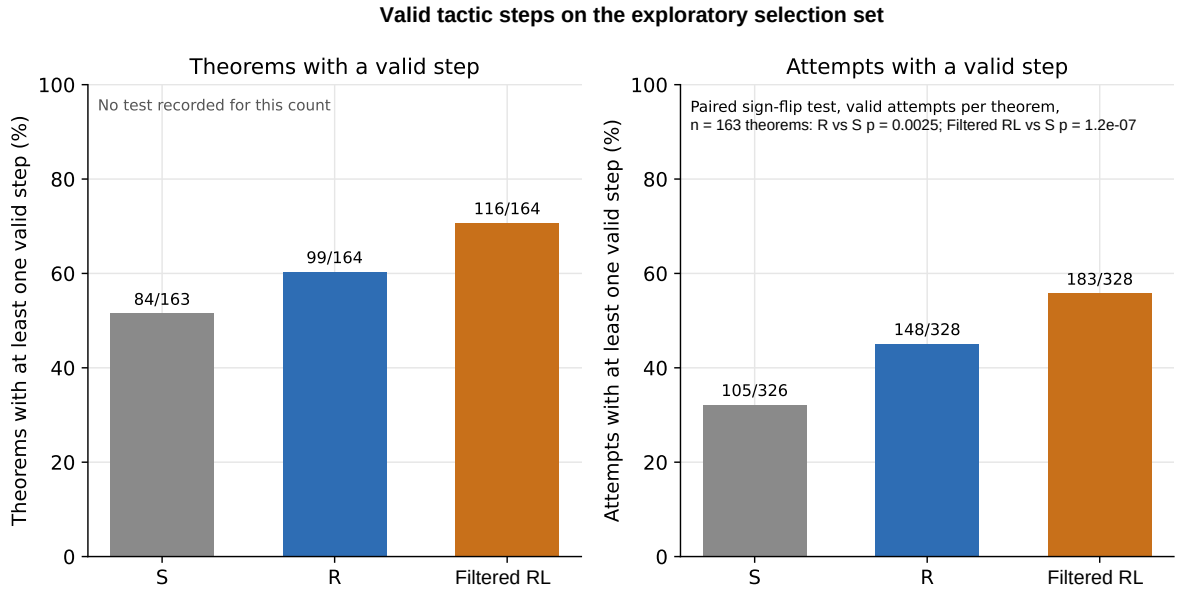



Figure 29: Valid steps on the selection set, 2 attempts per theorem. x-axis: model, S (E0), R (E7) and `RL_filtered` (E6). Left: y-axis is the percentage of theorems with at least one valid step; S 84/163, R 99/164, `RL_filtered` 116/164; no test. Right: y-axis is the percentage of attempts with a valid step; 105/326, 148/328 and 183/328; paired sign-flip test on valid attempts per theorem over 163 theorems, R against S $p = 0.0025$, `RL_filtered` against S $p = 1.2 \times 10^{-7}$. Source: `p7_heldout`, `selection`.

Table 9: Exploratory measurements of partial progress and training gains, with controls and limitations. All are exploratory and come from one run per arm. Files are in `out/`.

| Positive | Result and test | Control | Caveat |
|---|---|---|---|
| `RL_filtered` held-out valid attempts beat two S models | Against E0: 82 theorems up, 31 down, $p = 1.2 \times 10^{-7}$. Against E5: 76 up, 40 down, $p = 0.00068$. 163 paired theorems; `p12_positives` | E0 and E5. E5 against E0: 50 up, 31 down, $p = 0.030$ | Partial progress, not proofs. E6 proves 2 of 164 |
| `RL_filtered` has the most valid attempts of E0, E5 and E6 | 63 of the 98 theorems with a unique maximum, of 163; chance gives 32.667; exact $p = 3.8 \times 10^{-10}$ | E0 has the maximum on 12, E5 on 23 | Many valid steps are `simp_all` steps that leave the goal open |
| Repetition of actions with positive advantage | Share 0.286 against 0.023; 37 up, 1 down of 51 theorems; $p = 1.0 \times 10^{-10}$; `p9_update_follow` | Action with negative advantage in the same group | Association. R and C show no detected effect |
| Learned proofs are new text | 0 of 2,082 accepted attempts equal an SFT reference. On the 27 learned theorems, 69 of 69 late distinct texts differ from the references; `p5_recall`, `p12_positives` | S: 1 of 84 equal a reference | Distinct text is not a distinct method. Most texts are the template |
| New TRAIN proofs where S had none | Of 443 theorems, R proves 29 and C 54; C has 54 gains, 24 losses, McNemar $p = 0.0009$; `p12_positives`, `p11_regressions` | S with 4 attempts | R and C had 8 attempts, S had 4. R's gains equal its losses (29 and 29) |
| Task filtering gives more signal | Mixed groups: `RL_filtered` 207, R 81, C 103 of 1,012; `p6_signal` | R and C, same start model and group count | 62 theorems, not 506. No held-out proof gain |
| Fewer format errors in `RL_filtered` training | 142 of 4,480 actions in quarter 1, 10 of 3,149 in quarter 4; 48 theorems down, 2 up, sign test $p = 2.3 \times 10^{-12}$; `p12_positives` | The same theorems in quarter 1 | TRAIN only. R's format errors rise |
| Fewer unknown-name errors in C training | Share of non-format actions with an unknown-name error: 0.1514 at visit 1, 0.0549 at visit 2. Sign test on the per-theorem rate over all actions: 374 down, 66 up, $p = 2.7 \times 10^{-53}$ | The same theorems at visit 1 | Comes with the move to automation, which cannot invent a name |
| Fewer tokens per accepted attempt | `RL_filtered`: 388.2 in quarter 1, 141.4 in quarter 4; `p12_positives` | The same 62 theorems | Shorter outputs are part of the template collapse |
| Step search raises partial progress | E2 231 against E0 105 valid attempts; 99 up, 6 down, $p = 1.0 \times 10^{-23}$; `p10_remedies` | E0 on the same 163 theorems | E2 proves 1 of 163 |
| The selection set is provable | Kimina proves 56 of 164 with 16 samples each; `p6_coverage` | S proves 1 of 163 | Not under our action limits |
| Dense-reward weights can be inspected before training | 19,750 of 22,962 penalised R actions get a positive weight; `p9_counterfactual` | Recorded sparse advantages | Re-scoring alone does not establish an estimator error |

**Changes in training error categories.** These changes in error rates occur on TRAIN only. In R, the share of non-format actions with an unknown-name error falls from 0.146 at visit 1 to 0.0861 at visit 2. A sign test on the per-theorem rate over all actions gives 296 theorems that fall and 115 that rise ($p = 1.7 \times 10^{-19}$). R's format errors rise in the same comparison, from 0.164 to 0.2928 of actions (311 theorems rise, 138 fall, $p = 2.0 \times 10^{-16}$). `RL_filtered`'s unknown-name rate is mixed. Per action, it rises from 0.1741 to 0.2328 (Fisher $p = 3.4 \times 10^{-10}$). Per theorem, it falls in 40 and rises in 22 ($p = 0.030$). On held-out theorems, `RL_filtered` makes more unknown-name errors than S (Section A.3).

**Source records.** The archived inputs match their recorded hashes: 19 of 19 data members, 2 of 2 schedule members and 3 of 3 selection-set members (`.private/ops/abhinav_data_handoff_2026 1020_v1/member_checks.txt`). The file `runtime_order_manifest.json` has no recorded hash. One record chunk has an S3 expiry date of 18 November 2026. Without a copy, the records stop being reproducible after that date.

### A.5 Uncertainty and multiple comparisons

Table 10 lists the corrected tests. In the 278-test family, 86 tests survive Holm at 0.05 (`p8_findings`). All 11 tests of the headline family survive Holm at 0.05.

Table 10: Holm-corrected $p$-values for the tests of `out/p8_findings.json`. "Headline" is the 11-test family. "278" is the family of every test in that file. H1 is not in the 278-test family.

| H | Finding | Raw $p$ | Headline Holm $p$ | 278 Holm $p$ |
|---|---|---|---|---|
| *Survives both families* | | | | |
| 8 | `RL_filtered` against S, held-out valid attempts | $1.2 \times 10^{-7}$ | $1.2 \times 10^{-6}$ | $2.9 \times 10^{-5}$ |
| 10 | C, no-signal theorems, single-automation share, visit 2 against visit 1 | $2.8 \times 10^{-91}$ | $3.1 \times 10^{-90}$ | $7.7 \times 10^{-89}$ |
| 11 | `RL_filtered` template before against after the first acceptance | $2.3 \times 10^{-7}$ | $2.0 \times 10^{-6}$ | $5.7 \times 10^{-5}$ |
| 6 | Kimina proves theorems with shorter reference proofs | $7.0 \times 10^{-5}$ | $4.9 \times 10^{-4}$ | 0.014 |
| *Survives the headline family only* | | | | |
| 1 | `RL_filtered` TRAIN acceptance, last visit against first | $2.4 \times 10^{-6}$ | $1.9 \times 10^{-5}$ | not in family |
| 2 | C first visit against S | 0.0019 | 0.011 | 0.36 |
| 3 | C second visit against S, where the first visit had a proof | 0.011 | 0.019 | 1 |
| 4 | `RL_filtered` coverage, early half against late half | 0.0063 | 0.019 | 1 |
| 5 | `RL_filtered` pass@1, late half against early half | 0.0026 | 0.013 | 0.47 |
| 7 | C transfer band (direction opposite to transfer) | 0.0072 | 0.019 | 1 |
| 9 | R against S, held-out valid attempts | 0.0025 | 0.013 | 0.47 |

H7 contradicts transfer through similar theorems. C's theorems with an earlier accepted similar theorem gain less, not more: a mean of 0.040 against 0.209 (`p6_transfer`).

The following results are outside any correction:

- every result from `p9_*`, `p10_*`, `p11_regressions` and `p12_positives`, which includes the update-follow test, the error responses, the lemma reuse, the remedies, the dense-reward re-scoring, every per-theorem loss and gain, and the 48 tests of `p12_positives`;
- the breakdown cells of `p8_breakdown`;
- every DEV number, because DEV has 8 theorems.

Every arm is one training run with one seed. E5 against E0 gives 50 theorems up and 31 down ($p = 0.030$), so the seed noise of S is not small.

## A.6 Replication in EXP-023

EXP-023 is the curriculum run of 13 October (Section B.7), with a different start model and theorem set. Its records allow four of the harms to be checked (`p9_exp023`, first and last decile).

- The diversity loss replicates in a weaker form in arms B and C_seq, and not in arm A (Figure 30). Distinct first actions go from 94 of 120 to 97 of 112 in arm A, 107 of 120 to 93 of 112 in arm B, and 98 of 120 to 68 of 112 in arm C_seq. The EXP-023 arms lose at most 21 points, against 78 for C and 49 for `RL_filtered`.
- More identical repeats after an error replicate in two of three arms. Arm A goes from 20 of 345 to 31 of 301, and arm C_seq from 38 of 343 to 77 of 215. Arm B falls, from 20 of 452 to 11 of 164.
- The comment drop cannot be tested. EXP-023 has no comments in decile 1 (A 0 of 491, B 0 of 615, C_seq 0 of 503 actions).
- Acceptance does not fall late. Arm A goes from 69 of 120 to 63 of 112, arm B from 37 of 120 to 89 of 112, and arm C_seq from 65 of 120 to 70 of 112.

The EXP-023 deciles hold about 120 first actions, against about 404 in October. A smaller sample gives a higher distinct share, so only the trends are comparable.

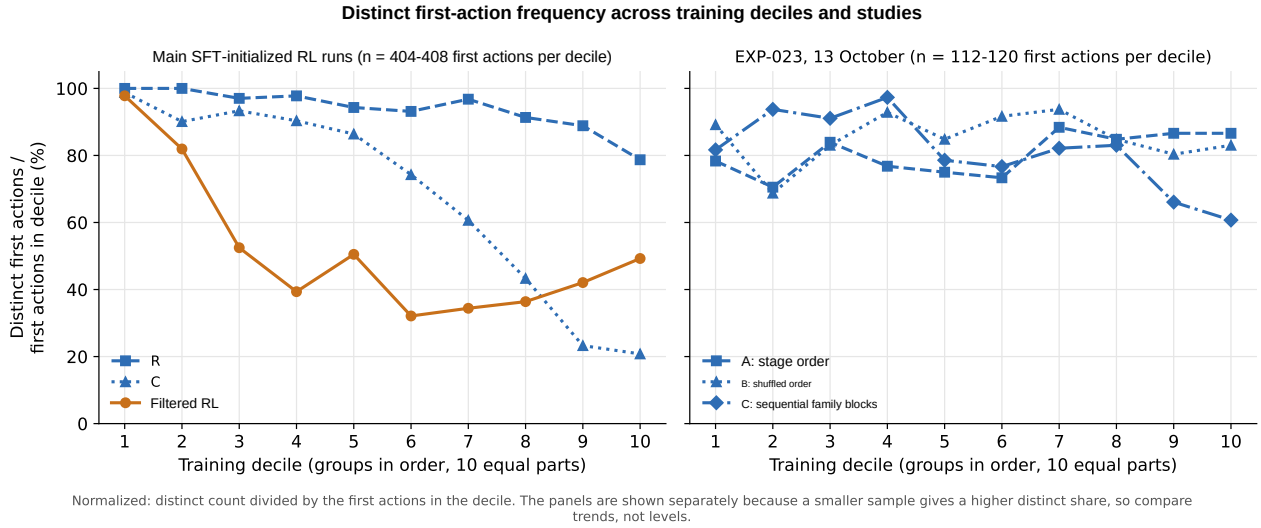



Figure 30: Distinct first actions as a percentage of the first actions in each training decile. x-axis: training decile, groups in order in 10 equal parts. Left: the October runs R, C and `RL_filtered`, 404 to 408 first actions per decile. Right: the EXP-023 arms of 13 October, 112 to 120 first actions per decile. The panels have different sample sizes, so compare trends, not levels. No error bars. Source: `p9_exp023` and `p5_diversity`.

## A.7 Retracted and weakened claims

- C's late acceptance drop is consistent with theorem order, not with a loss on the same theorems (Section A.3).

- Transfer through similar theorems is contradicted. H7 points in the opposite direction ($p = 0.0072$ raw; Table 10).
- Part of the held-out valid-step gain over E0 is seed noise. On the 163 paired theorems, E5, a second S, has a valid step on 94, against 84 for E0 (`p12_positives`).
- Lemma retrieval (E3) gives no detected change in unknown-name errors. E3 has 23.87 per 100 actions, against 23.32 for its control E5 ($p = 0.743$; `p10_remedies`).

### A.8 Open research questions and proposed tests

No new run is planned. Each item names the test that would answer it.

1. Does a KL or entropy term stop the template collapse? Test: rerun `RL_filtered` from S with a KL term to S, with the same theorems, budget and seed. Compare the template share and the late-half coverage.
2. Is the rare reward the main cause of the missing held-out gain? Test: start RL from a model that proves more TRAIN theorems, or train on theorems that S proves at a higher rate. Measure proved held-out theorems on a larger set.
3. Would relaxed action limits change what RL reinforces? Test: rerun R under 12 actions, 8 tactics and 384 tokens. Compare format errors and the closing action.
4. Did the template push out the proofs of the 11 lost theorems? Test: in a KL run, check whether these 11 theorems keep their early proofs.
5. Are the 16 TRAIN theorems that only S proves forgetting or sampling noise? Test: sample S and R 32 times each on these 16 theorems and compare the per-theorem pass rates.
6. Would retrieval of real lemma names lower the unknown-name errors after RL? Test: run E3-style retrieval on `RL_filtered` and compare against `RL_filtered` without retrieval.
7. Is there a held-out gain too small to detect? Test: evaluate S and `RL_filtered` on a held-out set large enough for the observed rates (Section 4.6 gives the power).
8. Does the dense reward help? Test: check the estimator against the intended discounted objective, then compare complete-proof acceptance under the registered matched-run protocol. The signs in `p9_counterfactual` alone do not determine whether the baseline needs changing.

### A.9 Software and reproduction notes

**Reproduction.** The README and script guide give setup, input requirements and reproduction commands. `scripts/reproduce.py` checks supplied files against recorded hashes and prepares SFT, DEV and RL command plans using `configs/reproduction_v1.json`. The settings are reconstructed from the documented recipe of 18–19 October; they are not the original private configuration files or evidence of a rerun. The locked training dependency group follows the documented 19 October library combination, while the original main-run package receipts remain external. Full reproduction still requires the archived data, later-study weights and Linux Lean runtime, and several dataset builders remain private. The helper does not execute a model or Lean, and its file checks do not establish tokenizer equivalence.

**Public dataset rebuilding.** `scripts/data/rebuild_nt_dataset.py` adds source fetching, corpus assembly, verifier identity receipts and task-order construction. `configs/data/nt_sources_v1.json` pins public files by full revision and SHA-256. The corpus retains proof candidates without claiming

Lean verification; the existing benchmark stages assign duplicate-aware splits, replay references, screen fixed tactics and export full-proof SFT and RL inputs. The strict screen excludes forbidden tactics, including `exact?`; old selections containing unsupported screen tactics require a new selection. New orders preserve task multiplicities, and prerequisite order requires explicit annotations. These commands produce a new exploratory dataset, not the historical filtered release or its interactive action examples. The recorded local checks use synthetic JSON and Parquet data and scripted verifier outcomes; they establish neither a full public-corpus rebuild nor real Lean admission.

**Proof-component reuse and search extensions.** `rl_components/environment.py` handles proof interaction and replay; `rl_components/observations.py` renders observations. Builders can import them directly. The training command retains compatibility exports. The host still supplies the pinned checker, REPL context, admitted tasks and run records; these modules are not a portable prover service. AlphaProof's policy/value-guided tree search [14] is an architectural reference, not a validated integration. The existing finite-catalogue candidate critic does not supply a general value model or AND–OR search over proof obligations. A prospective extension should first validate the pinned runtime, freeze data and budget controls, then register bounded-search and value-learning comparisons against independent sampling. Search-selected actions require an appropriate learning objective rather than being treated as on-policy samples. Only fresh whole-proof acceptance establishes a proof: goal-text progress remains heuristic, and live REPL state handles are not portable state identities. No new experiment or performance claim follows from this refactoring.

# B Earlier experiments

Figure 31 summarizes the earlier experiments through 13 October and the plan recorded at that time. A cause is *tested* when an experiment could have shown it to be false; otherwise it is a *candidate*.

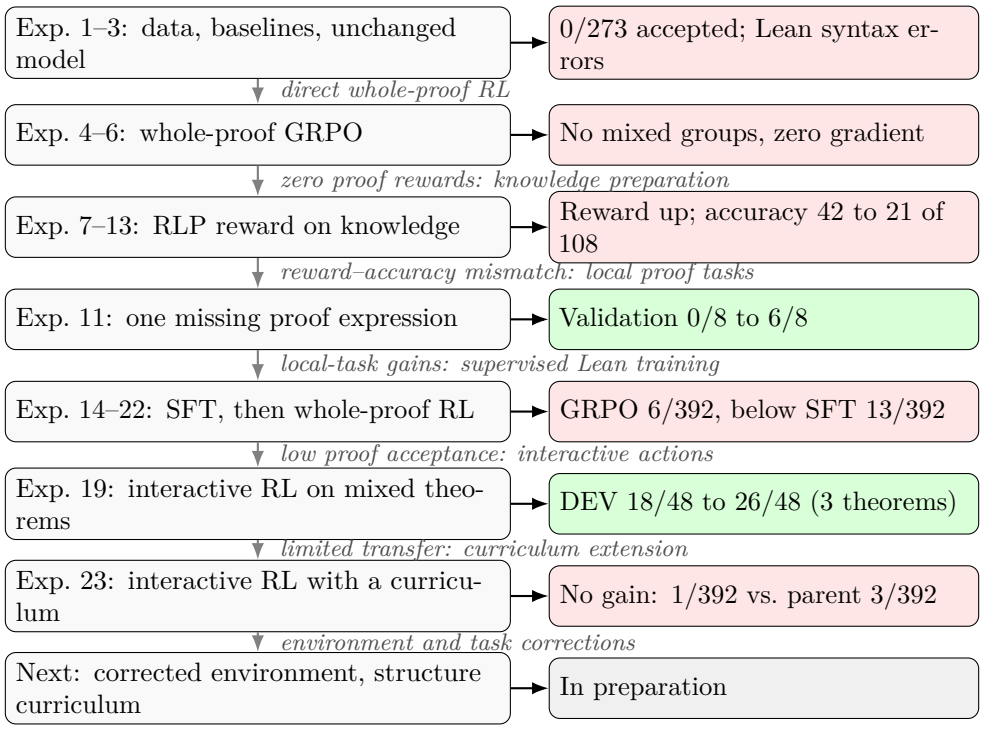



Figure 31: Historical experiment sequence through 13 October 2026. Red: failure. Green: RL gain. Grey: planned at that reporting cutoff. Later outcomes appear in Section E.1.

## B.1 Data, baseline, and the unchanged model (Experiments 1 to 3)

Experiment 1 selected a number-theory subset of LeanDojo. Its authors report that the random split overestimates performance [2], so we later built stricter splits (Figure 7). Experiment 2 tested the base model on school mathematics: 9 of 24 GSM8K and 3 of 20 MATH number-theory problems. A small fine-tune lowered the held-out loss, but the scores became 8 of 24 and 3 of 20. A lower loss did not mean better answers.

Experiment 3 asked the unchanged model to prove 21 statements. Lean accepted 0 of 63 local attempts and 0 of 273 GPU attempts. Figure 32 shows that no attempt reached the mathematics. A pattern match found old Lean 3 syntax in 33 of the 63 attempts. In 34 of the 63, the model wrote a new theorem when we asked only for a proof. The first barrier is the language (*tested*: the classification reads Lean's error messages).

## B.2 Whole-proof RL from the unchanged model (Experiments 4 to 6)

We ran GRPO in Formulation A with $G = 4$. In Experiment 4, Lean accepted no proof, no group was mixed, and the gradient was zero for all 32 optimizer updates. Validation stayed at 0 of 8 theorems. In Experiment 5, easier hand-written theorems gave 0 of 256 accepted proofs, so we did not train. In Experiment 6, with one expression missing from a given proof, Lean accepted 2 of 64 attempts, but only one theorem gave mixed groups, and held-out attempts stayed at 0 of 16.

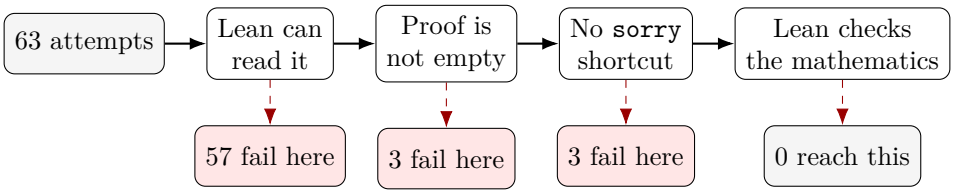



Figure 32: Where the 63 attempts of Experiment 3 failed.

*Tested explanation*: every group had reward 0, so every update was 0 (Section 2.2). Therefore Experiment 4 gives no information about curriculum order.

### B.3 RL with a reward that does not check correctness (Experiments 7 to 10, 12, 13)

Lean gave no signal, so we tried to teach knowledge first with the RLP reward.

- Experiment 7 (textbook text): held-out accuracy was 11/48 before and after.
- Experiment 8: only 8,557,971 of 494,032,768 weights had changed, because low-precision (BF16) rounding discards small updates. With full precision, accuracy stayed at 11/48.
- Experiment 9 (448 arithmetic questions): the reward increased, but held-out accuracy decreased from 42/108 to 21/108 (Figure 33). Answers without a readable number increased from 17 to 69, and answers at the length limit from 15 to 36.
- Experiment 10: an early checkpoint improved at recognizing concepts (7 to 14 of 32), not at using them. Two AI graders disagreed on 44 of 321 answers.
- Experiment 12: all 612 numeric answers in the data were correct, but 160 of 448 rules omitted a condition (for example, $\gcd(a, n) = 1$ for a modular inverse). A constant answer solved every proof-by-contradiction item.
- Experiment 13 (decreasing support): 0 of 18 answers were fully correct at all five checkpoints.

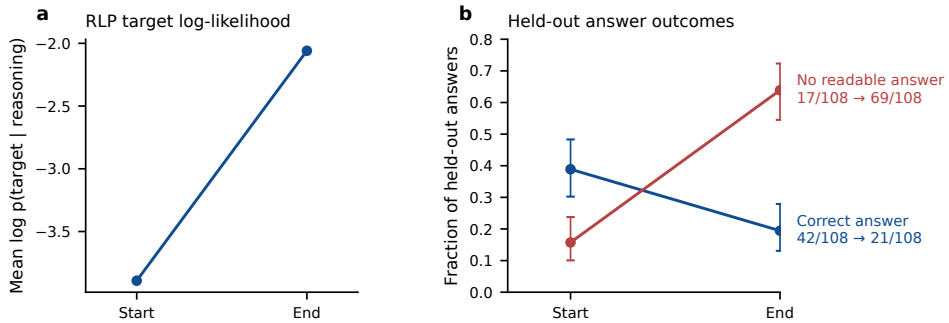



Figure 33: Experiment 9. (a) The quantity that the RLP reward pays for increased. (b) Held-out answers, $n = 108$. Error bars: 95% Wilson score intervals.

*Reward objective mismatch*: the reward pays for reasoning that makes the answer more probable, not for correct reasoning. This is *reward hacking*. In Experiment 9, 17 incorrect answers received a positive advantage. A deliberately false line of reasoning increased the reward in both inspected examples. A *candidate* reason for the accuracy decrease is that training supplied the start of the answer format, but the test did not.

## B.4 Lean-verified witness generation (EXP-011)

We reduced the proof to one missing expression with the mathematics supplied (Figure 34), and ran GRPO with $G = 8$. Validation accuracy increased from 0/8 to 6/8, and held-out accuracy from 0/4 to 2/4. It worked because some groups were mixed. The sets are small, and Lean checked only the expression: only 2 of 8 explanations had no mathematical error.

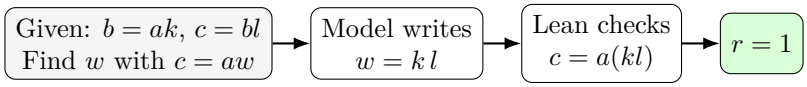



Figure 34: The one-expression goal of Experiment 11. Lean checks the expression, not the explanation.

## B.5 Supervised training, then whole-proof RL (Experiments 14 to 22)

After SFT v2, the model proved 0 of the 392 non-trivial validation statements (2 of all 420). On 200 training statements it proved 0 in 1,600 attempts. The most frequent first errors were an invented lemma name (582), a failed rewrite (329), and the length limit (253). The barrier moved from syntax to library knowledge.

Experiment 20 added a common SFT stage, then alternated SFT with whole-proof GRPO in one branch and with RLP in another (Figure 35). Common SFT increased the count from 0 to 13. The GRPO branch ended at 6, below its parent, and the RLP branch at 15. The GRPO model wrote the single tactic `simp_all` in 1,637 of 3,360 attempts; the parent did so in 62.

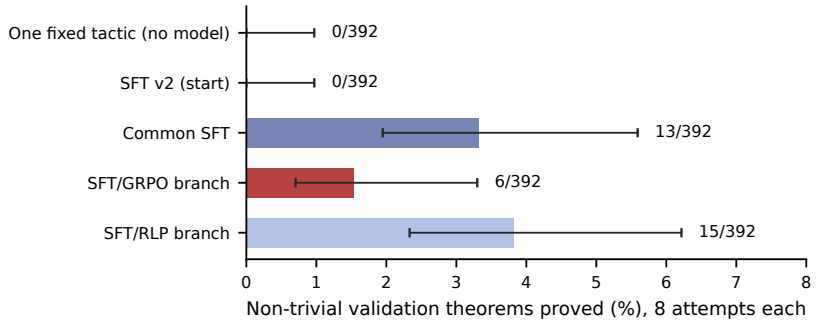



Figure 35: Experiment 20: non-trivial validation statements with an accepted proof in 8 attempts, $n = 392$. Error bars: 95% Wilson score intervals.

Experiment 21 used rejection-sampling fine-tuning (RFT: train on sampled proofs that Lean accepts [11]), then GRPO. On DEV (64 attempts per model), no model had an accepted proof. The first errors of the final model were proof construction (43), unknown names or types (17), syntax (3), and timeout (1). A parser fault rejected every action in a code block, so its interactive pilot made no update. Experiment 22 prepared PPO, but the GPU run failed before RL started.

*Candidate explanations*: the low acceptance rate gives few mixed groups; the GRPO model concentrated on `simp_all`, which closes only easy goals. The branches also had extra SFT stages and different data, so the comparison does not isolate RL.

## B.6 Interactive RL on familiar proofs (Experiment 19)

We used Formulation B from common SFT, with REINFORCE, RLOO, $G = 4$, and at most 4 actions. We trained only on three practice theorems that the model proved some of the time. The selection

rule, fixed before the run, required more accepted DEV attempts than the parent over two seeds and no lost DEV theorem. The rule selected the RL model (Figure 36). It proved 7 of 32 original theorems without help (parent: 4 of 32), and all 7 proofs passed a new Lean check. An earlier run of this type lost one DEV theorem, so the rule kept its parent. Not every theorem improved: `Nat.not_even_one` decreased from 4/4 to 2/4.

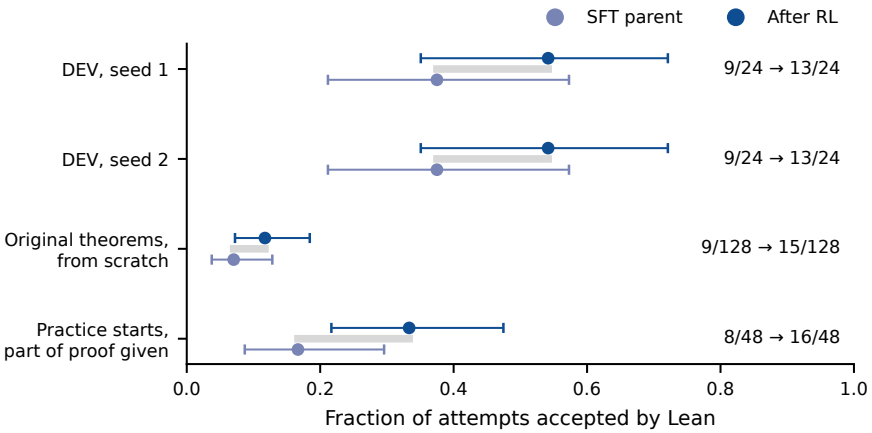



Figure 36: Experiment 19: SFT parent and RL model on the same theorems and seeds. Error bars: 95% Wilson score intervals, which treat correlated attempts as independent and so understate the uncertainty.

*Limits*: DEV has only 3 theorems, there is one training seed, and the 32 original theorems are from the training domain. RL made the model more reliable on proofs that it could already find; it did not show that RL teaches new proofs.

## B.7 Interactive RL with a curriculum (Experiment 23)

On 22 broader theorems, a short warm-start first increased accepted whole proofs on 16 training theorems from 0 to 13. Then an *ordered* arm (answer given, one lemma, combined steps, whole proof) and a *shuffled* arm trained on the same theorems. Neither arm improved on the parent. On broader DEV, all three models had 0 of 64. On elementary DEV retention, the parent had 24/48, ordered 20/48, and shuffled 21/48. On non-trivial validation, the parent proved 3 of 392 and ordered RL 1 of 392; one of the parent's accepted statements has the empty set as an answer. The ordered model made more valid first moves than the parent (88 against 40), but it completed no proof in its first move (Figure 37).

*Candidate explanations* (not yet tested):

1. A valid action is not progress. The model learned valid actions, not complete proofs.
2. The curriculum had gaps: the lemma stage covered 9 of 22 theorems, the combined stage 11 of 22, and the early stages contained the answer.
3. For one Lean error type, the interface lost the message and gave reward 0, while ordinary errors received a penalty. A failure with a lost message therefore scored better (186 actions).
4. The environment did not show the theorem or the remaining actions, and one error ended the attempt.

One common explanation is not supported: most repeated-state failures occurred without `simp_all`.

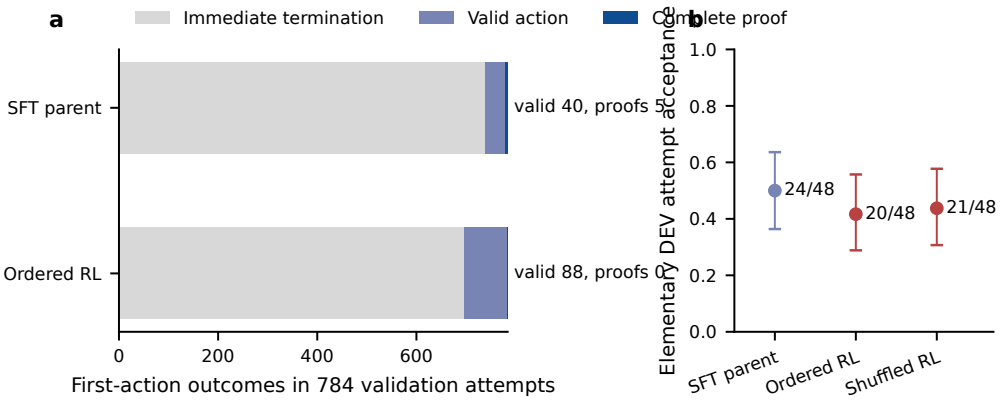



Figure 37: Experiment 23. (a) First action on the same 784 validation attempts. (b) Elementary DEV retention, $n = 48$. Error bars: 95% Wilson score intervals.

# C Reward-verifier exploits and data exposure

The verifier audit found six ways to obtain the Lean reward without proof skill (Figure 38).

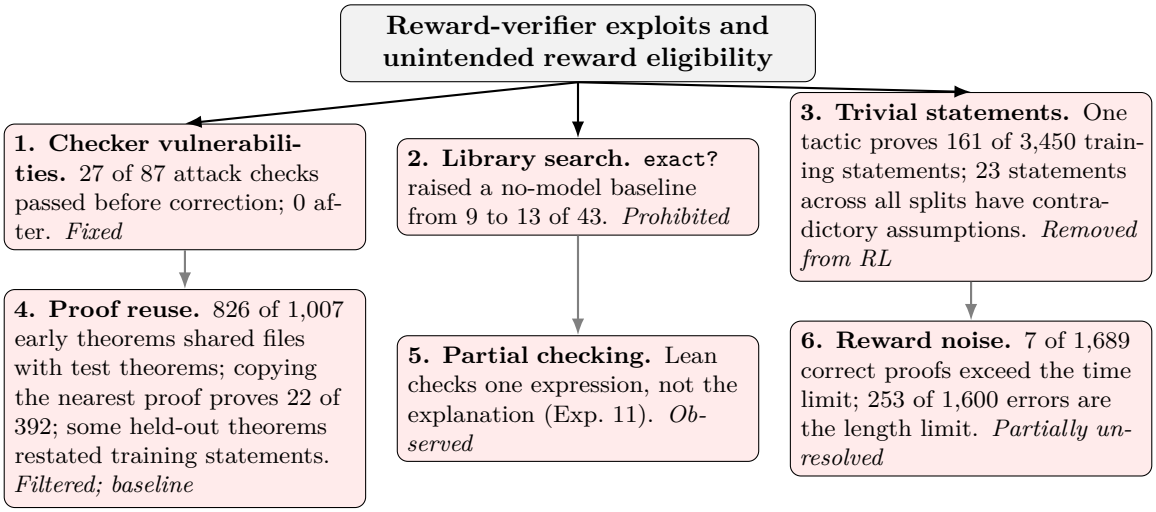



Figure 38: Six sources of unintended reward, with measurements and mitigation status.

The checker attacks included false "no axioms" messages, a replaced axiom report, and a command after the proof. A scan of 10,869 accepted proofs found no attack in real runs. Some statements are also formalized incorrectly; for example, a divisibility statement about $(5+\sqrt{35})^{2n-1}$ is about real numbers, where it is trivially true. Before we report an RL gain, we therefore separate trivial, contradictory, near-duplicate, and new statements, and report the new statements.

# D Training and verification implementation defects

Some failures were engineering faults, not scientific results (Table 11).

Table 11: Engineering faults that we found.

| Fault | Effect | Status |
|---|---|---|
| BF16 optimizer state | Small updates rounded away | Fixed (full precision) |
| Checker attacks | 27 of 87 attack checks accepted | Fixed (0 of 87) |
| Batch checking | One broken proof caused the rejection of the next proof; 15 of 40 correct proofs rejected in one test | Fixed |
| Action parser | Every action in a code block rejected before Lean | Fixed |
| Lost Lean error messages | One error type gave reward 0 instead of a penalty (186 actions) | Patch not yet verified |
| Proof export | Indentation of multi-line proofs changed, so exports failed | Fixed |
| Full-file compilation | DEV files exceeded the memory limit without a proof | Open |
| Prohibited-word filter | A string trick hid a prohibited command, and Lean accepted it | Fixed (version 2.2) |
| Run failures | Full disk at checkpoint save; missing PPO method; launch-script name conflict; wrong prompt-length count | Fixed |

# E Plan recorded on 13 October

**Corrected environment.** Version 2 replaced the observation, the failure transition, and the return of version 1 (Table 12). Version 2.1 then added tactic chunks and moved backtracking into the harness (Table 13). Version 2.2 fixed the deployed interface (Table 6). Integration tests with real Lean passed: 8 of 8 for version 2.1, and 86 of 86 for the version 2.2 trainer.

Table 12: The interactive MDP, environment version 2 (13 October 2026).

| Part | Definition |
|---|---|
| State | Theorem header, supplied proof lines, accepted policy steps, current goals, the last action result, and the steps remaining |
| Observation | The whole state as text. The oldest policy steps are hidden only when the prompt cap requires it |
| Action | Optional `--` comment lines, then one tactic block or `backtrack k` |
| Valid tactic | Append the step and read the new goals from Lean |
| Goals closed | Replay the full proof in file context with `verify_group`; the episode ends |
| Failure | A Lean error, unchanged goals, a cycle on the current path, a malformed or forbidden action, or an invalid backtrack gives a self-loop: the verified state stays, the diagnostic enters the observation, and one step is used |
| `backtrack k` | Return to the verified state after policy step $k$; later steps leave the proof |
| Timeout | The episode ends with reward 0, because the REPL closes |
| REPL crash | The group is censored and gets no update |
| Reward | 1 for a Lean-verified complete proof, otherwise 0. No heuristic term |
| Return | Discounted reward-to-go; the leave-one-out baseline uses the other trajectories' discounted returns from the start |
| Supplied lines | Shown and labelled in the observation; a success earns reward 1; records keep the task kind |

Version 2 rejects repair, the error-feedback reward, and a nonzero exploration bonus. A failed action is a self-loop, not a terminal event; its diagnostic enters the observation and costs one step. Supplied lines are shown and labelled, so an assisted success is recorded separately from an unassisted one. Evaluation stays undiscounted complete-proof acceptance without supplied lines.

Version 2.1 drops the policy `backtrack k` action from the default arm. No reviewed RL prover gives the policy a backtrack action, so the default puts search in the harness and keeps the policy

Table 13: Version 2.1 amendments to environment version 2.

| Part | Version 2.1 |
|---|---|
| Action | Up to `max_chunk_tactics` (default 4) top-level tactics, each optionally preceded by `--` comment lines |
| Transition | Lean runs the tactics in order and keeps the valid prefix. The first error, unchanged goals, or a cycle stops the action and enters the observation. A closing tactic ends the chunk |
| Self-loop | An action that applies no tactic, a malformed or forbidden action, or an invalid backtrack |
| Observation | A Lean file: header, labelled supplied lines, numbered policy steps, `/- tactic state: ... -/`, the last action result, the tactics that already failed at the current state, and the actions remaining |
| Backtracking, default | After `harness_backtrack_after` (default 2) distinct tactics fail at one state, the harness reverts one step and records it as failed at the parent. At most one revert per action |
| Backtracking, ablation | `backtrack_owner` set to `policy` restores the version 2 `backtrack k` action instead of the harness revert |
| Discount | $\gamma = 0.95$ at the configured value. Its role matches AlphaProof's step cost; that equivalence is an inference |
| Parser | A restated declaration through `:= by` is dropped without execution, block comments are treated as comments, and RL, evaluation and SFT builders share one renderer |

backtrack as an ablation.

**Rationale for environment revisions.** The first move, from Formulation A to version 1, was statistical: with a terminal reward, a group updates only if it contains both an accepted and a rejected attempt (Section 2.2). Experiments 4 to 6 produced almost no mixed groups. Version 1 made the action small enough to produce them, and Experiment 19 gave the first gain. The second move, from version 1 to version 2, was formal: the state was too thin, an error ended the episode, and the hidden horizon made the process non-Markov. Experiment 23 adds an empirical reason: valid first actions rose, but no proof completed. The report records four candidate causes; two of them are environment defects rather than learner defects: the lost diagnostic, and the missing state with the terminal error. The third move, to version 2.1, follows the same diagnosis: keep the valid prefix, put the diagnostic in the state, and let the harness backtrack. The fourth move, to version 2.2, came from the run records, not from a new hypothesis: the parser, the checker, batch checking, lost diagnostics, proof export, and the forbidden-command filter all failed in practice. The design also follows published interactive provers. DeepSeek-Prover-V1.5 truncates at the first error and puts the tactic state in a comment [11]. COPRA puts the error in the state [12]. InternLM2.5-StepProver keeps the preceding tactics [13]. AlphaProof uses a step cost [14]; the discount plays that role here, but the equivalence is an inference, not a result. No reviewed RL prover gives the policy a backtrack action. Table 14 summarizes the benefits and limitations of each revision.

**Curriculum ordered by proof structure.** The final curriculum has five stages, ordered by the type of reasoning that a proof needs (Figure 40). All theorems come from templates in families of 8, and Lean accepts the reference proof of every theorem. A family that one fixed tactic mostly proves (4 or more of 8) moves to S0. We defined this relabel rule after we saw the first probe results, so it was not chosen blind. Within the training set, exact duplicates were removed, which leaves 192 training theorems.

**Held-out set overlap exclusions.** A semantic check compared held-out theorems with the statements behind the SFT data. In the first panel, it found 26 held-out–training pairs with the same content and 3 near-transfers; for example, `ho_coprimechain_01` restates `lean_workbook_22701`. We

Table 14: Benefits and limitations of formulation revisions.

| Requirement | Change | Benefit | Limitation |
|---|---|---|---|
| Verifiable reward | Terminal Lean-only reward; forbidden constructs; full replay | Reward means proof | Very sparse signal |
| Mixed groups | Whole proof → tactic step; later tactic chunks | Signal becomes possible | Partial observability; longer horizon |
| Markov state | Full state, remaining actions, discount | Decisions use history | Prompt cap; hidden history; truncation ends episodes |
| Informative failure | Self-loop, first-error truncation, harness revert | Error recovery preserves the valid prefix | No-op loops; search controlled by the harness |
| Bounded cost | 4-tactic chunks; 192 tokens; 6 actions; 20 s | Fewer Lean calls; shorter horizon | 202 of 516 TRAIN reference proofs do not fit the limits |
| Clean measurement | Replay, censoring, status categories | Missing data is not counted as failure | Low statistical power at 164 theorems |
| Dense signal | EXP-030 per-action penalties (not run; Section G) | Gradient from errors | Heuristic; error avoidance is not proof skill |

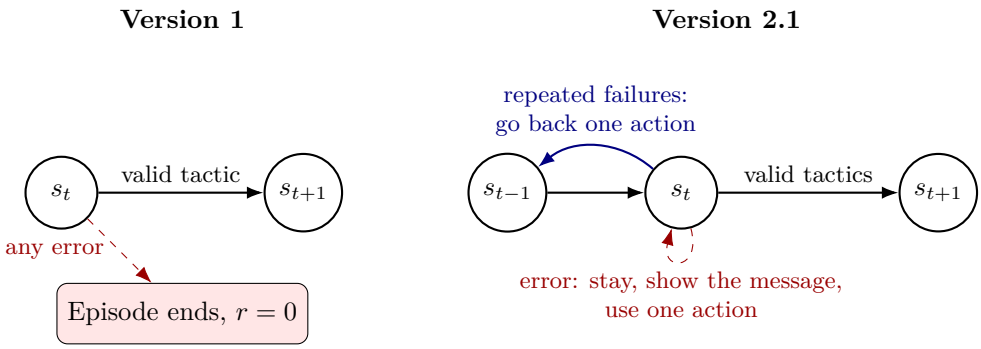



Figure 39: Treatment of a failed action in version 1 and version 2.1.

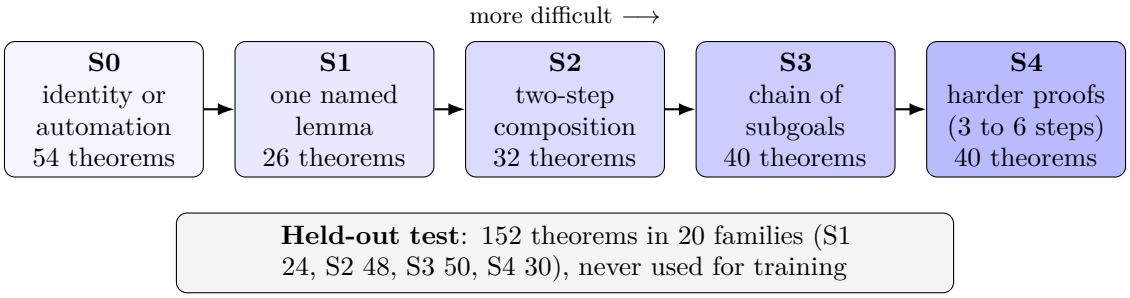



Figure 40: The final structure curriculum: 192 training theorems and 152 held-out theorems. Training excludes 670 names: every held-out, evaluation, reserve, demo, and DEV name.

removed 35 theorems, which left 61 clean theorems. A second set of new families went through the same check, which removed 7 contaminated held-out families and 2 training families. We then dropped the 5 remaining held-out theorems rated "related". The final held-out set has 152 theorems: the 61 original clean theorems and 91 new ones. On the 61 original clean theorems, the SFT parent (environment version 1) proved 6, with 8 of 488 attempts accepted.

**Curriculum-stage training readiness.** Before training, we ran the SFT parent 8 times on each S0 to S2 theorem of the first template set, with the version 1 prompt (Figure 41). With the relabelled stages, S0 had 187 of 448 attempts accepted (44 of 56 theorems mixed), S1 26 of 256 (14 of 32 mixed), and S2 10 of 256 (8 of 32 mixed). Every stage has mixed groups, so RL has a signal, but the signal is weak in S2. Under the original labels, one family that `omega` proves alone gave 32 of the 42 S2 accepted attempts, and the two real two-step families gave 8 of 64 and 0 of 64. With the version 2.1 prompt, 186 of 386 first actions of the parent failed on format. In 145 of these 186, the code block had a label other than `lean`, for example `liquorice`; no run has tested whether the label causes the failure. Therefore a short warm-start in the new format comes before RL. Its data has 1,969 training rows on 468 theorems, each traced again through the version 2.1 environment.

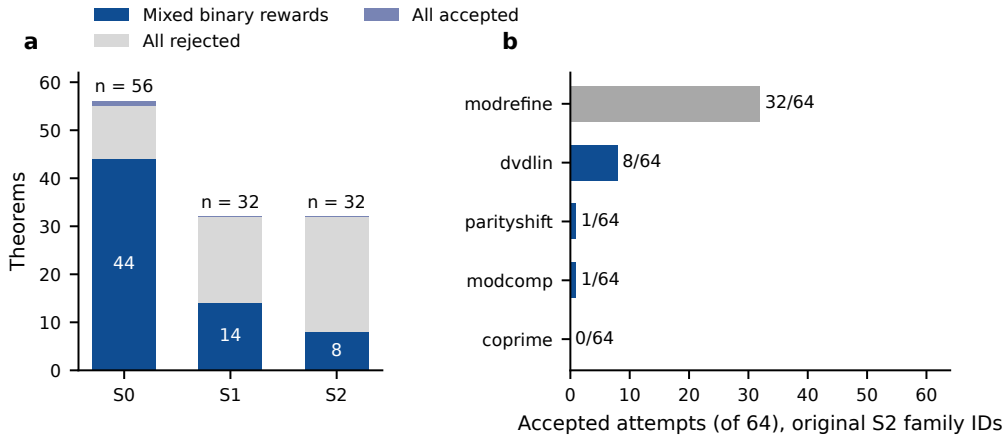



Figure 41: Readiness test of the SFT parent on the first template set, version 1 prompt, 8 attempts per theorem, one seed. (a) Theorems per relabelled stage by outcome. (b) Accepted attempts per family of the original S2 stage, $n = 64$; grey: the family that `omega` proves alone, later moved to S0.

**Registered final-run protocol.** We recorded this protocol before the launch, when no result was available. The subsequent outcome belongs to the EXP-023 completion handoff in `docs/EXPERIMENT_HISTORY.md`; the protocol below remains historical. The run tests two claims: RL improves the model over its SFT model W on held-out theorems, and a curriculum makes RL more efficient (Figure 42).

- **Settings.** Environment version 2.2, $\gamma = 0.95$, $G = 8$, at most 6 actions, REINFORCE with the leave-one-out baseline on discounted returns, learning rate $10^{-6}$, seed 20261013 for every arm, at most 150 groups per arm.
- **Admission.** W makes 8 attempts on each training theorem. A theorem enters training only if Lean accepts 1 to 6 of the 8 attempts, so every admitted theorem can give mixed groups. A stage with too few admitted theorems is dropped from both arms.
- **Primary measure.** The mean fraction of the 8 attempts that Lean accepts, per held-out theorem. The primary test permutes whole families, because theorems in one family are correlated; Holm

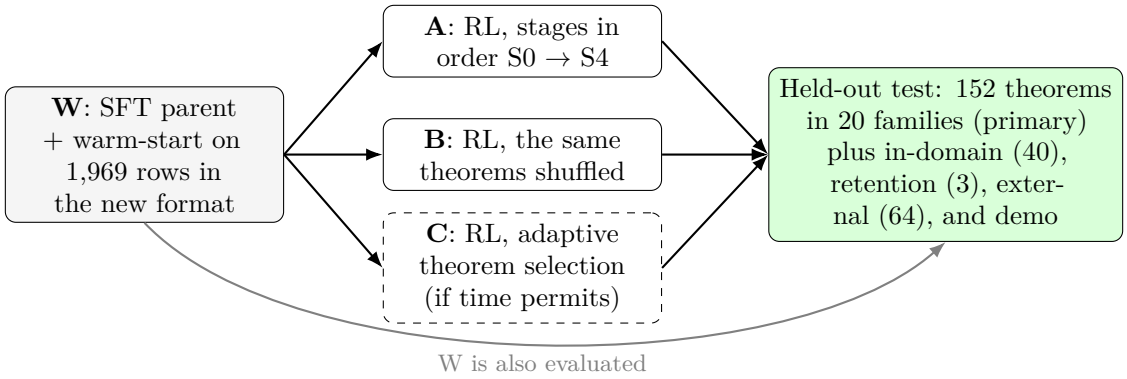



Figure 42: The frozen comparison. A and B against W test whether RL helps. A against B tests whether order helps. C tests whether theorems near a 50% acceptance rate give more mixed groups for the same compute.

correction controls the two comparisons of each claim.

- **Efficiency.** For each arm, the gain in proved theorems over W per million generated tokens, together with Lean calls, updates, and the share of mixed groups.
- **Validity gates.** A against B counts as confirmatory only if both arms complete the same number of groups, with updates within 10% and censored groups within 2. The efficiency claim also needs generated tokens within 10%. If training acceptance does not increase during RL, a null result is uninformative. No checkpoint, setting, or theorem is chosen from evaluation results.
- **Sample size.** In simulation, the McNemar test gave a false-positive rate of up to 0.14 against the intended 0.05; the family test stays at 0.05 or less. With the final 152 theorems, the family test detects 8 to 19 percentage points in the fraction of theorems proved, and 11 to 15 points at a baseline like the parent's. These values come from an assumed model, not from data.
- **Compute.** One L4 GPU for 12 hours, with a reservation of USD 37.01 and one attempt. A CPU fallback runs as a labelled deviation if the attempt fails.
- **Demo.** `demo_coprime` and one held-out theorem per level (a common divisor divides the gcd; odd $m$ and $n$ give an even $mn + 1$; $ab$ or $a + b$ is even), run on A if A proves at least as many held-out theorems as W, otherwise on W. Six earlier demo theorems matched statements in the training or SFT data and were withdrawn.

### E.1 Familiar-family reliability improves without an established transfer gain

**Results recounted on 18 October 2026.** The completed EXP-023 final run gives more accepted attempts on familiar number-theory constructions after RL (Table 15). These evaluations start from whole goals without supplied proof lines. They compare fixed final checkpoints with the SFT model W, using the same attempts per statement within each panel. Later data amendments leave a smaller held-out panel than the historical plan above.

More accepted attempts need not mean broader proof coverage. Every RL final proves fewer distinct held-out theorems than W in these attempts. The family-level tests find no overall held-out benefit after Holm correction; each adjusted $p$-value equals 1. The familiar-family tests support higher sampled reliability, but this is a secondary panel from a single training seed. It does not establish broad acquisition of number-theory proofs.

| Model | Familiar attempts | Familiar theorems | Held-out attempts | Held-out theorems |
|---|---|---|---|---|
| W: SFT model | 148/320 | 31/40 | 258/1,136 | 59/142 |
| A: ordered RL | 208/320 | 32/40 | 283/1,136 | 53/142 |
| B: shuffled RL | 213/320 | 34/40 | 302/1,136 | 54/142 |
| C: adaptive RL | 242/320 | 35/40 | 249/1,136 | 46/142 |

Table 15: Recorded Lean-accepted complete proofs in the final curriculum run. Each statement has eight attempts. A theorem counts as proved when Lean accepts at least one attempt. Familiar constructions and held-out constructions are separate populations; their denominators must not be pooled.

**Limits of the comparison.** The training record contains no group from the hardest stage S4. On held-out S4 statements, W accepts 14/240 attempts; A accepts 1/240, B accepts 7/240, and C accepts 0/240. Missing practice is a candidate explanation, but no causal ablation tests it. Actual generated training tokens differ across arms, so the registered token-budget criterion rejects the efficiency claim. Earlier supervised exposure required benchmark removals; remaining semantic and pretraining familiarity are not certified absent. These limits prevent an explanation that attributes the transfer result to a specific missing skill or optimizer property.

**Evidence boundary.** The recount reads saved strict-replay verdicts; it does not run fresh Lean replay or model inference. The raw inputs and counted denominators are recorded in `nt_wednesday_evidence_20261018_v1/recount.json`, with their source paths and planted counter controls. The original final-run results and amendments remain in `docs/EXPERIMENT_HISTORY.md`. This REINFORCE comparison does not measure PPO or the later candidate-selection policy.

# F Historical protocols: generalization and curriculum

**Revised framing, 16 October 2026.** The protocol separates generalization from curriculum effects. The generalization experiment compares the unchanged Qwen model, project SFT, and project SFT followed by RLVR. The curriculum experiment asks whether ordering RL practice improves familiar benchmark performance, generalization, or both. They share a benchmark and model lineage, but answer different questions. This revision supersedes the earlier curriculum-only framing. Both experiments were prospective at this reporting cutoff. Section 4 reports their subsequent execution and results.

Table 16: The two major experiments. Benchmark preparation and calibration support these comparisons.

| Experiment | Comparison | Question |
|---|---|---|
| Generalization, EXP-026 | Unchanged Qwen; SFT; SFT + RLVR | What does each training stage change on familiar and shifted theorems? |
| Curriculum, EXP-027 | Shuffled RL; prerequisite-ordered RL, from the same SFT checkpoint | Does curriculum improve familiar performance, generalization, or both? |

### F.1 Training-stage generalization (EXP-026)

The first experiment asks what SFT and RLVR add to the model's proof capability. Its conditions are B, the unchanged model; S, B after project SFT; and R, the exact S checkpoint after RLVR. Use one frozen, non-curricular RL recipe for R. S against B measures the effect of project SFT. R against S measures the effect of adding this RLVR stage. Evaluate every condition on the same frozen benchmark panels.

Our selected model is `Qwen2.5-0.5B-Instruct`. *Without SFT* means without additional project SFT. The official model card identifies it as instruction-tuned with upstream post-training. We therefore cannot describe B as a model that never received supervised instruction tuning. A raw Qwen Base checkpoint would define a different starting model. Do not substitute unrelated historical checkpoints for the B-to-S-to-R lineage.

DeepSeek-Prover-V1.5 evaluates capabilities across training stages [11]. That supports the comparison structure, not a prediction of our outcome. The stages receive different amounts of training. R against S therefore does not establish superiority over equal-cost continued SFT. Record training compute separately from the matched inference budget.

Generalization needs a specified change from training to evaluation. Lake and Baroni distinguish nearby generalization from systematic composition in SCAN [16]. This motivates separate panels, without making composition the only research question. Table 17 specifies the intended shifts. Familiar-family performance uses held-out theorems, not training accuracy.

Table 17: Shared evaluation panels. Claims depend on the relation between each panel and known training exposure.

| Panel | Change from training | Permitted conclusion |
|---|---|---|
| Familiar-family instances | New statements from represented constructions | Reliability within familiar constructions |
| Held-out families or topics | A reviewed construction or method absent from known training | Transfer under this specified shift |
| Unseen skill combinations | Components occur separately in training; their combination does not | Transfer to this combination, after exposure review |
| Longer reference dependencies | Longer chains of represented intermediate claims | Reference-structure extrapolation, not minimum proof depth |

Freeze primary panels and comparisons before sampling. Report B, S and R on each panel; do not hide different shifts inside one aggregate. A compatible established benchmark can provide an additional fixed panel after explicit adoption and exposure review. Its name alone does not establish generalization. Its relation to training determines what the result means.

EXP-025 supplies candidate inputs, not a certified structural benchmark. Manas reviews intended meaning, semantic families, component skills, reference strategies and known SFT/RL exposure. Reference dependency depth and tactic length remain separate from minimum mathematical difficulty. Keep current TRAIN, DEV and TEST membership unchanged. Any required expansion needs a separately versioned release. If a panel cannot be supported, report that limitation and retain an explicitly labelled pilot.

### F.2 Curriculum effects on proof completion (EXP-027)

The second experiment holds the SFT checkpoint S and RLVR implementation fixed. Its conditions are R, shuffled RL practice, and C, prerequisite-ordered RL practice. Both start from the exact S

checkpoint. The first intervention changes RL order while keeping SFT unchanged. This measures the total effect of the schedule, without assuming a transfer benefit.

R and C receive permutations of the same TRAIN task-ID multiset, including identical repeat counts. Reviewed skill prerequisites determine ordered stages; freeze within-stage tie-breaking and permutation seeds. Keep frozen-S acceptance as a separate empirical difficulty measure. If calibration determines TRAIN admission, apply one common rule to both arms. Retain exclusions and never filter TEST by its acceptance rate. Use whole theorems without supplied proof prefixes.

R can serve both experiments when every resolved setting and seed matches. Reusing the run avoids duplicate training, but does not add an independent replication. Adaptive sampling, changed theorem allocation and supplied-prefix assistance remain distinct later curriculum interventions. DeepSeek-Prover-V2 incorporates subgoal-derived problems into a curriculum [15]. Our first comparison fixes the theorem multiset and tests order.

Measure C minus R on familiar benchmark theorems and separately on each generalization panel. Table 18 gives the possible interpretations. A familiar-panel gain does not establish transfer. No detected gain does not establish absence of an effect. Report uncertainty and assessment completion alongside every contrast.

Table 18: Interpretation of curriculum results, conditional on a valid comparison and the stated uncertainty analysis.

| Familiar-panel result | Generalization-panel result | Interpretation |
|---|---|---|
| Gain detected | Gain detected | Evidence for benefits on both named panels |
| Gain detected | No gain detected | Familiar performance improves; transfer benefit remains unestablished |
| No gain detected | Gain detected | Evidence for transfer without a detected familiar-panel benefit |
| No gain detected | No gain detected | No measured curriculum benefit under this protocol |

Efficiency is a secondary outcome of the curriculum experiment. Plot prescheduled DEV measurements against actual generated tokens, alongside physical Lean calls and GPU time. Equal group counts do not imply equal cost. Freeze any target score before training and keep an unreached target as unreached. The earlier EXP-023 null order result and failed efficiency gate remain unchanged.

### F.3 Common controls and interpretation limits

At evaluation, all conditions receive the same interface, statements, library access, feedback and per-attempt limits. An interface warm-up that updates B's weights is additional training and must not be hidden. Keep evaluation proofs outside SFT, RL, demonstrations and retrieval. Choose checkpoints and settings on DEV; evaluate TEST after those choices freeze. SFT validation remains a TRAIN-only monitor.

Report complete-proof accepted-attempt fraction and proved theorems within the same fixed attempt budget. Acceptance requires an unchanged statement and complete replay under the pinned verifier. Every planned theorem–attempt pair stays in the operational denominator. Retain rejection, timeout, infrastructure failure, truncation and missing output as separate categories. Material infrastructure failures make the scientific comparison inconclusive. Report completion coverage and resources for failed attempts as well as accepted attempts.

Use paired RL seeds for R against C and retain semantic-family dependence in uncertainty calculations. Freeze primary contrasts and any multiplicity correction before evaluation. Repeated generations of one model are not independent training replications. Few seed pairs or families support descriptive conclusions.

For the curriculum contrast, freeze the optimizer, reward, discount, interaction protocol and hard resource caps. Use no extra shaping or exploration bonus. Compare planned, consumed and uncensored theorem multisets; a cap stop can prevent equal realized exposure. Preserve incomplete runs without raising caps or adding selected retries. Topic clustering and recency remain alternatives to a prerequisite-learning mechanism. Mixed-reward groups do not by themselves explain generalization.

The RLVR condition must name its optimizer. REINFORCE and GRPO can be method arms within generalization, rather than additional major experiments. Their current interactive and whole-proof paths change feedback as well as optimization. Begin with one fixed RLVR recipe; an optimizer comparison requires matched interaction and budget. This avoids a full optimizer-by-curriculum grid in the first study.

### F.4 Ownership and shared handoff

Manas owns the curriculum experiment: its design, prerequisite schedules, C training and R/C analysis. Abhinav owns the generalization experiment: its design, B/S/R training and evaluation. Anwesha owns Lean interaction and replay. Abhinav binds B, derives S with the frozen SFT recipe, and derives R from that exact S with the frozen RLVR recipe. Historical checkpoints need verified lineage before reuse. He delivers a B/S/R table for generalization, with separate benchmark-panel columns.

For curriculum, Manas derives C from the same S and compares it with R. Abhinav supplies the immutable S checkpoint, R schedule, resolved settings and R outcomes. The existing `TaskSchedule.frozen` and trainer `--task-order` accept explicit theorem repetitions. The study manifest must check whole-theorem membership and equal theorem multiplicities across schedules. The scheduler does not enforce those cross-arm conditions. Manas delivers the R/C table with familiar performance, generalization and measured cost.

Use optimizer-disabled TRAIN/DEV calibration before fixing training settings. The existing `--assessment-only` interface uses `--assessment-split validation` for DEV and `test` for TEST. Preserve configurations, source identities, raw attempts, consumed schedules, failures and checkpoint provenance. Resolve panel counts, model revisions, learning settings, seeds, resource caps and launch costs before execution. The guides are in EXP-026 and EXP-027 of `docs/EXPERIMENT_HISTORY.md`. These arm definitions do not themselves authorize training or new source admission.

### F.5 Execution plan for the number theory study

The immediate study asks whether RL improves independent number theory proofs and whether prerequisite order changes that improvement. The primary sequence is B, one fixed SFT checkpoint S, then matched shuffled R and curriculum C branches. Use the interactive REINFORCE implementation for both branches, with whole-theorem starts and the same theorem repetitions. The separate small PPO comparison tests optimizer feasibility under its own interface. It cannot supply R for this curriculum study.

**Dataset validity before difficulty assessment.** Broad prerequisite coverage takes priority over selecting theorems the model already proves. Keep the checked interactive action corpus separate from the checked complete-proof corpus. Action SFT targets responses to live states, including format and Lean-error recovery. Complete-proof SFT targets complete reference constructions under a different prompt contract. Do not concatenate these formats or change their mixture without registering the training recipe. Repeated SFT examples do not count as additional mathematical statements. Keep one RL theorem per admitted statement, started from the whole theorem, with an alias ledger that preserves its source proofs.

The current consumer release is `nt_curriculum_data_20261018_v1/validated_v2/`. It supersedes the initial common release for new training. Fresh strict replay accepts all 580 source proofs under the pinned Lean and mathlib pair. All 536 source theorems reconstruct through the selected interaction environment. Full-proof acceptance does not certify the extracted SFT examples. Their separate live checks reject examples that start after closure, continue after closure, or fail the required transition checks.

This release retains 1,880 TRAIN SFT examples. The separate SFT validation retains 125 SFT examples from 32 TRAIN-only theorem names. The primary RL bank contains 506 theorems with whole-theorem starts. Its alias ledger preserves 516 source proof variants. Every source record has a retained or quarantine disposition; no record remains pending. All retained action messages pass exact v2 rendering, parsing and tokenizer checks without truncation. The limits remain 2,048 tokens for context and 192 tokens for an action completion, including its chat terminator. The evidence is in `validated_v2/data_consumer_receipt.json` and `model_contract_v2/validated_v2_fit.json` on the remote artifact store.

Table 19: Source topic labels in the filtered TRAIN bank. The denominator is 506 theorems. These lexical labels do not certify mathematical skill coverage.

| Source label | Theorems |
|---|---|
| Divisibility | 217 |
| Primes and factorization | 180 |
| Congruences | 77 |
| Bézout and coprimality | 32 |

The semantic review reads all 536 canonical TRAIN statements and all 548 TRAIN proof variants. The primary policy excludes 19 valid foundational controls into a separate source pack. It excludes ten AI-reviewed redundancy cases without claiming certified equivalence. It excludes one theorem whose reference invokes a reserved DEV target lemma. These policy exclusions are distinct from rejected action transitions and contradictory assumptions. The source admission certificates remain bound to the actual proof identities and statements. Their scope is formal premise satisfiability, not informal source alignment or unknown pretraining exposure.

Both runtime schedules contain 1,012 groups with identical theorem multiplicities. The reviewed candidate order retains all 54 proposed pedagogical edges and their providers. No annotation has human confirmation. These assignments describe source operations and a proposed teaching order; they do not measure model mastery or establish prerequisite necessity. The current comparison can measure an order effect under these fixed assignments. It cannot isolate prerequisite learning from recency or topic clustering.

Training gaps and reserved transfer concepts require different actions. For a missing TRAIN-eligible prerequisite, add source-bound examples through the existing admission and replay path in a new data revision. Do not relax admission, copy evaluation solutions or move reserved concepts into TRAIN to fill the coverage table. The existing undergraduate syllabus reserves quadratic residues and arithmetic functions for topic transfer. Its capability labels and the informal school-level corpus do not certify formal skill coverage. The review therefore distinguishes observed source coverage, missing annotations, missing examples and deliberately held-out material. The fixed order comparison does not consume a curriculum-validation set or use adaptive stage promotion. A distinct curriculum-validation reservation remains required before an adaptive extension.

**SFT checkpoint identity and comparison controls.** The historical SFT checkpoint is ineligible for the clean primary comparison. Its recorded training examples explicitly invoke `Nat.factor`

ial_mul_factorial_dvd_factorial, which proves a reserved DEV target up to product order. Removing the source examples cannot remove their prior effect on those weights. Train a fresh S from the pinned upstream B using the SFT examples of the validated_v2 release, then derive both RL branches from that exact final checkpoint. The historical checkpoint remains available for a labelled familiar diagnostic. Direct-name and exact-statement scans do not establish complete semantic alias independence. Upstream pretraining exposure remains unknown.

The base and historical tokenizers produce identical token IDs for every retained message and theorem prompt. Both model files pass a bounded finite-forward check without generation or updates. These checks establish file and input compatibility; they do not establish proof capability or validate a future GPU training run. Future SFT workers record exact consumed source bytes, selected rows, configuration and tokenizer metadata before loading weights. At this planning cutoff, the fresh S checkpoint does not yet exist. Section 4 reports the completed main run.

Before training, use a bounded optimizer-disabled TRAIN/DEV calibration to inspect reward variation and failures. Do not select theorems from TEST or treat loss, valid local tactics or supplied-prefix acceptance as complete-proof capability. In the selected environment version, intermediate goal changes and Lean diagnostics receive zero reward. Terminal reward requires accepted complete replay; the legacy goal_change_v1 setting does not add shaping in this version. Freeze prompts, action and context limits, discount, optimizer, schedule repetitions, evaluation attempts and hard resource caps together. Bind the completed TRAIN theorem replay and reconstruct the fixed DEV theorems in the actual consumer environment before loading the training model.

**Delivery and failure interpretation.** Complete semantic annotation, input binding and the costed execution package first. After exact-cost approval, run the S checks and matched RL branches early enough to finish evaluation before Tuesday close. Reserve Wednesday, 21 October 2026, for the counted comparison and its limitations. Use the fixed final checkpoints and preserve every planned theorem–attempt pair, failure and cap stop. Report familiar performance, reviewed transfer panels, completion coverage and physical resource use separately. If semantic review remains incomplete, report exploratory order results without a prerequisite mechanism claim. If rewards remain constant or an arm stops before matched exposure, report that limit without extending caps or selecting replacement theorems. The historical help, null and retention results remain available even if the new comparison is incomplete. The prepared optimizer pilot and this broader program each require their own concrete launch approval. The staged broad-bank package has a quoted reservation of USD 102.31. Its proposed ceiling is USD 110. Its maximum duration is 36 hours. The earlier fallback requires an admitted historical S checkpoint. The confirmed target exposure makes that fallback ineligible for the clean study. These planning envelopes do not authorize spend or guarantee completion.

# G Follow-up designs that did not run

Two follow-up designs were registered on 19 and 20 October and prepared up to a dry run. On 20 October, Manas stopped all experiments before any launch. Neither design ran, so neither has a result or a GPU cost. This section records their motivation and setup, because they state the open questions. Their programs, configurations and dry-run quotes are in the EXP-029 and EXP-030 records of `docs/EXPERIMENT_HISTORY.md`.

**EXP-029: more attempts, screened strata and the action limits.** Three observations motivate EXP-029:

- Two attempts per theorem put every arm at the measurement floor (Section 4.7).
- The data audit of 19 October found three unchecked risks in the selection set. At least 5 of the 22 statements with quantified hypotheses that were read by hand have contradictory premises; Lean has not checked this. No screen tried short automation chains. The near-duplicate screen compares text only, so it does not flag theorems close to TRAIN proofs.
- The selection set has already been used to choose arms, so any further result on it is exploratory.

EXP-029 evaluates S, C, R and `RL_filtered` with 8 attempts per theorem on the 164 selection theorems. Three screens split the theorems into four strata: vacuous hypotheses, automation chains, near TRAIN, and "other". The "other" stratum is the headline. EXP-029 also evaluates S and `RL_filtered` with 8 attempts on the 62 `RL_filtered` theorems. The paired comparisons, the bootstrap and the 5-point promotion rule are those of EXP-028.

The amendment of 20 October fits the program on one L4 GPU within about USD 50. Its dry-run quote is USD 49.51 for 28 hours, against USD 118.75 for the first program on one L40S. It adds an exploratory comparison on the 62 theorems: S with 12 actions, 384 action tokens and 8 tactics per action, against S at the registered limits. The limits recount of 20 October on TRAIN (`pass_rates`) motivates it:

- 1,935 of S's 2,024 attempts use all 6 actions without a proof.
- S's 84 proofs close at every action from 1 to 6 (Figure 16c).
- The token and tactic limits cause 1,358 of S's 1,416 format errors.
- At $\gamma = 0.95$, a proof closed by the last action earns 0.774 with 6 actions and 0.569 with 12.

The relaxed configuration also changes the instruction from "up to 4 tactics" to "up to 8 tactics". The comparison therefore measures the new limits and the new wording together.

**EXP-030: a dense per-action reward.** Most first visits in R have no proof, and a group with no proof gives no update (Section 4.6). Manas proposed a dense reward. EXP-030 asks whether per-action penalties for Lean errors make REINFORCE close more TRAIN proofs than the terminal proof reward alone. R_dense starts from the same S and uses R's seed and the first 506 groups of R's frozen task order, which cover 377 distinct TRAIN theorems. Its configuration differs from R's only in the reward mode, the run budgets and the run identifiers. Table 20 gives the reward. The control is R's existing record of the same 506 groups, in which R closes 87 of 2,024 attempts. The test is an exact two-sided sign-flip test on the per-group difference in accepted attempts, with groups paired by position. R_dense passes only if it closes more proofs than R with $p < 0.05$. A higher return or fewer errors without more proofs does not pass. A gaming audit compares the shares of valid actions and of no-progress actions between the arms. R trained on an L40S, and R_dense would train on an L4, so their sampled attempts would differ even with the same seed.

Table 20: The per-action reward of EXP-030 (`error_feedback_v1` in environment version 2). A partial action is scored by the failure that stopped it. An action whose next prompt exceeds the limit ends the attempt and keeps the reward of its outcome. The discount stays $\gamma = 0.95$.

| Event after an action | Reward | Label |
|---|---|---|
| Lean accepts the complete proof | $+1$ | Lean-verified |
| Format error, syntax error, forbidden code, unresolved name, or invalid backtrack | $-0.2$ | Heuristic |
| Type or tactic error, unchanged goals, repeated state, unclassified Lean rejection, or rejected replay | $-0.1$ | Heuristic |
| Valid action, timeout, or crash | 0 | None |

The registration of `error_feedback_v1` on 11 October gave 0 to unchanged goals, repeated states and unclassified rejections. EXP-030 gives them $-0.1$. Errors do not end a version 2 attempt, so a reward of 0 would let a policy avoid every penalty by repeating an action that changes nothing. The penalties are heuristic: they score Lean errors, not proofs, so a lower error rate does not show more proof skill. `RL_filtered` makes 18 format errors in 1,948 held-out actions and proves 2 of 164 theorems (Section 4.6), so fewer format errors alone did not give held-out proofs. The dense reward also penalizes Lean rejections, which are 1,636 of `RL_filtered`'s 1,948 held-out actions. No record shows what that penalty would change; this is an inference, not a result of EXP-030.

# H Evaluation review pack: statements and expected answers

This catalogue describes the encoded theorems from the evaluation review pack prepared on 17 October 2026. It preserves their source order, mathematical domains and task IDs. The source contains 34 DEV theorems and 82 TEST theorems. A description explains the statement; it does not repair or certify its intended mathematical meaning. Existing semantic concerns remain next to the affected theorem. Theorems without a warning still need the source-intent and exposure reviews recorded in `docs/EXPERIMENT_HISTORY.md`.

Every expected answer is a complete Lean proof of the unchanged statement under the pinned verifier. Existence theorems require witnesses with proofs of their conditions. Equivalences require both directions; conjunctions require every conjunct. A negated universal claim requires its negation, rather than a proof of the positive claim. The proof need not match the private reference. No proof solutions, curriculum order, difficulty ranking or model outcomes are supplied here.

**Notation.** $\mathbb{N}$ includes zero, $\mathbb{Z}$ denotes integers, and $\mathbb{Q}$ denotes rationals. $\varphi$ denotes Euler's totient function; $J$ and the parenthesized symbol $\left(\frac{a}{n}\right)$ denote the Jacobi symbol. ord denotes Lean's multiplicative order, and $\mathbb{Z}/m\mathbb{Z}$ denotes the residue ring used by `ZMod m`. Divisor sums range over the finite natural divisor sets in the source. Natural subtraction truncates at zero; division domains are specified where they affect interpretation. The original statement and initial state remain in the remote review pack.

## H.1 DEV theorems

DEV 1. **Odd powers and factorials modulo 16**

Prove all three claims: every odd integer $y$ satisfies $y^8 \equiv 1 \pmod{16}$; every natural $x \geq 6$ satisfies $x! + 1 \equiv 1 \pmod{16}$ after casting $x!$ to the integers; and for natural $x \geq 6$ and integer $y$, $x! + 1 = y^8$ implies that $y$ is odd.

*Expected answer.* A complete proof of all three universally quantified claims, including the parity conclusion from $x! + 1 = y^8$.

*Task ID.* `ntb_0849ff82c82325aedf67ae3d:whole:0`

DEV 2. **Decimal length after appending zeros**

For natural numbers $a, n$ with $1 \leq a < 10$ and $0 < n$, prove that the decimal digit list of $a10^n$ has length $n + 1$.

*Expected answer.* A proof that $\text{length}(\text{digits}_{10}(a10^n)) = n + 1$ under both bounds on $a$ and positivity of $n$.

*Task ID.* `ntb_0b44211e960985affabb0e96:whole:0`

DEV 3. **Prime divisor and final nine digits**

Let $p$ be a natural prime with $10^{50} < p$. Assume $(10^{294} - 1) \bmod p = 0$, and assume $(10^n - 1) \bmod p \neq 0$ for every natural $n < 294$, including zero. Prove $p \bmod 10^9 = 572857143$.

*Expected answer.* A proof of the stated remainder 572857143 from all the encoded assumptions.

*Review note.* Existing review flags contradictory assumptions: the universal nonzero-remainder hypothesis includes exponent zero. Semantic clearance remains withheld; this description does not certify the other assumptions.

*Task ID.* `ntb_0c317e1120b03cf847d9597c:whole:0`

DEV 4. **Negation of a multiplicative-order conjunction**

Prove that the following conjunction is false. Its first part says that every natural prime $p$ with equal multiplicative orders of 2 and 3 in $\mathbb{Z}/p\mathbb{Z}$, and with some natural $k > 1$ satisfying $p \mid 2^k - 3^k$ in the integers, satisfies $7 \leq p$. Its second part says that these same order-equality and existence conditions hold at $p = 7$.

*Expected answer.* A proof of the negation of the entire conjunction, not a proof that 7 is a least qualifying prime.

*Task ID.* `ntb_12c42551bcbe67163eb49a45:whole:0`

DEV 5. **Divisibility equivalence under square-sum premises**

Let $n \in \mathbb{N}$ satisfy $n \geq 4$. Assume $i^2 > 0$ for every natural $i \leq n$, and $\sum_{i=0}^{n} i^2 = 2n(n^2 - n + 2)$. Prove $n \mid n(5n - 3)$ if and only if $n = 4$, with all subtraction in $\mathbb{N}$ truncated at zero.

*Expected answer.* A proof of both directions of $n \mid n(5n - 3) \iff n = 4$ under every stated premise.

*Review note.* Existing review flags a contradictory premise: positive squares are required even at zero. A saved Lean check confirms that premise contradiction; this description does not provide semantic clearance.

*Task ID.* `ntb_24495954b801bd41cc09d7a9:whole:0`

DEV 6. **Factorial product divisibility**

For all natural numbers $n, k$ with $k \leq n$, prove that $(n - k)!\,k!$ divides $n!$. Subtraction takes place in $\mathbb{N}$.

*Expected answer.* A universally quantified proof of $(n - k)!\,k! \mid n!$ whenever $k \leq n$.

*Task ID.* `ntb_26c13028f7860afcbd1b1330:whole:0`

DEV 7. **A rational sum is not one**

For a natural prime $p$, prove that $\sum_{k=0}^{p-1} pk \neq 1$, where each summand and the sum are rational numbers.

*Expected answer.* A proof that the indicated rational sum differs from 1 for every natural prime $p$.

*Task ID.* `ntb_28ce24a682cba3a80079ac05:whole:0`

DEV 8. **No square completion for odd inputs**

For every pair of odd integers $x, y$, prove that no integer $z$ satisfies $x^2 + y^2 + z^2 = x^2y^2$.

*Expected answer.* A universally quantified nonexistence proof for integer $z$ under both oddness assumptions.

*Task ID.* `ntb_2ac8f84bb1611299a7fdbc33:whole:0`

DEV 9. **Order characterizes powers equal to one**

Let $a \in \mathbb{Z}$ and $n \in \mathbb{N}$ satisfy $\gcd(a, n) = 1$ and $n > 0$. Let $d \in \mathbb{N}$ equal the multiplicative order of the residue of $a$ in $\mathbb{Z}/n\mathbb{Z}$. For every natural $k$, prove $a^k = 1$ in that ring if and only if $d \mid k$.

*Expected answer.* A proof of both implications between the residue equation $a^k = 1$ and natural divisibility $d \mid k$.

*Task ID.* `ntb_2f522c6a2246200d9d42f7d4:whole:0`

DEV 10. **Integer divisibility implication**

For an integer $x$, assume $x \mid 39x - 24$ and $328x - 4 \mid x^3 - 4x^2 + 4x$. Prove $x \mid 4x^2 - 16x(x - 1)$, using integer arithmetic throughout.

*Expected answer.* A proof of the final divisibility statement under the two encoded divisibility premises.

*Task ID.* `ntb_34759c85c220863a9b934698:whole:0`

DEV 11. **No mutually adjacent prime divisors**

Prove that there are no natural primes $p, q$ with $p > 2$, $q > 2$, $p \mid q + 1$, and $q \mid p + 1$.

*Expected answer.* A proof excluding every natural pair satisfying all six conditions.

*Task ID.* `ntb_42221e0e76e026e91ccae67b:whole:0`

DEV 12. **Prime powers modulo the predecessor**

Let $p, n \in \mathbb{N}$, with $p$ prime and $n > 1$. Assume there exist natural $a, b$ with $ab = p$ and $b \equiv 1 \pmod{p}$ in the integers. Prove $p^n \equiv 1 \pmod{p - 1}$, where the modulus is the integer cast of $p$ minus 1.

*Expected answer.* A proof of the integer congruence $p^n \equiv 1 \pmod{p - 1}$ under the prime, exponent, and existential assumptions.

*Task ID.* `ntb_4678ae628e2cd00da60bb8e9:whole:0`

DEV 13. **Digit-sum congruence and a missing digit**

Prove both claims. For every positive natural $N$, its decimal digit sum is congruent to $N$ modulo 9 in $\mathbb{N}$; and there exists a natural $x \leq 9$ such that $9 \mid 123456789 \cdot 10^{10} + x10^9 + 123456789$.

*Expected answer.* A proof of the universal digit-sum congruence together with a digit witness $x \leq 9$ and a proof of the specified divisibility.

*Review note.* Existing review records cross-split semantic overlap: the first conjunct overlaps the TEST decimal digit-sum congruence. This is not evidence of independent generalization.

*Task ID.* `ntb_540f08afe6aea5bca8e7f52f:whole:0`

DEV 14. **Integer-division sum bound**

Let $p \in \mathbb{N}$ be prime, and assume $p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{4}$ in the integers. Prove $\sum_{i=0}^{p-1} (1 \,/_{\mathbb{Z}}\, i) \leq 2p^p - 1$, with $p$ and each $i$ cast to integers and each quotient evaluated using Lean integer division, including the term at $i = 0$.

*Expected answer.* A proof of the encoded integer sum inequality, not a rational harmonic-sum bound.

*Review note.* Existing review flags an intent mismatch: the summand uses integer division, not rational division. Semantic clearance remains withheld.

*Task ID.* `ntb_5623ea7db1926fecb82805a4:whole:0`

DEV 15. **An implicitly quantified exponent**

Let $m, p \in \mathbb{N}$, with $p$ prime. Assume there exists a nonzero natural $a$ such that $2a = p - 1$, where that subtraction is natural subtraction. Prove $p^m \equiv 1 \pmod{p-1}$ after casting $p$ to the integers in the congruence and its modulus.

*Expected answer.* A proof of the congruence for arbitrary natural exponent $m$, which is an implicit parameter in the statement.

*Task ID.* `ntb_57b89c0669597337e6ce5691:whole:0`

DEV 16. **Negation of a factorial-divisibility count**

Prove that the following universal counting assertion is false: for every $n, m \in \mathbb{N}$ with $1 \le m \le n - 1$, the set $\{k \in \mathbb{N} : m \mid k!,\ 1 \le k \le n - 1\}$ has cardinality $n - m$. Both subtractions are natural subtraction, truncated at zero.

*Expected answer.* A proof of the negation of the universal cardinality formula; a counterexample with a checked failure of the formula is acceptable.

*Task ID.* `ntb_5ce565368939b4275182fc8c:whole:0`

DEV 17. **Prime-exponent premise and a nested power**

Let $p \in \mathbb{N}$ be prime with $p \neq 2$, and let $a \in \mathbb{N}$ satisfy $a^p = 2$. Prove $a^{(2^p)} \equiv 2 \pmod{p}$ after casting $a, p$ to integers in the congruence; exponentiation is right-associative.

*Expected answer.* A proof of the stated nested-power congruence from all encoded assumptions, including $a^p = 2$.

*Review note.* Existing review flags contradictory assumptions: no natural base has a natural prime exponent different from two and power equal to two. This is a review finding, not fresh Lean verification.

*Task ID.* `ntb_60328a061f4b90dcae8c1b6b:whole:0`

DEV 18. **Absolute value of a square minus one**

Let $n \in \mathbb{N}$ satisfy $2 \le n$ and $3 < n$. Prove that if the integer cast of $n$ is at least 2 and $n$ is not prime, then the natural absolute value of the integer $n^2 - 1$ is not 1.

*Expected answer.* A proof of the conditional inequality $\mathrm{natAbs}((n : \mathbb{Z})^2 - 1) \neq 1$ under all the encoded bounds and the nonprimality premise.

*Task ID.* `ntb_66128a7673636dccc946e79c:whole:0`

DEV 19. **No odd multiple as a fourth-power difference**

For every odd integer $k$, prove that there are no integers $x, y$ with $x^4 - y^4 = 1223334444k$.

*Expected answer.* A universally quantified nonexistence proof for integer solutions $x, y$ whenever $k$ is odd.

*Task ID.* `ntb_6795aa26f8089aa7c1f49e08:whole:0`

DEV 20. **Power of an element with known order**

Let $p \in \mathbb{N}$ be prime, let $a, b \in \mathbb{Z}/p\mathbb{Z}$, and let $m, k \in \mathbb{N}$. Assume the multiplicative order of $a$ is $m$ and $a^k = b$. Prove $b^m = 1$ in $\mathbb{Z}/p\mathbb{Z}$.

*Expected answer.* A proof of $b^m = 1$ under the stated order and power relation, without adding a nonzero assumption on $a$.

*Task ID.* `ntb_6bf87afbdba942d3d665e9d7:whole:0`

DEV 21. **Digit-sum exclusion on an interval**

Prove that no natural number $x$ in the inclusive interval $200 \leq x \leq 288$ has decimal digit sum divisible by 19.

*Expected answer.* A proof of nonexistence of such an $x$, including both interval bounds and the digit-sum divisibility condition.

*Task ID.* `ntb_8cadc503b92a3294a3770016:whole:0`

DEV 22. **Negation of a decimal-remainder set equality**

Let $S$ be the set of natural remainders $(100000A + 10000B + 1000C + 100D + 48) \bmod 72$, where $A, B, C, D \in \mathbb{N}$ are each less than 10. Prove $S \neq \{12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68\}$. Leading digit $A = 0$ is permitted.

*Expected answer.* A proof that the two sets differ, not a proof that the displayed set lists all possible remainders.

*Task ID.* `ntb_8fb14729a7d6626bd3782104:whole:0`

DEV 23. **Remainder after division by multiplicative order**

Let $a, m, n, k, q, r \in \mathbb{N}$ satisfy $m > 1$, $\gcd(a, m) = 1$, and $k = \mathrm{ord}_m(a)$. Assume $a^n = 1$ in $\mathbb{Z}/m\mathbb{Z}$, $n = qk + r$, and $r < k$. Prove $r = 0$.

*Expected answer.* A proof that the specified remainder $r$ is zero under the order, residue-power, decomposition, and bound assumptions.

*Task ID.* `ntb_92a8e6497c8f664aa1696ccf:whole:0`

DEV 24. **Zero residue and zero remainder**

Let $n \in \mathbb{N}$ satisfy $6 < n$, $10 < n$, and $\gcd(n, 3) = 1$. Prove that its residue in $\mathbb{Z}/3\mathbb{Z}$ is zero if and only if its natural remainder $n \bmod 3$ is zero.

*Expected answer.* A proof of both directions of $(n : \mathbb{Z}/3\mathbb{Z}) = 0 \iff n \bmod 3 = 0$ under every given assumption.

*Task ID.* `ntb_9c107931350d9e67fa24c1f8:whole:0`

DEV 25. **Decimal digit sum modulo nine**

For every natural $n$, including zero, prove that $n$ is congruent modulo 9 to its decimal digit sum, with both values cast to integers.

*Expected answer.* A proof of the integer congruence $n \equiv \sum \mathrm{digits}_{10}(n) \pmod 9$ for all natural $n$.

*Review note.* Existing review records cross-split semantic overlap with the TEST decimal digit-sum congruence, despite different positivity assumptions and notation.

*Task ID.* `ntb_9f2790ee0b18fb114caa8b87:whole:0`

DEV 26. **Order divides a positive exponent**

Let $q \in \mathbb{N}$ be prime and $a \in \mathbb{Z}$ with $q \nmid a$ in the integers. Let $k \in \mathbb{N}$ be the multiplicative order of the residue of $a$ in $\mathbb{Z}/q\mathbb{Z}$. If natural $m > 0$ satisfies $a^m = 1$ in that ring, prove $k \mid m$.

*Expected answer.* A proof that the natural order $k$ divides the positive natural exponent $m$ under the nondivisibility and residue-power assumptions.

*Task ID.* `ntb_be9c16ca3754ba8a9ab01583:whole:0`

DEV 27. **A decimal quotient with an implicit parameter**

For an arbitrary fixed natural parameter $N$, prove that no natural $n$ satisfies $k \leq N \,/_{\mathbb{N}}\, 10^{\mathrm{length}(\mathrm{digits}_{10}(N))}$ for every natural $k$ with $1 \leq k \leq 9$. The bound variable $n$ does not occur in the quotient.

*Expected answer.* A proof of the stated nonexistence claim for each natural $N$, using natural division and the inclusive range from 1 through 9.

*Review note.* New possible intent issue: the initial state declares a natural parameter whose identifier is n!, so the repeated n! tokens denote that parameter rather than a factorial of the bound n. The description renames this parameter N and preserves the unused existential variable; no fresh Lean check or source repair is claimed.

*Task ID.* `ntb_bea1d7078c9528575e9bf318:whole:0`

DEV 28. **A larger prime does not divide a factorial**

Let $m, p \in \mathbb{N}$ satisfy $m > 0$, $p$ prime, and $p > m$. Prove $p \nmid m!$.

*Expected answer.* A proof that the prime $p$ does not divide the factorial of $m$ under both positivity and size assumptions.

*Task ID.* `ntb_cb242a8876b8745c69ac06ff:whole:0`

DEV 29. **A greatest-common-divisor bound attained**

Prove that every positive natural $n$ satisfies $\gcd(9n + 4, 5n + 2) \leq 2$. Also prove that some positive natural $n$ attains equality $\gcd(9n + 4, 5n + 2) = 2$.

*Expected answer.* A proof of the universal upper bound and a positive natural witness attaining the bound, with its equality proof.

*Task ID.* `ntb_d62af373edae6d51dcf617f2:whole:0`

DEV 30. **Nonempty decimal digits and reconstruction bound**

Let $b, n \in \mathbb{N}$ satisfy $1 < b$ and $0 < n$. Prove that the decimal digit list of $nb$ is nonempty and that interpreting that list in base 10 gives a natural number at most $2nb^2$.

*Expected answer.* A proof of both nonemptiness and $\mathrm{ofDigits}_{10}(\mathrm{digits}_{10}(nb)) \leq 2nb^2$.

*Task ID.* `ntb_d6d6dd394cac5461186aa8d4:whole:0`

DEV 31. **A factorial plus a residue is not a square**

Let $c \in \mathbb{Z}$ satisfy $c \equiv 3 \pmod 4$. Prove that for every natural $m \geq 4$, no integer $k$ satisfies $k^2 = m! + c$, with the factorial cast to $\mathbb{Z}$.

*Expected answer.* A proof excluding all integer square roots of $m! + c$ for every natural $m \geq 4$ under the congruence assumption on $c$.

*Task ID.* `ntb_e244bf908dd21056c803610d:whole:0`

DEV 32. **No solution with an even integer input**

For every even integer $x$, prove that there is no integer $y$ satisfying $14x^2 + y^2 = 727$.

*Expected answer.* A universally quantified nonexistence proof for integer $y$ whenever integer $x$ is even.

*Task ID.* `ntb_ed3f3886e6597e95184acb2d:whole:0`

DEV 33. **Contradiction below multiplicative order**

Let $p \in \mathbb{N}$ be prime, and let $b \neq 0$ in $\mathbb{Z}/p\mathbb{Z}$. Let natural $d$ be the multiplicative order of $b$. Assume natural $r$ satisfies $0 < r < d$ and $b^r = 1$. Derive a contradiction.

*Expected answer.* A proof of False from the encoded assumptions about $p, b, d, r$, rather than an independently chosen numerical answer.

*Task ID.* `ntb_ee0e013b74d5f43c8a8e241a:whole:0`

DEV 34. **Four cubes realize every residue modulo nine**

For every natural $r < 9$, prove that there exist natural $a, b, c, d$ such that $(a^3 + b^3 + c^3 + d^3) \bmod 9 = r$.

*Expected answer.* For each permitted residue $r$, provide four natural witnesses and prove that their cube sum has remainder $r$ modulo 9.

*Task ID.* `ntb_f995e1335705564e3d7db684:whole:0`

## H.2 TEST theorems

TEST 1. **Negated least-prime Jacobi claim**

Prove that the following combined claim is false: 43 is prime; the Jacobi symbols $J(2, 43)$, $J(3, 43)$ and $J(7, 43)$ all equal $-1$; and every natural prime $q < 43$ fails to have all three symbols equal to $-1$. Here $J(a, n)$ denotes the Jacobi symbol.

*Expected answer.* A complete Lean proof of the negation of the entire conjunction, including its least-prime condition.

*Review note.* The target negates the combined claim; it does not separately negate each conjunct.

*Task ID.* `ntb_05fad5360cfbe30eebe17b91:whole:0`

TEST 2. **Natural-power inequality**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and suppose $n$ is not prime. Prove $n^n \neq n^2 + (1 \operatorname{div} 2)$, where the division is natural-number division.

*Expected answer.* A complete Lean proof of the stated inequality under both assumptions.

*Review note.* Recorded intent warning: natural division makes $1/2$ zero. Preserve the formal target; do not replace it with a rational half.

*Task ID.* `ntb_07b7569a02e930619d9f4d6b:whole:0`

TEST 3. **Parity of two variables**

Let $x, y, n \in \mathbb{N}$ with $x > 0$, $y > 0$, $n$ odd and $n \geq 1$. If $3x^2 + 5y^2 = 2^n$, prove that $x, y$ are both even or both odd.

*Expected answer.* A complete Lean proof of $(\mathrm{Even}(x) \wedge \mathrm{Even}(y)) \vee (\mathrm{Odd}(x) \wedge \mathrm{Odd}(y))$.

*Task ID.* `ntb_08d43e44ee8db479c3f3713e:whole:0`

TEST 4. **Factors of a prime power**

Let $p, a, b, k \in \mathbb{N}$. Assume $p$ is prime, $a > 0$, $b > 0$, $ab = p^k$ and $\gcd(a, b) = 1$. Prove that $a$ and $b$ are each powers of $p$ with natural exponents.

*Expected answer.* A complete Lean proof supplying $i, j \in \mathbb{N}$ with $a = p^i$ and $b = p^j$. No relation between $i, j$ beyond these equalities is requested.

*Task ID.* `ntb_1413c3ebe87c42aaf582105d:whole:0`

TEST 5. **Order modulo three**

In $\mathbb{Z}/3\mathbb{Z}$, prove that 10 has multiplicative order 1. Also prove that some positive natural $k$ satisfies $10^k = 1$ and no natural $j$ with $0 < j < k$ satisfies $10^j = 1$.

*Expected answer.* A complete Lean proof of the order equality and an existential witness $k$ with positivity, power equality and minimality.

*Review note.* All powers and equalities involving 10 occur in ZMod 3.

*Task ID.* `ntb_16aa8a2ecd721698a83e6389:whole:0`

TEST 6. **Divisor totient sum bound**

Let $n \in \mathbb{N}$ with $1 < n$. Prove $\sum_{d|n} \varphi(d) \leq n^2 - 1$, summing over the natural divisors of $n$. The subtraction is natural-number subtraction.

*Expected answer.* A complete Lean proof of the divisor-sum inequality under $1 < n$.

*Task ID.* `ntb_16d0b14eb4f46af62451c672:whole:0`

TEST 7. **Prime divisibility and a square**

Let $n \in \mathbb{Z}$ and let $p \in \mathbb{N}$ be prime. Prove both $p \mid n \iff n^2 \bmod p = 0$ and $p \mid ((p+1)^2 - 1)$, interpreting $p$ as an integer in both claims.

*Expected answer.* A complete Lean proof of both the divisibility equivalence and the second divisibility statement.

*Review note.* The remainder and subtraction occur in the integers.

*Task ID.* `ntb_17574657fab0aeeab228be04:whole:0`

TEST 8. **Negated Fibonacci congruence**

Prove that it is false that every natural prime $p$ satisfies $\sum_{i=1}^{p} F_i \equiv J(p, 5) \pmod{p}$. Here $F_i$ is the natural Fibonacci number, cast to an integer in the sum, and $J$ is the Jacobi symbol.

*Expected answer.* A complete Lean proof negating the universal prime congruence, rather than proving that congruence for all primes.

*Task ID.* `ntb_1a7891d49276f2fdb7643bdf:whole:0`

TEST 9. **Decimal digit-sum remainder**

For every positive $n \in \mathbb{N}$, prove that the sum of its base-ten digits has the same remainder modulo 9 as $n$.

*Expected answer.* A complete Lean proof of $(\sum \mathrm{digits}_{10}(n)) \bmod 9 = n \bmod 9$ for every $n > 0$.

*Review note.* Recorded semantic overlap: DEV theorem 25 states the digit-sum congruence, and the first conjunct of DEV theorem 13 includes it. This is not a certified independent family.

*Task ID.* `ntb_1aa39c678607b996200182fc:whole:0`

TEST 10. **Order and prime exponent equivalence**

Let $n \in \mathbb{N}$ be prime and let $a \in \mathbb{Z}/n\mathbb{Z}$ be nonzero. Prove $(a^{n-1} \neq 0 \wedge a^n = 1) \iff (\mathrm{ord}(a) = 1 \vee \mathrm{ord}(a) = n)$, where ord is multiplicative order.

*Expected answer.* A complete Lean proof of the equivalence, preserving the conjunction on its left and the disjunction on its right.

*Task ID.* `ntb_1af4833b840d0b0fbec15e8c:whole:0`

TEST 11. **Cancel a coprime congruence factor**

Let $a, b, m, x, y \in \mathbb{Z}$ satisfy $ax \equiv b \pmod{m}$, $ay \equiv b \pmod{m}$ and $\gcd(a, m) = 1$. Prove $x \equiv y \pmod{m}$.

*Expected answer.* A complete Lean proof of the final congruence under all three assumptions.

*Review note.* No positivity or nonzero assumption on the integer modulus is stated.

*Task ID.* `ntb_1d4a55a1396005c641f975de:whole:0`

TEST 12. **Odd square divisibility**

Let $n \in \mathbb{N}$ be odd. Assuming $3 \neq 0$ in $\mathbb{N}$ and $2 < n$, prove both $8 \mid (n^2 - 1)$ and $((n^2)^2 \bmod 8) \leq 9$. The subtraction is natural-number subtraction.

*Expected answer.* A complete Lean proof of the implication from the stated conjunction to both target claims.

*Review note.* The second target is the stated remainder bound; do not replace it with a sharper congruence.

*Task ID.* `ntb_20e25d7da730561b7776bb3e:whole:0`

TEST 13. **Divisibility equivalence from a product**

Let $a, b, c \in \mathbb{Z}$ and assume $ab \mid c$. Prove $a \mid c \iff b \mid c$.

*Expected answer.* A complete Lean proof of both directions of the divisibility equivalence under the product-divisibility assumption.

*Task ID.* `ntb_21c9319bec71cef5f8c5a55d:whole:0`

TEST 14. **Square root of a field quotient**

Let $p \in \mathbb{N}$ be an odd prime and let $a, b, k, l \in \mathbb{Z}/p\mathbb{Z}$. Assume $a \neq 0$, $b \neq 0$, $ka^2 + lb^2 = 0$, $l$ is a square and $k$ is not a square. Prove that a nonzero $x \in \mathbb{Z}/p\mathbb{Z}$ satisfies $x^2 = -(l/k)$.

*Expected answer.* A complete Lean proof supplying $x$ and proving both $x \neq 0$ and $x^2 = -(l/k)$.

*Review note.* Division is field division in ZMod p, not integer or natural division.

*Task ID.* `ntb_2b2b05cfd427f7eccfae7a28:whole:0`

TEST 15. **Odd integer positive powers**

Let $a \in \mathbb{Z}$ satisfy $a \equiv 1 \pmod 2$. For every positive natural exponent $k$, prove $a^k \equiv 1 \pmod 2$.

*Expected answer.* A complete Lean proof of the congruence for every exponent in the positive naturals.

*Task ID.* `ntb_2c9caaaff14dc84be05601b1:whole:0`

TEST 16. **Natural quotient lower bound**

Let $a, b, k \in \mathbb{N}$ satisfy $b \geq 3$, $b$ odd, $a > 0$, $k > 0$ and $2kb(b-1) \geq a(b-1)^2$. Prove $k \operatorname{div} a \geq (b-1) \operatorname{div} (2b)$. Both quotients and all subtractions are natural-number operations.

*Expected answer.* A complete Lean proof of the inequality between the natural quotients under every stated assumption.

*Review note.* Recorded intent warning: for natural b at least three, the right quotient is zero, so the ratio bound degenerates. This is not a rational-ratio statement.

*Task ID.* `ntb_317a357e031c74d97e1ba8f4:whole:0`

TEST 17. **Negated Jacobi residue formula**

Prove that it is false that every odd natural prime $p$ satisfies $J(-3, p) = 1$ when $p \bmod 6 = 1$ and $J(-3, p) = -1$ otherwise. Here $J$ is the Jacobi symbol with integer numerator.

*Expected answer.* A complete Lean proof negating the universally quantified conditional formula for odd primes.

*Review note.* The outer negation covers the whole universal statement; the otherwise branch includes every residue other than one.

*Task ID.* `ntb_3260c7ed4ee56866de06e982:whole:0`

TEST 18. **No coprime square representation**

Prove that there are no positive natural numbers $a, b$ satisfying $\gcd(a, b) = 1$ and $b^2 = 5a$.

*Expected answer.* A complete Lean proof of the nonexistence of $a, b$ satisfying all four conditions.

*Task ID.* `ntb_36c456b2166347d83b0ed3f3:whole:0`

TEST 19. **Prime divisors and nonsquare two**

Prove three claims: every natural prime $p$ has exactly two natural divisors; 2 has exactly two natural divisors; and no natural number $k$ satisfies $k^2 = 2$.

*Expected answer.* A complete Lean proof of the full conjunction, including the universal prime claim, the concrete divisor count and the nonexistence claim.

*Task ID.* `ntb_38d4e0e8d6c4a9e630133c64:whole:0`

TEST 20. **Existence of prime product witnesses**

Prove that natural numbers $p, q, r, n$ exist such that $p, q, r$ are prime, $pqr = n$ and $(p+1)(q+1)r = n + 138$.

*Expected answer.* A complete Lean proof supplying all four witnesses and proving all three primality claims and both equalities.

*Review note.* The primes are not required to be distinct.

*Task ID.* `ntb_398249d36acc8ef9a5e1ca40:whole:0`

TEST 21. **Empty set of nonsquare conditions**

Prove that no $n \in \mathbb{N}$ has this property: for every natural prime $p$, both $n^2 + 4n + 1 - p$ and $2n^2 + n + 2 - p$ are nonsquares in $\mathbb{N}$. Both subtractions truncate at zero.

*Expected answer.* A complete Lean proof that the set of natural $n$ satisfying the stated universal conjunction is empty.

*Review note.* Natural truncated subtraction is part of the statement; do not read these as integer polynomial differences.

*Task ID.* `ntb_3ca1d1b12b5dfddc44f1290e:whole:0`

TEST 22. **Remainder of prime-square minus one**

Let $p \in \mathbb{N}$ be prime, and let $a \in \mathbb{N}$ be arbitrary. Prove $(p^2 - 1) \bmod (p^2 + 1) = p^2 - 1$, with natural-number subtraction and remainder.

*Expected answer.* A complete Lean proof of the remainder identity in the stated context.

*Review note.* The arbitrary parameter a appears in the statement but not in the assumptions or target.

*Task ID.* `ntb_3fd7520aa33f27729d8a7a07:whole:0`

TEST 23. **Negated quadratic representation theorem**

Prove that the following universal claim is false. Every natural prime $p > 5$ for which $-5$ is a square in $\mathbb{Z}/p\mathbb{Z}$ has positive natural $m, n, k$ with $p^2 = m^2 + 5n^2 + 6k^2$; every such positive representation has $m + n + k \geq 6$; and some such representation has $m + n + k = 6$.

*Expected answer.* A complete Lean proof negating the universal conjunction of representation existence, the lower bound for every positive representation, and attainment of that bound.

*Review note.* The target is the negation of the entire universal claim, not a request to construct those representations for every qualifying prime.

*Task ID.* `ntb_43bce97fd20bfa368ddf2b30:whole:0`

TEST 24. **Nonprime exponential with an odd factor**

Let $N \in \mathbb{N}$ be positive. Suppose some odd natural prime divides $N$. Prove that $3^N + 1$ is not prime.

*Expected answer.* A complete Lean proof of nonprimality under the positivity and odd-prime-divisor assumptions.

*Task ID.* `ntb_44ecc8d8d9a04797f8633215:whole:0`

TEST 25. **Totient of a prime power**

Let $p \in \mathbb{N}$ be prime and let $k \in \mathbb{N}$ be positive. Prove $\varphi(p^k) = p^k - p^{k-1}$, where $\varphi$ is Euler totient and the subtractions occur in $\mathbb{N}$.

*Expected answer.* A complete Lean proof of the totient identity under primality and positive exponent assumptions.

*Review note.* Recorded semantic overlap: TEST theorem 40 states the same prime-power totient identity.

*Task ID.* `ntb_45847c8fcf7e4ce252bf9497:whole:0`

TEST 26. **Least prime square residue modulo seven**

Prove that 11 is prime, $11 > 7$ and its residue in $\mathbb{Z}/7\mathbb{Z}$ is a square. Also prove $11 \leq q$ for every natural prime $q > 7$ whose residue modulo 7 is a square.

*Expected answer.* A complete Lean proof of all three properties of 11 and the universal minimality statement.

*Task ID.* `ntb_4613017e2c6d8b3546380196:whole:0`

TEST 27. **Reciprocal-sum disjunction**

Prove that positive natural numbers $a, b, c$ exist for which at least one branch holds: they are pairwise coprime and $1/a+1/b+1/c = 9/(a+b+c)$; or $1/a+1/b+1/c = 10/(a+b+c)$; or $1/a + 1/b + 1/c = 11/(a + b + c)$. All fractions are rational numbers.

*Expected answer.* A complete Lean proof supplying positive witnesses and proving a branch of the exact disjunction. Pairwise coprimality is required only for the branch with numerator 9.

*Review note.* Recorded intent warning: pairwise coprimality constrains only the first branch, not the branches with numerators 10 or 11.

*Task ID.* `ntb_49e6cc8d82bba05f96425e21:whole:0`

TEST 28. **Sum of prime proper divisors**

Let $p \in \mathbb{N}$ be prime. Prove that the sum of its natural divisors strictly less than $p$ equals 1.

*Expected answer.* A complete Lean proof of $\sum_{d|p,\ d<p} d = 1$.

*Task ID.* `ntb_4b422de08469fc142a8bd3f2:whole:0`

TEST 29. **Even-term sum bound by a prime power**

Let $p \in \mathbb{N}$ be prime with $p \neq 2$ and assume $p^p \geq p + 1$. Prove $\sum_{k=0}^{p-1}(2k + 2) \leq p^p$.

*Expected answer.* A complete Lean proof of the finite-sum bound under primality, exclusion of two and the stated power bound.

*Task ID.* `ntb_4c9e096311c691a10fc18f4c:whole:0`

TEST 30. **Prime exceeding a square-root threshold**

Let $n \in \mathbb{N}$ satisfy $1 < n$. Prove that some natural prime $p \neq n$ satisfies $\lfloor\sqrt{n}\rfloor < p^2$, where the square root is the natural integer square root.

*Expected answer.* A complete Lean proof supplying $p$ with primality, $p \neq n$ and the square-root inequality.

*Review note.* The conclusion does not require p to divide n or to be less than n.

*Task ID.* `ntb_507de6b94de228dd2f022572:whole:0`

TEST 31. **Divisor sum of a prime square**

Let $p \in \mathbb{N}$ be prime, and let $S$ be the natural sum of the divisors of $p^2$. Prove both $S = 1 + p + p^2$ and, after casting to $\mathbb{Q}$, $S/(p^2 + p) = 1 + 1/(p(p + 1))$.

*Expected answer.* A complete Lean proof of the natural divisor-sum equality and the rational quotient equality.

*Review note.* The second equality uses rational division after casts; it is not a natural quotient.

*Task ID.* `ntb_549962e77307dd430ec3b45c:whole:0`

TEST 32. **Negated weighted Jacobi-sum identity**

Prove that it is false that every natural prime $p$ with $p \bmod 4 = 3$ satisfies $\sum_{k=1}^{p-1} k^3 J(k, p) - p^2 \sum_{k=1}^{p-1} kJ(k, p) = 0$. Both sums and the subtraction are integer-valued, and $J$ denotes the Jacobi symbol.

*Expected answer.* A complete Lean proof negating the universal weighted-sum identity under the stated prime and residue conditions.

*Task ID.* `ntb_554b2dc790fab84a168853ef:whole:0`

TEST 33. **Integer function determined at primes**

Let $x : \mathbb{N} \to \mathbb{Z}$ satisfy $n = \prod_{d|n} x(d)$ for every natural $n \geq 1$, with $n$ cast to an integer. Prove both $x(1) = 1$ and $x(p) = p$ for every natural prime $p$, again casting $p$ to an integer.

*Expected answer.* A complete Lean proof of the value at one and the universal prime-value claim under the divisor-product hypothesis.

*Task ID.* `ntb_58cf341d56782a157b4f37da:whole:0`

TEST 34. **Totient ratio at a prime**

Let $p \in \mathbb{N}$ be prime. After casting $p$ and $\varphi(p)$ to $\mathbb{Q}$, prove $\varphi(p)/p = 1 - 1/p$.

*Expected answer.* A complete Lean proof of the rational identity under the prime assumption.

*Task ID.* `ntb_5a32ab19491895603fc1af22:whole:0`

TEST 35. **Even modulus fails a totient divisibility**

Let $n \in \mathbb{N}$ be even with $n > 2$. Prove $n \nmid (2^{\varphi(n)} - 1)$, where the subtraction is natural-number subtraction.

*Expected answer.* A complete Lean proof that the stated natural divisibility does not hold.

*Task ID.* `ntb_6be76276083ed05501bfa715:whole:0`

TEST 36. **Prime powers differ from a truncated difference**

Let $p, m, n \in \mathbb{N}$ with $p$ prime and assume $m \geq 0 \wedge n \geq 0$. Prove $p^{m+n} \neq p^m - p^n$, where subtraction truncates at zero.

*Expected answer.* A complete Lean proof of the inequality under all stated assumptions.

*Review note.* The nonnegativity assumptions are explicit but already follow from the natural-number domains.

*Task ID.* `ntb_73b60c27b218a377c1d1c37e:whole:0`

TEST 37. **Congruent exponents modulo a totient**

Let $a, b \in \mathbb{N}$ satisfy $a \equiv b \pmod{\varphi(125)}$. Prove $2^a \equiv 2^b \pmod{125}$ using natural-number congruence.

*Expected answer.* A complete Lean proof of the power congruence under the exponent congruence assumption.

*Task ID.* `ntb_767e128f330effc40b628557:whole:0`

TEST 38. **Nondivisibility of a sum of squares**

Let $p \in \mathbb{N}$ be prime with $p \bmod 4 = 3$, and let $a, b \in \mathbb{Z}$. Assume $p \mid a$ and $p \nmid b$, casting $p$ to an integer. Prove $p \nmid (a^2 + b^2)$.

*Expected answer.* A complete Lean proof of nondivisibility under all four assumptions.

*Task ID.* `ntb_771a85cbc730dded6e9f1fe8:whole:0`

TEST 39. **Residue class with two nonsquares**

For arbitrary $a, b \in \mathbb{Z}$, prove that $m \in \mathbb{N}$ and $c \in \mathbb{Z}$ exist with $m > 0$ such that every natural prime $p \equiv c \pmod{m}$ has both $-a$ and $-b$ nonsquare in $\mathbb{Z}/p\mathbb{Z}$.

*Expected answer.* A complete Lean proof supplying $m, c$, proving $m > 0$, and proving the universal pair of nonsquare claims for primes in the class.

*Review note.* Recorded intent warning: the residue class need not contain a prime, so the universal claim can be vacuous. No coprimality of c and m or prime existence is required.

*Task ID.* `ntb_78234cf56468fef0944a0a2a:whole:0`

TEST 40. **Repeated prime-power totient identity**

Let $p, k \in \mathbb{N}$ with $p$ prime and $k > 0$. Prove $\varphi(p^k) = p^k - p^{k-1}$, where $\varphi$ is Euler totient and the subtractions occur in $\mathbb{N}$.

*Expected answer.* A complete Lean proof of the totient identity under both assumptions.

*Review note.* Recorded semantic overlap: TEST theorem 25 states the same prime-power totient identity.

*Task ID.* `ntb_797a92fae369277d867f92aa:whole:0`

TEST 41. **Rational inequality for an even sequence sum**

Let $p \in \mathbb{N}$ be prime with $p \neq 2$. Let $a : \mathbb{N} \to \mathbb{N}$ satisfy $a(i) = 2i$ for every natural $i$. If $S = \sum_{k=1}^{100} a(k)$, prove $S \neq 2S/p$ after casting the sums and $p$ to $\mathbb{Q}$.

*Expected answer.* A complete Lean proof of the rational inequality under the prime, exclusion and sequence assumptions.

*Review note.* The division and comparison are rational after casts; the sum has natural terms.

*Task ID.* `ntb_79908f21d3a04c4fdaa934ac:whole:0`

TEST 42. **Prime divisor of adjacent factorial sum**

Prove two claims about natural numbers. First, there exist $n > 0$ and a prime $p$ such that $p \nmid (5n)!$, $p \mid (5n+1)! + (5n+2)!$, and $p \bmod 5 = 1$. Second, every positive $n$ and prime $p$ satisfying those two divisibility conditions has $p \bmod 5 \geq 1$.

*Expected answer.* Supply witnesses for the existence claim and prove the universal remainder bound. Both conjuncts require complete Lean proofs.

*Task ID.* `ntb_7a12343ef5f6b7715d2b8dd7:whole:0`

TEST 43. **Unequal powers for a nonprime exponent**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and assume $n$ is not prime. Prove $2^{n^2-n} \neq 3^n$, with the exponent subtraction performed in natural numbers.

*Expected answer.* Prove the stated inequality under both hypotheses.

*Task ID.* `ntb_7a8b6877ed0dbbf567a789ba:whole:0`

TEST 44. **Maximum digit product under divisibility constraints**

For natural-number digits $A, B < 10$ with $A \geq 1$, impose divisibility of $100000A + 20210 + B$ by both 4 and 9. Prove that an admissible pair has $AB = 42$, and that every admissible pair has $AB \leq 42$.

*Expected answer.* Provide admissible digit witnesses attaining 42 and prove the upper bound for every pair satisfying all digit and remainder conditions.

*Task ID.* `ntb_7b9304b912534907d37cabf6:whole:0`

TEST 45. **Number of divisors of a prime**

Let $n \in \mathbb{N}$ be prime. Prove that its finite set of positive divisors has cardinality 2.

*Expected answer.* Prove $|\text{divisors}(n)| = 2$.

*Task ID.* `ntb_7ed9af586fa024354709e2f9:whole:0`

TEST 46. **A power differs from its natural quotient**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and assume $n$ is even. Prove $2^n \neq \lfloor 2^n/4 \rfloor$, where the quotient is Lean natural-number division.

*Expected answer.* Prove the inequality of these two natural numbers.

*Review note.* The division is natural-number division, not rational division.

*Task ID.* `ntb_819c891a3cb42e3ed8f8c31c:whole:0`

TEST 47. **Different prime-base powers after subtraction**

Let $p, n \in \mathbb{N}$. Assume $p$ is prime, $4 < p$, $1 < n$, and $n$ is not prime. Prove $p^n - 1 \neq 2^n - 1$, with both subtractions performed in natural numbers.

*Expected answer.* Prove the inequality while retaining all four assumptions.

*Task ID.* `ntb_81a6446c304a5842e1b56484:whole:0`

TEST 48. **Inverse of an assumed quadratic equivalence**

Let $n \in \mathbb{N}$ be positive and not prime. Assume $\sigma : \mathbb{N} \to \mathbb{N}$ is a bijection satisfying $\sigma(x) = x^2+2$ for every natural $x$. Prove $\sigma^{-1}(2^3 n) \neq n$.

*Expected answer.* Prove the inverse-value inequality from the exact stated assumptions.

*Review note.* Existing review flags contradictory assumptions: a bijection must attain zero, whereas $x^2 + 2$ cannot. This is a source warning from manual review, not fresh Lean verification.

*Task ID.* `ntb_8a04bce03beb48fdc3862d35:whole:0`

TEST 49. **Negation of a Jacobi sum identity**

Prove that the following identity does not hold for every natural prime $p$ with $p \bmod 4 = 3$: $\sum_{k=1}^{p-1} k^4 \left(\frac{k}{p}\right) - p \sum_{k=1}^{p-1} k^2 \left(\frac{k}{p}\right) = 0$. Here $\left(\frac{k}{p}\right)$ is the Jacobi symbol, and both sums are integer-valued.

*Expected answer.* Refute the universally quantified identity. The target is its negation; it does not request a proof that every eligible prime violates it.

*Review note.* Preserve the outer negation and the scope of the universal quantifier.

*Task ID.* `ntb_8da859b76deb65034cb08063:whole:0`

TEST 50. **A prime divisor-square sum is not fourteen**

Let $n \in \mathbb{N}$ be prime and assume $n \neq 2$. Prove $\sum_{k \mid n,\ k>0} k^2 \neq 14$, where the sum ranges over the finite set of positive divisors of $n$.

*Expected answer.* Prove the inequality for the natural-number divisor sum.

*Task ID.* `ntb_8fa705e94721f2d74586f12b:whole:0`

TEST 51. **Bound on a sum of squared natural quotients**

Let $n \in \mathbb{N}$ satisfy $1 < n \leq 100$ and $3 \mid n$. Prove $\sum_{k=1}^{n} (2\lfloor n/k \rfloor)^2 \leq 2n(n^2 - 1)$. The quotients and subtraction are natural-number operations.

*Expected answer.* Prove the sum bound for every $n$ satisfying all three hypotheses.

*Review note.* The summands contain natural-number division, not rational fractions.

*Task ID.* `ntb_947d302024239a0c66ed6c03:whole:0`

TEST 52. **Equivalence of a remainder and divisibility claim**

Let $a \in \mathbb{N}$ be prime with $2 < a$. Prove $a^{a-1} \bmod a \neq 0 \iff a \mid a^{a-1} + 1$, with a natural-number exponent and remainder.

*Expected answer.* Prove both directions of the exact biconditional.

*Review note.* An equivalence does not assert that either side is true.

*Task ID.* `ntb_966e922f641f91f7ef67f209:whole:0`

TEST 53. **Prime square divisibility equivalence**

Let $a \in \mathbb{N}$ be prime with $2 < a$. Prove $a \mid a^2 + 1 \iff a = a^2 + 1$.

*Expected answer.* Prove both implications under the prime and size hypotheses.

*Review note.* An equivalence does not assert that either side is true.

*Task ID.* `ntb_a0396886f1b22e0a21fd7e72:whole:0`

TEST 54. **Least totient witness with two prime factors**

Prove that there exists a positive $n \in \mathbb{N}$ of the form $n = 2^x 3^y$ for positive natural $x, y$, with $\varphi(n) = \lfloor n/3 \rfloor$. Require $n$ to be no larger than every other positive $m$ of that form satisfying the same totient equation. Also require the sum of the distinct prime factors of $n$ to equal 5, where $\varphi$ is Euler's totient function.

*Expected answer.* Supply $n$ and its positive exponent witnesses; prove the totient equality, the universal minimality condition, and the distinct-prime-factor sum.

*Review note.* The factor list is converted to a finite set before summation, so multiplicities are removed. Division is natural-number division.

*Task ID.* `ntb_a53c82259ef05c3220f305af:whole:0`

TEST 55. **One is a least positive totient divisibility exponent**

Prove $1 > 0$ and $2 \cdot 1 \mid \varphi(2^1 + 3^1)$. Also prove that every positive natural $x$ satisfying $2x \mid \varphi(2^x + 3^x)$ obeys $1 \leq x$, where $\varphi$ is Euler's totient function.

*Expected answer.* Prove the two concrete claims and the universally quantified lower bound.

*Task ID.* `ntb_a77aff91cab6f59d7fe10c04:whole:0`

TEST 56. **Nonzero residue power equivalence**

Let $n, a \in \mathbb{N}$. Assume $1 < n$, $2 < n$, and $a \neq 0$. Prove $(a \bmod n)^{n-1} \neq 0 \iff a \bmod n \neq 0$, with natural-number subtraction in the exponent.

*Expected answer.* Prove both directions while retaining both stated bounds on $n$ and the assumption on $a$.

*Task ID.* `ntb_a8321a47269f6f807338a3be:whole:0`

TEST 57. **Zero solutions for a nonsquare coefficient**

Let $p \in \mathbb{N}$ be prime and $a \in \mathbb{Z}$. Assume the integer $p$ does not divide $a$, and the residue of $a$ is not a square in $\mathbb{Z}/p\mathbb{Z}$. Prove that every $x, y \in \mathbb{Z}/p\mathbb{Z}$ satisfying $ax^2 = y^2$ has $x = 0$ and $y = 0$, with $a$ cast into that residue ring.

*Expected answer.* Prove the universal implication and both zero conclusions in $\mathbb{Z}/p\mathbb{Z}$.

*Task ID.* `ntb_a980417194ef95918cb0bf7d:whole:0`

TEST 58. **Identify a multiple from its divisor sum**

Let $x \in \mathbb{N}$. Assume the sum of its positive divisors is 24 and $15 \mid x$. Prove $x = 15$.

*Expected answer.* Prove the exact value of $x$ from both hypotheses.

*Task ID.* `ntb_add27f3592c556b6b88b8c09:whole:0`

TEST 59. **Congruence equivalence under a negative-square premise**

Let $a, b, p \in \mathbb{N}$, with $p$ prime. Assume there exists an integer $x$ with $x^2 = -p$, where $p$ is cast into $\mathbb{Z}$. Prove $(p \mid a \vee p \mid b) \iff (a \equiv 0 \pmod p \vee b \equiv 0 \pmod p \vee a \equiv b \pmod p)$, where the congruences are integer congruences after casting $a, b, p$.

*Expected answer.* Prove both directions of the full biconditional, preserving all three alternatives on the right.

*Review note.* Existing review flags contradictory assumptions: an integer square cannot equal the negative of a natural prime. The initial state reveals that implicit $a, b$ are natural numbers. The contradiction warning is manual, not fresh Lean verification.

*Task ID.* `ntb_b23977f4ad063101726eb32a:whole:0`

TEST 60. **Square-sum inequality for a constrained integer sequence**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and $1 < n$, and assume $n$ is not prime. Let $a : \mathbb{N} \to \mathbb{Z}$ satisfy $a(k) = n\,a(k)$ for every $k$. Writing $S = \sum_{k=0}^{n-1} a(k)^2$, prove $Sn \geq (nS)/(n-1)$, where $n$ is cast to an integer and the division and subtraction on the right are integer operations.

*Expected answer.* Prove the integer inequality under the duplicated size hypothesis and the universal sequence constraint.

*Review note.* Existing intent review notes that the sequence constraint forces every $a(k)$ to zero when $1 < n$. Division is integer division. The source warning is manual, not fresh Lean verification.

*Task ID.* `ntb_b6e5df954bbe811b2595e721:whole:0`

TEST 61. **Negation of a Jacobi residue characterization**

Prove it is not true for every natural prime $p$ that $\left(\frac{-75}{p}\right) = 1 \iff p \bmod 12 = 1$, where the fraction notation denotes the Jacobi symbol.

*Expected answer.* Refute the universal biconditional. The goal is not to prove the negated biconditional for every prime.

*Review note.* Preserve the outer negation and its scope over all primes.

*Task ID.* `ntb_b967b0cf42b2c29b96f24210:whole:0`

TEST 62. **Sum bound from least common multiple and gcd**

Let $a, b \in \mathbb{N}$ be positive. Assume $\operatorname{lcm}(a, b) = 140$ and $\gcd(a, b) = 5$. Prove $a + b \leq 145$.

*Expected answer.* Prove the upper bound using the positivity, least-common-multiple, and greatest-common-divisor hypotheses.

*Task ID.* `ntb_ba0da67564e4aee3b28f8bc4:whole:0`

TEST 63. **A nonsquare product has a nonsquare factor**

Let $p \in \mathbb{N}$ be prime and odd. Assume 39 is not a square in $\mathbb{Z}/p\mathbb{Z}$. Prove that 3 is not a square in that ring or 13 is not a square in that ring.

*Expected answer.* Prove the disjunction of the two nonsquare claims in the same residue ring.

*Task ID.* `ntb_bb24d44ab7f1b559a280beda:whole:0`

TEST 64. **Finite-set sum under an interval equivalence**

Let $S$ be a finite set of natural numbers. Assume that for every natural $p$, the conjunction '$p$ is prime and $p \in S$' is equivalent to $1 < p \leq 10$. Prove $\sum_{k \in S} k = 17$.

*Expected answer.* Prove the finite-set sum equality from the exact universal biconditional.

*Review note.* Existing review flags contradictory assumptions: substituting $p = 4$ forces 4 to be prime. This does not encode the usual set of primes at most 10. The warning is manual, not fresh Lean verification.

*Task ID.* `ntb_bb8109462d7df8e0ab8a17eb:whole:0`

TEST 65. **Least square satisfying an iterated divisor sum**

For positive natural $u$, write $\sigma(u)$ for the sum of its positive divisors. Prove that there exist natural $n, k$ with $n > 0$, an odd natural $m$ satisfying $n = 2^k m$, a natural $s$ satisfying $s^2 = n$, and $\sigma(\sigma(n)) = 2n$. Require that no positive $n' < n$ simultaneously admits an odd-part representation $n' = 2^{k'} m'$, is a square, and satisfies $\sigma(\sigma(n')) = 2n'$.

*Expected answer.* Supply the witnesses and prove every existence condition, the iterated divisor-sum equality, and the exclusion of all smaller positive candidates.

*Review note.* No positivity condition is imposed on $k$ or $k'$. The oddness condition is the natural remainder equation $m \bmod 2 = 1$.

*Task ID.* `ntb_beedbd080221783debb0b0ff:whole:0`

TEST 66. **Totient congruence and positivity for a coprime base**

Let $p, a \in \mathbb{N}$, with $p$ prime and $\gcd(a, p) = 1$. Prove $a^{\varphi(p)} \equiv 1 \pmod{p}$, $a^p \neq 0$, and $a > 0$, where $\varphi$ is Euler's totient function. The congruence is over integers after casting $a$ and $p$; the other two conclusions are over naturals.

*Expected answer.* Prove all three conjuncts, including nonvanishing and positivity.

*Task ID.* `ntb_bf724e76836b4276f40bbe4d:whole:0`

TEST 67. **Divisor sum bound for a perfect nonprime**

Let $n \in \mathbb{N}$ satisfy $1 < n$ and assume $n$ is not prime. Assume its positive proper divisors sum to $n$. Prove that the sum of all its positive divisors is greater than $\lfloor n^2/(n+1) \rfloor$.

*Expected answer.* Prove the strict inequality of natural numbers under all three hypotheses.

*Review note.* The quotient on the right is natural-number division.

*Task ID.* `ntb_c0570ab949614d32f731fda0:whole:0`

TEST 68. **Power-product congruence modulo four**

Let $a, b, k \in \mathbb{N}$. Assume $a \bmod 2 = 1$ or $a \bmod 2 = 0$, assume the corresponding disjunction for $b$, and assume $(ab) \bmod 4 = 0$ or $(ab) \bmod 4 = 1$. Prove $a^k b^k \equiv (ab)^k \pmod{4}$ using natural-number congruence.

*Expected answer.* Prove the congruence while preserving all three disjunctive hypotheses.

*Task ID.* `ntb_c150b80beff0c40596c26203:whole:0`

TEST 69. **Base-seven digit length of a power minus one**

Let $k \in \mathbb{N}$ satisfy $k \geq 1$. Prove that the base-7 digit list of $7^k - 1$ has length $k$, where subtraction is over natural numbers.

*Expected answer.* Prove the exact digit-list length equality.

*Task ID.* `ntb_c40ed915e8bcab761ac71f09:whole:0`

TEST 70. **Negation of a squarefree counting identity**

Prove that the following equality does not hold for every positive natural $n$. Its left side counts natural $k$ with $0 \le k \le n$, $k > 0$, $k$ squarefree, and the natural quotient $\lfloor n/k \rfloor$ odd. Its right side is $\sum_{d \mid n,\ d>0} \mu(d)((n : \mathbb{Z})/(d : \mathbb{Z}))$, with integer division and the integer-valued Moebius function $\mu$; the count is cast to an integer.

*Expected answer.* Refute the universally quantified equality with these exact counting and arithmetic conventions.

*Review note.* The left-hand filter does not require $k \mid n$. Its quotient is natural-number division, whereas the right-hand quotient is integer division. Preserve the outer negation.

*Task ID.* `ntb_c881044f36da98a364d84d31:whole:0`

TEST 71. **Bound for a totient sum over divisors of a prime minus one**

Let $p \in \mathbb{N}$ be prime, and let $n \in \mathbb{N}$ satisfy $n \neq 0$. Prove $\sum_{k \mid p-1,\ k>0} \varphi(k) \le p^{p-1} + p - 1$, where $\varphi$ is Euler's totient function and subtraction is over natural numbers.

*Expected answer.* Prove the natural-number upper bound while retaining the stated nonzero parameter $n$.

*Review note.* The parameter $n$ and its hypothesis do not occur in the target expression.

*Task ID.* `ntb_cb9923d415db6eaf0b10d6bd:whole:0`

TEST 72. **Odd totient of a power of two**

For every $k \in \mathbb{N}$, prove that $\varphi(2^k)$ is odd if and only if $k \le 1$, where $\varphi$ is Euler's totient function.

*Expected answer.* Prove both directions, including the case $k = 0$.

*Task ID.* `ntb_d703b92567dd399885123b13:whole:0`

TEST 73. **Prime conclusion under a universal quadratic bound**

Let $p \in \mathbb{N}$ be prime with $p \equiv 1 \pmod 2$ and $4 < p$. Assume $2x^2 \le p^2$ for every natural $x$. Prove $p^4 = 1$ or $p = 2$.

*Expected answer.* Prove the exact disjunction under the prime, congruence, size, and universal-bound hypotheses.

*Review note.* Existing review flags contradictory assumptions: the universal bound at $x = p$ conflicts with $4 < p$. The warning is manual, not fresh Lean verification.

*Task ID.* `ntb_e17448422e4f2cbf449a13be:whole:0`

TEST 74. **Bound for a square with a square decimal prefix**

For every $N \in \mathbb{N}$, assume $N$ is a square, $N \bmod 10 \neq 0$, and $\lfloor N/100 \rfloor$ is also a square. Prove $N \le 1681$.

*Expected answer.* Prove the universal upper bound with all three premises.

*Review note.* The quotient $N/100$ is natural-number division, so it removes the final two decimal digits.

*Task ID.* `ntb_e5cc8ba57ce19eaf086ba3dd:whole:0`

TEST 75. **Divisor counts for a prime and its square**

For every natural prime $p$, prove that $p$ has exactly two positive divisors and $p^2$ has exactly three positive divisors.

*Expected answer.* Prove both cardinality equalities for the corresponding finite divisor sets.

*Task ID.* `ntb_e5f06a50a73ffcbedc223452:whole:0`

TEST 76. **Integer witnesses for reduced rational fractions**

Find integers $a, b, c, d$ such that $a, b, c \geq 0$, $\gcd(a, b) = 1$, $\gcd(c, d) = 1$, $b \neq 0$, $d \neq 0$, and $a/b + c/d = 1$ in $\mathbb{Q}$. Also require $a + b = c$ or $a + c = b$ or $b + c = a$, together with $a + c = 2b$ and $a + b + c = 6$.

*Expected answer.* Supply all four integer witnesses and prove every sign, gcd, nonzero-denominator, rational-equality, disjunction, and linear-equality condition.

*Review note.* The source does not require $d \geq 0$. Division is rational after casting the integer numerators and denominators.

*Task ID.* `ntb_f098a3f400e5643c0856230f:whole:0`

TEST 77. **Transfer of nonsquare residues through two congruences**

Let $p, q \in \mathbb{N}$ be distinct odd primes, and let $a, b, c \in \mathbb{Z}$. Assume $a \equiv b \pmod{p}$ and $a \equiv c \pmod{q}$, with $b$ nonsquare in $\mathbb{Z}/p\mathbb{Z}$ and $c$ nonsquare in $\mathbb{Z}/q\mathbb{Z}$. Prove that $a$ is nonsquare in both respective residue rings.

*Expected answer.* Prove the two nonsquare conclusions, one modulo $p$ and one modulo $q$.

*Review note.* The displayed initial state suppresses the two residue moduli; the statement distinguishes them.

*Task ID.* `ntb_f0ed5c021d48de136cceb5c7:whole:0`

TEST 78. **Power inequality from square divisibility**

Let $a, b \in \mathbb{Z}$ satisfy $1 < a$ and $1 < b$, and let $n \in \mathbb{N}$. Assume $a^2 \mid b^2$ over integers. Prove $a^{2n} \leq (b^n)^4$.

*Expected answer.* Prove the integer inequality for every natural exponent $n$, including zero.

*Task ID.* `ntb_f4cbc48f0a66be7e4c5910e7:whole:0`

TEST 79. **Least positive multiple with digit sum twelve**

Prove that there exists $s \in \mathbb{N}$ with $s = 48$, $s > 0$, decimal digit sum 12, and $s \bmod 12 = 0$. Also require $s \leq t$ for every positive natural $t$ whose decimal digit sum is 12 and whose remainder modulo 12 is zero.

*Expected answer.* Supply $s$ and prove its fixed value, positivity, digit sum, divisibility, and universal minimality.

*Task ID.* `ntb_f571b2ea2148e532749d00e3:whole:0`

TEST 80. **Prime divisors and Jacobi symbol at one hundred sixty-three**

Prove two claims. For every natural $p$, if $p$ is prime, $p \mid 163$, and 163 is prime, then $p = 163$. Also prove $\left(\frac{-163}{163}\right) = 0$, where the notation denotes the Jacobi symbol.

*Expected answer.* Prove the universally quantified implication and the concrete Jacobi-symbol equality.

*Review note.* Primality of 163 is an antecedent in the first conjunct; the statement does not separately request its proof.

*Task ID.* `ntb_f61adf6ee5fe0552363aedb5:whole:0`

TEST 81. **Integer gcd with a prime**

Let $p \in \mathbb{N}$ be prime and $k \in \mathbb{Z}$. Let $g$ be the nonnegative natural-valued gcd of the integers $k$ and $p$. Prove $g = 1$ or $g = p$; the second source comparison casts $g$ and $p$ to integers.

*Expected answer.* Prove one of the two stated gcd equalities with the source casts respected.

*Task ID.* `ntb_fc84491591d39fa8112b24ae:whole:0`

TEST 82. **Fixed square-divisor biconditional**

Prove that there exists a natural $k$ with $2024 = k^2$ if and only if every natural square $d$ dividing $2024^3$ satisfies $d \leq 2024^2$. The square premise on $d$ means that some natural $m$ satisfies $d = m^2$.

*Expected answer.* Prove both directions of the exact biconditional at the fixed number 2024.

*Review note.* The goal does not assert either side separately and is not a universally quantified characterization for arbitrary base numbers.

*Task ID.* `ntb_fdb9aa58f002498f2660c655:whole:0`